SUPPLEMENTARY MATERIAL

The following supplement contains the complete proofs, retained established results, numerical definitions, and additional validation details. The main paper and this supplement are also provided as one continuously referenced manuscript.

APPENDIX I
PROOFS OF THE SHARP GLOBAL ASYMPTOTIC AND ITS CONVERSES

Throughout this appendix the assumptions are the non-acceleration hypotheses of Theorem 1: $p_0 \in \mathcal{P}_2(\mathbb{R}^d)$, $u \in C^2$, u_t is globally L-Lipschitz with uniformly bounded $u_t(0)$, $\mathbf{a}_t$ is globally L_a-Lipschitz, and the stated uniform $L^2(p_t)$ bounds on acceleration and material jerk hold. The acceleration condition is not imposed. All constants below may depend on this fixed field and on T, but not on the grid. The proof does not assume a positive density, a Poincaré inequality, a fourth moment, or a smooth weighted-gradient projection.

A. Reference coordinates and transport of solenoidal fields

For $z \in \mathbb{R}^d$, set

$$X_t(z) = F_{0,t}(z), \qquad J_t(z) = DF_{0,t}(z),$$
$$c_t(z) = J_t(z)^{-1}\mathbf{a}_t(X_t(z)), \qquad C_t^\sigma(z) = \int_0^t \sigma(s)c_s(z)\,\mathrm{d}s.$$

Then

$$B_t^\sigma(X_t(z)) = J_t(z)C_t^\sigma(z). \tag{18}$$

Standard flow bounds, used here directly, give constants $C < \infty$ such that, uniformly in $t \in [0,T]$ and z,

$$\begin{aligned} |X_t(z)| + |\mathbf{a}_t(X_t(z))| &\le C(1+|z|), \\ \|J_t(z)\|_{\mathrm{op}} + \|J_t(z)^{-1}\|_{\mathrm{op}} &\le 2e^{LT}. \end{aligned} \tag{19}$$

Indeed $\sup_t |\mathbf{a}_t(0)| < \infty$ follows from continuity and compactness in time, so the assumed Lipschitz bound gives linear growth of acceleration. The spatial derivative satisfies $\partial_t J_t = Du_t(X_t)J_t$. In particular, as multiplication operators on $\mathcal{H} = L^2(p_0;\mathbb{R}^d)$,

$$\|J_t - J_r\|_{\mathcal{H}\to\mathcal{H}} \le Le^{LT}|t-r|. \tag{20}$$

Moreover c_t is continuous in $\mathcal{H}$ by dominated convergence. In fact it is Lipschitz there: differentiating along exact trajectories gives

$$\partial_t c_t = J_t^{-1}\{\mathbf{j}_t(X_t) - Du_t(X_t)\mathbf{a}_t(X_t)\},$$

whose $\mathcal{H}$ norm is at most $e^{LT}(C_j + LC_a)$. Thus C_t^σ is Lipschitz in $\mathcal{H}$ for every bounded measurable σ. The field B_t^σ is continuous in (t,x), locally C^1 in x with spatial derivatives bounded uniformly on compact sets for $0 \le t \le T$, and has uniformly linear growth. These claims follow by differentiating the reference formula locally: $u \in C^2$ makes J_t locally C^1 in z, $\mathbf{a} \in C^1$, and all flow images and inverse images of a fixed compact set remain in a common compact set. Differentiation under the bounded-time integral is then justified by its local continuous bounds.

Lemma S1 (Transport of the weighted solenoidal space)**.** *Let F be a C^1 bi-Lipschitz diffeomorphism, $q = F_\# p$, and define*

$$(F_* w)(y) = DF(F^{-1}y)w(F^{-1}y).$$

This is a bounded invertible map from $L^2(p)$ to $L^2(q)$ and maps $\mathcal{G}_p^\perp$ onto $\mathcal{G}_q^\perp$.

Proof. The norm statements follow from the derivative bounds for F and its inverse. For a compactly supported smooth test ϕ,

$$\int \nabla\phi \cdot F_* w \,\mathrm{d}q = \int \nabla(\phi\circ F)\cdot w \,\mathrm{d}p.$$

The composition is compactly supported and C^1. Such tests can be approximated by smooth compactly supported tests with uniformly convergent gradients, so the right side vanishes if $w \in \mathcal{G}_p^\perp$. Apply the same argument to F^{-1} for the converse. □

Lemma S2 (Continuity of the projected global coefficient)**.** *For bounded σ, the function $K(t) = \|\Pi_{p_t} B_t^\sigma\|_{L^2(p_t)}$ is Lipschitz on $[0,T]$.*

Proof. Put $S = \mathcal{G}_{p_0}^\perp$, a fixed closed subspace of $\mathcal{H}$. Lemma S1 and Eq. (18) give

$$K(t) = \inf_{z\in S} \|J_t(C_t^\sigma - z)\|_{\mathcal{H}}. \tag{21}$$

The norms $q_t(v) = \|J_t v\|_{\mathcal{H}}$ are uniformly equivalent to the norm of $\mathcal{H}$: with $m = e^{-LT}$ and $M = e^{LT}$, $m\|v\| \le q_t(v) \le M\|v\|$. The infimum has a minimizer in S because q_t is an equivalent Hilbert norm. If z_r is a minimizer at r, then

$$\begin{aligned} K(t) - K(r) &\le M\|C_t^\sigma - C_r^\sigma\|_{\mathcal{H}} + \|J_t - J_r\|_{\mathcal{H}\to\mathcal{H}}\|C_r^\sigma - z_r\|_{\mathcal{H}} \\ &\le M\|C_t^\sigma - C_r^\sigma\|_{\mathcal{H}} + m^{-1}\|J_t - J_r\|_{\mathcal{H}\to\mathcal{H}} K(r). \end{aligned}$$

The same bound with t, r exchanged controls the other sign. C_t^σ and J_t are Lipschitz in the indicated norms and $K(t) \le M \sup_r \|C_r^\sigma\|_{\mathcal{H}} < \infty$. This proves the assertion without differentiating Π_{p_t}. □

B. A tangent expansion uniform over the probability path

The next lemma supplies the exact norm needed after the classical trajectory expansion. Its single-measure version is the standard Wasserstein tangent norm [4, Chapter 8]; the proof below records the uniformity and domination needed here.

Lemma S3 (Uniform tangent expansion)**.** *Let $p_t = (X_t)_{\#}p_0$ be the flow family above. Suppose $b_t(x)$ is continuous in (t, x), locally C^1 in x with derivatives bounded uniformly on compact sets in time and space, and $|b_t(x)| \le C(1 + |x|)$. Suppose also that $k(t) = \|\Pi_{p_t} b_t\|_{L^2(p_t)}$ is continuous. Then*

$$\sup_{t\in[0,T]} \left| \frac{W_2((\mathrm{Id} + \epsilon b_t)_{\#}p_t, p_t)}{|\epsilon|} - k(t) \right| \longrightarrow 0 \quad \text{as } \epsilon \to 0. \tag{22}$$

Proof. The uniform linear-growth bounds imply $M_b = \sup_t \|b_t\|_{L^2(p_t)} < \infty$ and uniform tail control. Explicitly, $|X_t(z)| \le C_0(1 + |z|)$ implies

$$\begin{aligned} &\sup_t \int_{|x|>R} |b_t(x)|^2 \, p_t(\,\mathrm{d}x) \\ &\qquad \le C_1 \int_{|z|>R/C_0-1} (1 + |z|)^2 \, p_0(\,\mathrm{d}z) \longrightarrow 0. \end{aligned}$$

Let χ_R be a smooth spatial cutoff, equal to one on the ball of radius R and zero outside the ball of radius $2R$, and put $b_t^R = \chi_R b_t$. For fixed R the fields b_t^R are globally Lipschitz with a common constant L_R and

$$d_R := \sup_t \|b_t - b_t^R\|_{L^2(p_t)} \longrightarrow 0.$$

Apply Lemma S15 to b_t^R or $-b_t^R$ according to the sign of ϵ. The difference between its frozen flow and the map $\mathrm{Id} + \epsilon b_t^R$ has $L^2(p_t)$ norm at most $L_R \epsilon^2 e^{L_R|\epsilon|}\|b_t^R\|_{L^2(p_t)}/2$. Coupling the two Euler maps, and using contraction of Π_{p_t}, therefore yields, uniformly in t,

$$\frac{W_2((\mathrm{Id} + \epsilon b_t)_{\#}p_t, p_t)}{|\epsilon|} \le k(t) + 2d_R + C_R|\epsilon| \tag{23}$$

for $|\epsilon| \le 1$, where C_R is finite. First let $\epsilon \to 0$ and then $R \to \infty$ for the uniform upper bound.

For the lower bound, write $d_{t,\epsilon} = W_2((\mathrm{Id} + \epsilon b_t)_{\#}p_t, p_t)$. The same-start coupling gives $d_{t,\epsilon} \le |\epsilon| M_b$. Taylor expansion first in this coupling and then in an optimal coupling gives, for every $\phi \in C_c^\infty$,

$$|\langle b_t, \nabla\phi\rangle_{p_t}| \le \|\nabla\phi\|_{L^2(p_t)} \frac{d_{t,\epsilon}}{|\epsilon|} + |\epsilon|\|D^2\phi\|_\infty M_b^2. \tag{24}$$

For each fixed test both its numerator and its gradient norm are continuous in t: compose with $X_t(z)$ and use the stated continuity and the integrable growth bound. Fix $\delta > 0$. At each t with $k(t) \ge \delta$, the dual definition of projection supplies a test with positive gradient norm and ratio at least $k(t) - \delta$. Continuity gives a neighborhood in which that norm stays bounded away from zero and its ratio is at least $k(s) - 2\delta$. The compact set $\{t : k(t) \ge \delta\}$ admits a finite subcover by such neighborhoods. Equation (24) for this finite collection consequently gives $d_{t,\epsilon}/|\epsilon| \ge k(t) - 2\delta - C_\delta|\epsilon|$ on this set. On its complement nonnegativity gives a lower bound of $k(t) - \delta$. Let $\epsilon \to 0$ and then $\delta \to 0$. □

C. The global trajectory expansion with only a second moment

Lemma S4 (Variational expansion of recursive Euler)**.** *Under the grids and profiles of Theorem 2, initialize Euler and the exact trajectory at the same z. Then*

$$\left\| \sup_{0\le t\le T} \left| \frac{\widehat{X}_t^n - X_t}{\eta_n} + \frac{1}{2} J_t C_t^\sigma \right| \right\|_{L^2(p_0)} \longrightarrow 0. \tag{25}$$

Proof. Suppress n temporarily, and write $e_k(z) = \widehat{X}_k(z) - X_{t_k}(z)$. The exact local defect before its second-order expansion is

$$X_{t_{k+1}} - X_{t_k} - h_k u_{t_k}(X_{t_k}) = \int_0^{h_k} (h_k - s)\mathbf{a}_{t_k+s}(X_{t_k+s}) \, \mathrm{d}s.$$

Equation (19) and discrete Gronwall therefore give the pointwise estimate

$$\max_k |e_k(z)| \le CH(1+|z|). \tag{26}$$

Taylor expansion along the exact trajectory also gives

$$X_{t_{k+1}} = X_{t_k} + h_k u_{t_k}(X_{t_k}) + \tfrac{1}{2}h_k^2 \mathbf{a}_{t_k}(X_{t_k}) + r_k,$$
$$\|r_k\|_{L^2(p_0)} \le C_j h_k^3/6.$$

Put $D_k(z) = Du_{t_k}(X_{t_k}(z))$ and

$$n_k = u_{t_k}(X_{t_k} + e_k) - u_{t_k}(X_{t_k}) - D_k e_k.$$

The exact recurrence is

$$e_{k+1} = (I + h_k D_k)e_k - \tfrac{1}{2}h_k^2 \mathbf{a}_{t_k}(X_{t_k}) + h_k n_k - r_k. \tag{27}$$

The last two terms are negligible after propagation and division by η_n. For the Taylor remainder, Minkowski's inequality gives

$$\frac{1}{\eta_n}\sum_k \|r_k\|_{L^2(p_0)} \le \frac{C_j T H_n^2}{6\eta_n} \longrightarrow 0.$$

For the nonlinear remainder, fix z. All its exact and numerical positions for small H_n lie in one compact set depending on z. On that set Du is uniformly continuous in (t,x). If ω_z is a corresponding modulus in the spatial variable, then

$$\frac{1}{\eta_n}\sum_k h_k |n_k(z)| \le C\frac{H_n}{\eta_n}(1+|z|)\omega_z(CH_n(1+|z|)) \longrightarrow 0.$$

The left side is bounded by $C'(1+|z|)$, since $\|Du\|_{\mathrm{op}} \le L$ and H_n/η_n is bounded. Dominated convergence in $L^2(p_0)$ is therefore valid. This is the step that avoids a fourth-moment assumption or a global bound on D^2u.

The fundamental matrix products of $I + h_k D_k(z)$ are bounded by e^{LT}. For each fixed z they converge, uniformly over pairs of grid nodes, to the fundamental matrix of $\dot v = Du_t(X_t(z))v$, namely $J_t(z)J_s(z)^{-1}$. Here is an explicit consistency justification: the coefficient is continuous on $[0,T]$ and bounded by L; comparing one exact linear step with $I + h_k D_k$ gives a defect bounded by $Ch_k\{\omega_{D,z}(H_n) + H_n\}$ in operator norm. Summing and applying the uniform stability bound makes the total error tend to zero. Replacing the starting node t_{k+1} of a propagation product by t_k adds $O(H_n)$ by the same flow bound. Variation of constants in Eq. (27) now yields

$$\left\| \max_k \left| \frac{e_k}{\eta_n} + \frac{1}{2} J_{t_k} \sum_{j<k} \frac{h_{n,j}^2}{\eta_n} c_{t_{n,j}} \right| \right\|_{L^2(p_0)} \longrightarrow 0. \tag{28}$$

Indeed every pointwise matrix-replacement error tends to zero and is bounded by $C(1+|z|)$ after summation, because $\sum_j h_{n,j}^2/\eta_n \le TH_n/\eta_n$. Thus these replacements also pass to $L^2(p_0)$ by dominated convergence.

For each fixed z, continuity of $c_s(z)$ shows uniformly in the upper grid node that

$$\sum_{j<k} \frac{h_{n,j}^2}{\eta_n} c_{t_{n,j}}(z) - \int_0^{t_{n,k}} \sigma_n(s) c_s(z)\, \mathrm{d}s \longrightarrow 0.$$

The difference is bounded by $(TH_n/\eta_n)\sup_{|s-r|\le H_n} |c_s(z) - c_r(z)|$. Weak-star convergence of σ_n gives, for every fixed z, t,

$$\int_0^t \sigma_n(s) c_s(z)\, \mathrm{d}s \longrightarrow \int_0^t \sigma(s) c_s(z)\, \mathrm{d}s.$$

The primitives are equi-Lipschitz in t for this fixed z, so the convergence is uniform over $t \in [0,T]$. Both differences are dominated by $C(1+|z|)$, allowing one more application of dominated convergence in $L^2(p_0)$. This proves the node version of Eq. (25).

Finally, Euler's continuous interpolation is the linear interpolation of its node values. The difference between the chord of the exact trajectory and that trajectory is bounded by $CH_n^2(1+|z|)$, using its second derivative $\mathbf{a}_t(X_t(z))$. After division by η_n this vanishes in L^2. The continuous field $t \mapsto J_t(z)C_t^\sigma(z)$ has a uniformly vanishing interpolation modulus for each fixed z, dominated by $C(1+|z|)$. These two facts extend the node result uniformly to all times. □

Proof of Theorem 2. Let $b_t = -B_t^\sigma/2$. Equations (18) and (25), using the common starting point as a coupling, imply

$$\sup_t W_2\big(\widehat{p}_t^{\,n}, (\mathrm{Id} + \eta_n b_t)_{\#} p_t\big) = o(\eta_n).$$

The regularity and growth properties of b_t were proved above, and Lemma S2 gives continuity of its projected norm. Apply Lemma S3 and the triangle inequality to obtain Eq. (4). Since $H_n \to 0$, the maximum of the continuous coefficient over grid nodes converges to its maximum over $[0,T]$. This proves Eq. (5). □

D. Quantile schedules and the characterization theorem

For a positive C^1 function ρ with integral one, the quantile nodes obey, by the mean value theorem, $Nh_k = 1/\rho(\xi_k)$ for some $\xi_k \in [t_k, t_{k+1}]$. The positive minimum of ρ gives $H_N = O(N^{-1})$, and uniform continuity gives $\sigma_N = Nh_k \to 1/\rho$ uniformly on the intervals. Taking $t = T$ in Theorem 2 proves Eq. (6). The projection can be moved inside the integral because it is a bounded linear operator on the fixed Hilbert space $L^2(p_T)$. Also $g_s^T = \Pi_{p_T} A_{s,T}$ is continuous in that space: after composition with X_T, $A_{s,T}$ is $J_T c_s$, a continuous $L^2(p_0)$ curve.

Proof of Theorem 3. Item 1 implies item 2 by Theorem 1; item 2 implies item 3 immediately. For item 3, take $\eta_n = h_n = T/N_n$, so $\sigma_n = 1$. Equation (5) forces $B_t^1 \in \mathcal{G}_{p_t}^{\perp}$ at every t. Lemma S1 says equivalently

$$C_t^1 = \int_0^t c_s \, \mathrm{d}s \in S = \mathcal{G}_{p_0}^{\perp} \quad \text{for every } t.$$

The curve is continuously differentiable in $L^2(p_0)$ with derivative c_t. Difference quotients lie in the closed subspace S, so $c_t \in S$ for all t, including the endpoints by one-sided limits. Transporting back gives $\mathbf{a}_t \in \mathcal{G}_{p_t}^{\perp}$, which is item 1.

Item 1 also implies item 4, because each fixed positive allocation density has $H_N = O(N^{-1})$ and item 2 applies. Conversely suppose item 4 holds, even only for densities in a prescribed C^1 neighborhood of $1/T$. Equation (6) gives $\int g_s^T/\rho(s) \, \mathrm{d}s = 0$ for each such density. For an arbitrary smooth real function f, set

$$\sigma_\lambda(s) = 1 + \lambda f(s), \qquad Z_\lambda = \int_0^T \frac{\mathrm{d}s}{\sigma_\lambda(s)}, \qquad \rho_\lambda = \frac{\sigma_\lambda^{-1}}{Z_\lambda}.$$

For sufficiently small $|\lambda|$, these are positive smooth probability densities and $\rho_\lambda \to 1/T$ in C^1. Since $1/\rho_\lambda = Z_\lambda(1 + \lambda f)$, subtracting the $\lambda = 0$ equality gives $\int_0^T f(s) g_s^T \, \mathrm{d}s = 0$. Testing against every vector in $L^2(p_T)$ and then using continuity in s gives $g_s^T = 0$ for every s. Thus $A_{s,T}$ is solenoidal under p_T; the inverse of the transport in Lemma S1 makes $\mathbf{a}_s$ solenoidal under p_s. This is item 1. □

For clarity, the positive first-order constant in the non-preserving case on uniform grids is

$$c_u = \frac{1}{2} \max_{0 \le t \le T} \left\| \Pi_{p_t} \int_0^t A_{s,t} \, \mathrm{d}s \right\|_{L^2(p_t)}. \tag{29}$$

If it vanished, the preceding proof would establish the acceleration condition. It is finite by the flow bounds and strictly positive otherwise. Equation (5) therefore gives $\max_k W_2 = c_u h + o(h)$, which proves the claimed $\Theta(h)$ conclusion.

If the uniform terminal coefficient vanishes but the acceleration condition fails, then $\int_0^T g_s^T \, \mathrm{d}s = 0$ but the continuous Hilbert-valued curve g_s^T is not identically zero. Choose a smooth f such that $V = \int_0^T f(s) g_s^T \, \mathrm{d}s \neq 0$. The densities ρ_λ just constructed have exact terminal coefficient $Z_\lambda |\lambda| \|V\|/2 > 0$ for every sufficiently small nonzero λ. This proves the stated sensitivity to arbitrarily small schedule changes. The same conclusion holds near any positive smooth density ρ_0 whose terminal coefficient vanishes: replace $1 + \lambda f$ by $1/\rho_0 + \lambda f$ and normalize its reciprocal. At $\lambda = 0$ the normalizing integral is $\int \rho_0 = 1$, and the first-order coefficient is again a positive scalar times $|\lambda| \|V\|/2$.

Proof of Corollary 4. Theorem 6 excludes the acceleration condition for every generator in the stated class when m or g is nonaffine. Equation (29) and Theorem 2 then give the strictly positive first-order constant for each fixed generator. When both functions are affine, the same rigidity theorem identifies the unique acceleration-preserving generator and proves its acceleration is identically zero. Its exact trajectories are straight with constant velocity, so Euler is exact on every grid. Every other fixed generator fails the acceleration condition and is covered by the same positive-constant conclusion. □

E. Finite schedule tests and the Gaussian counterexample

Fix $T = 1$, $a > 0$, and a smooth real function f with $\|f\|_\infty < a^2$. Put $b = \sqrt{a^2 - f}$ and $M_t = aI + b(t)J$. The flow of $u_t(x) = M_t x$ is

$$F_{s,t} = e^{a(t-s)} \exp\left(J \int_s^t b(r) \, \mathrm{d}r \right).$$

It therefore generates $p_t = \mathcal{N}(0, e^{2at} I_2)$ from the standard Gaussian. Direct differentiation gives

$$\mathbf{a}_t(x) = \{ f(t) I + [b'(t) + 2ab(t)] J \} x, \qquad \Pi_{p_t} \mathbf{a}_t = f(t) x. \tag{30}$$

Indeed Jx is solenoidal for every circular Gaussian, whereas x is in its gradient closure by smooth cutoff approximation of $|x|^2/2$. All hypotheses of Theorem S11 hold: the fields are smooth and linear in x with bounded matrix coefficients on the compact time interval, and Gaussian moments are finite. The acceleration condition holds exactly when $f = 0$.

Every Euler update is a scalar expansion followed by a rotation. Writing its isotropic standard deviation after k steps as r_k, we have the exact recurrences and distance

$$r_0 = 1, \qquad r_{k+1} = r_k\sqrt{(1+ah_k)^2 + b(t_k)^2 h_k^2},$$
$$W_2(\widehat{p}_k, p_{t_k}) = \sqrt{2}\,|r_k - e^{at_k}|. \tag{31}$$

For sufficiently small H, uniformly in t and $0 \le h \le H$,

$$\begin{aligned}\frac{1}{2}\log\{1 + 2ah + (2a^2 - f(t))h^2\} - ah \\ = -\frac{1}{2}f(t)h^2 + \left(-\frac{2}{3}a^3 + af(t)\right)h^3 \\ + \left(a^4 - a^2 f(t) - \frac{1}{4}f(t)^2\right)h^4 + O(h^5).\end{aligned} \tag{32}$$

The remainder is uniform because f is bounded and the logarithm's argument stays in a fixed neighborhood of one. In particular, for any refining grid,

$$\log(r_N/e^a) = -\frac{1}{2}\sum_k f(t_k)h_k^2 + O\left(\sum_k h_k^3\right). \tag{33}$$

For a fixed positive C^1 density ρ, its quantile grid satisfies

$$\sum_k f(t_k)h_k^2 = \frac{1}{N}\int_0^1 \frac{f(s)}{\rho(s)}\,\mathrm{d}s + O(N^{-2}). \tag{34}$$

To see the order of the remainder, write the sum as $N^{-1}\sum_k f(t_k)(Nh_k)h_k$. The mean value formula and the C^1 bound on $1/\rho$ give $Nh_k = 1/\rho(t_k) + O(H_N)$ uniformly. The remaining left Riemann sum for f/ρ has error $O(H_N)$, and $H_N = O(N^{-1})$. Also $\sum h_k^3 = O(N^{-2})$. Thus vanishing of the integral in Eq. (34) gives an actual $O(N^{-2})$ endpoint error through Eq. (31), rather than merely the $o(N^{-1})$ conclusion supplied by a zero first-order coefficient.

Proof of Proposition 5. Choose $m+1$ nonzero smooth bump functions with disjoint compact supports in $(0,1)$. The m constraints $\int_0^1 f/\rho_i = 0$ form a homogeneous linear system in their $m+1$ coefficients, so there is a nonzero solution. Disjoint supports make the resulting f nonzero. Multiplying it by a sufficiently small nonzero scalar gives $\|f\|_\infty < a^2$ and arbitrarily small size in any fixed finite number of derivative norms. The same is true of $\sqrt{a^2 - f} - a$ by smoothness of the square root near a^2. Equations (33) and (34) prove all the endpoint $O(N^{-2})$ bounds. Equation (30) shows failure of the acceleration condition, and Theorem 3 gives $\Theta(N^{-1})$ maximum error on uniform grids.

For finitely many observation times τ_j, impose the additional linear constraints $\int_0^{\tau_j} f/\rho_i = 0$ and choose more disjoint bumps than constraints. If the time is not a grid node, insert it into the quantile grid or use the Euler continuous interpolation. Splitting the one interval containing a fixed time changes the sum in Eq. (33) by $O(N^{-2})$. The same endpoint argument at τ_j applies. This is a finite collection of observation times, not a dense family whose size grows with N. □

F. The explicit cosine example and its exact coefficients

Take $f(t) = \epsilon\cos(2\pi t)$, with $0 < \epsilon < a^2$. For uniform steps $h = 1/N$ and $N \ge 3$, discrete Fourier sums give

$$\sum_{k=0}^{N-1} f(k/N) = 0, \qquad \sum_{k=0}^{N-1} f(k/N)^2 = N\epsilon^2/2.$$

Summing Eq. (32) gives the stronger expansion

$$\log(r_N/e^a) = -\frac{2a^3}{3N^2} + \frac{a^4 - \epsilon^2/8}{N^3} + O(N^{-4}). \tag{35}$$

Since $W_2 = \sqrt{2}e^a|\exp(\log(r_N/e^a)) - 1|$, this proves Eq. (8). Setting $\epsilon = 0$ gives the preserving field $a(I+J)x$ and exactly the same leading second-order coefficient.

Because all involved matrices commute with the rotation flow,

$$A_{s,t}(x) = \{f(s)I + [b'(s) + 2ab(s)]J\}x.$$

Projection under p_t removes its Jx component. Thus the entire first-order error profile for uniform grids, including continuous interpolation, is

$$\lim_{N\to\infty} NW_2(\widehat{p}_t^N, p_t) = \frac{\sqrt{2}\,e^{at}\epsilon}{4\pi}|\sin(2\pi t)|, \tag{36}$$

uniformly in t. At $t = 1/4$, taking N divisible by four identifies the interpolation with the numerical node and proves the displayed intermediate-time coefficient in the main text. The maximum coefficient is positive and, if an explicit value is desired, its maximizer and value are

$$t_* = 1 - \frac{1}{2\pi}\arctan(2\pi/a),$$
$$C_{\max} = \frac{\sqrt{2}\,\epsilon}{4\pi} e^{at_*} \sin\big(\arctan(2\pi/a)\big).$$

There is one local maximum in each half interval; the second is larger by the factor $e^{a/2} > 1$.

Finally let $\rho_\delta(t) = 1 + \delta\cos(2\pi t)$ with $0 < |\delta| < 1$. Its quantile nodes are the unique solutions of

$$t_k + \frac{\delta}{2\pi}\sin(2\pi t_k) = k/N.$$

The classical integral $\int_0^1 (1+\delta\cos(2\pi t))^{-1}\,\mathrm{d}t = (1-\delta^2)^{-1/2}$ follows, for example, by the substitution $z = \tan(\pi t)$ on the two half intervals. Since $\cos\theta/(1+\delta\cos\theta) = \delta^{-1}\{1-(1+\delta\cos\theta)^{-1}\}$,

$$\int_0^1 \frac{f(t)}{\rho_\delta(t)}\,\mathrm{d}t = \frac{\epsilon}{\delta}\left(1 - \frac{1}{\sqrt{1-\delta^2}}\right).$$

Equation (6), or directly Eqs. (33)–(34), now gives Eq. (9). Its coefficient is strictly positive for every nonzero δ, however small.

APPENDIX II
NONLINEAR PARITY, RIGIDITY, AND QUANTITATIVE RESIDUAL GAPS

A. A pointwise identity for every fixed-shape dilation

First consider a centered scaling path $p_t(x) = g(t)^{-d}\rho(x/g(t))$, and abbreviate $\alpha = g'/g$ and $\kappa = g''/g = \alpha' + \alpha^2$. Every generating field can be written $u = \alpha x + r$ with $\nabla\cdot(p_t r) = 0$. Distributionally,

$$\begin{aligned}\partial_t p_t &= -\alpha\nabla\cdot(p_t x),\\ \partial_{tt} p_t &= \alpha^2\nabla\cdot\nabla\cdot(p_t x x^\top) - \kappa\nabla\cdot(p_t x).\end{aligned} \tag{37}$$

For clarity, the double divergence of a tensor S is $\sum_{ij}\partial_i\partial_j S_{ij}$. The identity

$$\nabla\cdot\nabla\cdot(p_t x x^\top) = \nabla\cdot(p_t x) + \nabla\cdot[x\nabla\cdot(p_t x)]$$

proves the second formula directly. Differentiating the continuity equation of u gives

$$\partial_{tt} p_t = \nabla\cdot\nabla\cdot(p_t u u^\top) - \nabla\cdot(p_t \mathbf{a}_u).$$

Moreover $\nabla\cdot\nabla\cdot(p_t x r^\top) = \nabla\cdot(p_t r) = 0$, and the transposed cross tensor has the same double divergence. Acceleration preservation therefore implies

$$\nabla\cdot\nabla\cdot(p_t r r^\top) = -\kappa\nabla\cdot(p_t x). \tag{38}$$

All identities follow either pointwise with the stated smoothness or by testing against compactly supported smooth functions.

Fix time and write $p \propto e^{-V_t}$. Since $\operatorname{div} r = \nabla V_t\cdot r$,

$$\begin{aligned}p^{-1}\operatorname{div}\operatorname{div}(p r r^\top) &= p^{-1}\operatorname{div}(p(Dr)r)\\ &= \operatorname{tr}((Dr)^2) + r\cdot\nabla(\operatorname{div} r) - \nabla V_t\cdot(Dr)r\\ &= \operatorname{tr}((Dr)^2) + r^\top D^2 V_t\, r.\end{aligned}$$

Dividing (38) by p proves (12). The trace is that of the matrix square, not the squared Frobenius norm; it can be negative.

B. Large level sets and the all-generator rigidity theorem

Proof of Theorem 6. The tail condition implies Gaussian upper tails. In fact, on every ray outside R_0, $\partial_r V(re) \ge cr$; integration gives $V(re) \ge cr^2/2 - C$ uniformly in unit e. Thus all polynomial moments are finite. The affine-mean restriction already proved in Corollary S26 gives $m'' = 0$. Removing this affine mean does not change material acceleration or its weighted divergence, so we work with $m = 0$.

Let $M_\rho = \mathbb{E}_\rho\|Z\|^2 > 0$. The gradient base field $v = \alpha x$ belongs to the weighted-gradient closure, and $r = u - v$ is orthogonal to it. The second-moment identity therefore gives

$$\|r_t\|^2_{L^2(p_t)} = \tfrac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2}(g(t)^2 M_\rho) - (g'(t))^2 M_\rho = g(t)g''(t)M_\rho. \tag{39}$$

Hence $g'' \geq 0$, and $g'' = 0$ forces $r = 0$ almost everywhere, then everywhere by continuity and positivity of p_t.

Suppose $g'' > 0$ at some time. Freeze that time, let $L = \mathrm{Lip}(r)$, and put $W(x) = V(x/g)$, dropping an irrelevant additive constant. For sufficiently large $\|x\|$ there are positive c_t, M_t with $x \cdot \nabla W \geq c_t \|x\|^2$ and $D^2 W \preceq M_t I$. Choose R_1 beyond this region and a level $s > \max_{\|x\| \leq R_1} W(x)$. On every ray from zero, $W(re)$ is strictly increasing for $r \geq R_1$ and tends to infinity. It therefore crosses s at exactly one radius $R_s(e) > R_1$. The implicit function theorem shows that R_s is smooth. Thus $\{W = s\}$ is a compact smooth radial graph diffeomorphic to S^{d-1}. As $s \to \infty$, its minimum Euclidean radius tends to infinity. Also $\|\nabla W(x)\| \geq c_t \|x\|$ there.

Project r onto the tangent bundle of this level set. Because $d-1$ is even, the hairy-ball theorem [9, Section 2.2] gives a point x_s where that projection vanishes. For $d = 1$, the tangent space of a level point is zero and the same conclusion is immediate. At x_s, r is parallel to ∇W. Weighted divergence then gives

$$\|r(x_s)\| = \frac{|\nabla W(x_s) \cdot r(x_s)|}{\|\nabla W(x_s)\|} = \frac{|\operatorname{div} r(x_s)|}{\|\nabla W(x_s)\|} \leq \frac{dL}{\|\nabla W(x_s)\|}. \tag{40}$$

But the pointwise identity requires

$$\kappa(x_s \cdot \nabla W(x_s) - d) \leq dL^2 + \frac{M_t d^2 L^2}{\|\nabla W(x_s)\|^2}.$$

The left side tends to infinity and the right side remains bounded. This contradiction proves that $g'' = 0$ everywhere. Equation (39) proves uniqueness of the generator.

Conversely, if m and positive g are affine, the field (11) has zero material acceleration. Its exact flow from s to t is $m(t) + (g(t)/g(s))(x - m(s))$; since both scalar and mean increments are affine, this map equals its Euler step on every interval. Its global regularity and moment assumptions follow from $\min_{[0,T]} g > 0$ and the source moments. □

The topological fact used here can be seen directly. A nowhere-zero continuous tangent field on S^{d-1} would normalize to $\tau(e)$ with $\|\tau(e)\| = 1$ and $e \cdot \tau(e) = 0$. Then $H_s(e) = \cos(\pi s)e + \sin(\pi s)\tau(e)$ is a sphere-valued homotopy from the identity to the antipodal map. Their degrees are 1 and $(-1)^d$, which differ for odd d. This argument is classical; the new use is to force an obstruction on the unconstrained nonlinear generator through its weighted-divergence and acceleration identities.

a) Finite Gaussian mixtures.: Let $\rho = \sum_j \pi_j \rho_j$ with nondegenerate Gaussian components, and set $w_j = \pi_j \rho_j / \rho$, $Q_j = \Sigma_j^{-1}$, $s_j(x) = Q_j(x - \mu_j)$. Differentiating the logarithm yields $\nabla V = \sum_j w_j s_j$ and $D^2 V = \sum_j w_j Q_j - \mathrm{Cov}_w(s_j)$. The covariance matrix is positive semidefinite, so $D^2 V \preceq \Lambda I$. The displayed radial lower bound in the main text gives $x \cdot \nabla V \geq (\lambda/2)\|x\|^2$ whenever $\|x\| \geq 2B/\lambda$. These prove every hypothesis of Theorem 6. No separation, equal covariance, radial symmetry, or log-concavity of the mixture is required.

b) Gaussian specialization.: For $p = \mathcal{N}(0, g^2 I)$, the pointwise identity reduces to

$$\|r(x)\|^2 + g^2 \operatorname{tr}((Dr(x))^2) = \frac{g''}{g}\|x\|^2 - dgg''. \tag{41}$$

When $g'' > 0$, bounded Dr would force asymptotically nonzero tangential speed on every sufficiently large sphere. Equation (40) is the topological contradiction. This is not the eigenvalue pairing argument for linear skew matrices.

Proposition S5 (Necessary Jacobian growth and polynomial rigidity)**.** *Fix $p = \mathcal{N}(0, g^2 I_d)$, $g > 0$, and $\kappa > 0$. Let r be a C^2 field satisfying* $\operatorname{div}(pr) = 0$ *and the stress equation* (38)*. No global Lipschitz or polynomial growth assumption is imposed here. If d is odd, then, for every $R > g\sqrt{d}$,*

$$\sup_{\|x\| = R} \|Dr(x)\|_{\mathrm{op}} \geq \frac{\sqrt{\kappa}\, R}{g\sqrt{d}} \sqrt{\frac{R^2 - dg^2}{R^2 + dg^2}}. \tag{42}$$

Consequently the liminf of this supremum divided by R is at least $\sqrt{\kappa}/(g\sqrt{d})$. In every dimension, if r is polynomial, then $r(x) = Kx$, where $K^\top = -K$ and $K^2 = -\kappa I_d$. Thus no polynomial correction is possible in odd dimension.

Proof. Write $M_R = \sup_{\|x\| = R} \|Dr(x)\|_{\mathrm{op}}$. At a hairy-ball pole Re of this sphere, $r(Re)$ is parallel to e. Weighted divergence therefore gives $\|r(Re)\| \leq dg^2 M_R / R$. The Gaussian identity (41) implies

$$\kappa(R^2 - dg^2) \leq \frac{d^2 g^4 M_R^2}{R^2} + dg^2 M_R^2.$$

Solving for M_R proves (42).

If r is a nonzero polynomial of degree $m \geq 2$, the leading term of $\|r\|^2$ has degree $2m$ and is a nonzero sum of squares. The derivative term in (41) has degree at most $2m - 2$, while the right side has degree two. This is impossible, so $r(x) = Kx + b$ is affine. The polynomial identity $\operatorname{tr} K = g^{-2} x \cdot (Kx + b)$ forces $K^\top = -K$ and $b = 0$. The quadratic terms in (41) then give $K^\top K = \kappa I_d$, equivalently $K^2 = -\kappa I_d$. An invertible real skew-symmetric matrix cannot have odd size. These conclusions are instantaneous; any time-dependent polynomial generator of the Gaussian scaling path must meet them at every time. □

C. Why all convex-observable tests can pass

Proof of Corollary 7. Let $X_t = g(t)Z$, with Z centrally symmetric. Differentiating along the base flow $v = \alpha x$ gives

$$F_\phi'' = \mathbb{E}[v^\top D^2\phi(X_t)v] + \kappa\mathbb{E}[X_t \cdot \nabla\phi(X_t)].$$

Central symmetry shows

$$\mathbb{E}[X \cdot \nabla\phi(X)] = \tfrac{1}{2}\mathbb{E}\big[X \cdot (\nabla\phi(X) - \nabla\phi(-X))\big] \geq 0,$$

because the gradient of a convex differentiable function is monotone. For $g'' \geq 0$ the second term is nonnegative. The first term is at least K_ϕ, since the base field is one admissible velocity in that infimum. This proves all the convex and weighted-energy inequalities, under the stated integrability for each test.

If $C_\rho = \mathrm{Cov}(Z)$ is positive definite, then $C = g^2 C_\rho$ and the choice $B = gg'C_\rho$ in the covariance test has Schur complement $gg''C_\rho \succeq 0$. For singular C_ρ, the same positive semidefinite block matrix follows by a direct tensor factorization on its support. The odd-dimensional exponential Gaussian has strictly convex scale, so Theorem 6 nevertheless excludes every admissible generator. □

D. Conditional variance identities and the Gaussian residual floor

Fix a time for a Gaussian scaling path and write $g = g(t)$, $\alpha = g'/g$, $\kappa = g''/g$, $c_t = gg''$. Let $X \sim p_t$ and $r = u - \alpha x$. For every unit vector e, write $Y = e \cdot X$, $q_g(y) = (2\pi g^2)^{-1/2} e^{-y^2/(2g^2)}$, and $Q_e(y) = \mathbb{E}[(e \cdot r(X))^2 \mid Y = y]$. Projecting $\mathrm{div}(p_t r) = 0$ onto this coordinate gives

$$\mathbb{E}[e \cdot r(X) \mid Y = y] = 0 \quad \text{for every } y. \tag{43}$$

Indeed the derivative of $q_g(y)$ times this conditional mean is zero; the only constant compatible with Gaussian tails and linear field growth is zero. The conditional mean and variance have continuous versions, by dominated convergence and Lipschitz growth, so these conclusions are available at every chosen y, not merely almost everywhere.

For any twice differentiable scalar test with bounded first and second derivatives, the scaling identity and the physical chain rule give

$$\int \nabla[\phi(e \cdot x)] \cdot \mathbf{a}_u(x)\, p_t(\,\mathrm{d}x) = \int_{\mathbb{R}} \phi''(y)\{c_t - Q_e(y)\} q_g(y)\,\,\mathrm{d}y. \tag{44}$$

To verify this, if $s(t) = g(t)^2$ then $\partial_t q_g = (s'/2)\partial_{yy} q_g$ and $\partial_{tt} q_g = (s''/2)\partial_{yy} q_g + (s'^2/4)\partial_{yyyy} q_g$. The latter equals $\partial_{yy}[q_g(\alpha^2 y^2 + c_t)]$. By (43), the conditional squared velocity is $\alpha^2 y^2 + Q_e(y)$. Subtracting the velocity-Hessian term from the second-derivative chain rule proves (44). If acceleration preservation were imposed, this also forces $Q_e(y) = c_t$: a function whose second distributional derivative is zero is affine, and integrability of $q_g(Q_e - c_t)$ excludes nonzero affine terms.

Proof of Theorem 8. Let $L = \mathrm{Lip}(r)$. Pointwise weighted divergence gives

$$|x \cdot r(x)| = g^2 |\,\mathrm{div}\, r(x)| \leq C, \qquad C = dg^2 L. \tag{45}$$

For every radius $R > 0$, the tangential projection of r on the sphere of radius R has a zero Re_R. Hence $r(Re_R)$ is parallel to e_R. For $y = R + s$, $0 \leq s \leq g$, condition on $e_R \cdot X = y$ and write $X = ye_R + Z_\perp$, where $Z_\perp \sim \mathcal{N}(0, g^2 I_{d-1})$. Since $Z_\perp \cdot r(Re_R) = 0$, (45) and Lipschitzness give

$$|e_R \cdot r(ye_R + Z_\perp)| \leq \frac{C + L\|Z_\perp\|\sqrt{s^2 + \|Z_\perp\|^2}}{R}.$$

Minkowski's inequality, $\mathbb{E}\|Z_\perp\|^2 = (d-1)g^2$, and $\mathbb{E}\|Z_\perp\|^4 = (d-1)(d+1)g^4$ imply

$$\sup_{R \leq y \leq R+g} \sqrt{Q_{e_R}(y)} \leq \frac{Lg^2}{R}\{d + \sqrt{d^2 + d - 2}\} = \frac{Lg^2 A_d}{R}. \tag{46}$$

Choose $R = g\max\{1, 2A_d L/\sqrt{\kappa}\}$. Then $Q_{e_R} \leq c_t/4$ throughout this strip.

Choose a scalar convex $C^{1,1}$ test with $\phi'' = g^{-1}\mathbf{1}_{[R,R+g]}$, and $\phi' = 0$ below R, so $0 \leq \phi' \leq 1$. Formula (44) and the dual definition of the projected norm yield

$$\varepsilon_t \geq \frac{3c_t}{4g}\Pr\{R \leq Y \leq R + g\} \geq \frac{3g''}{4\sqrt{2\pi}}\exp[-(R/g + 1)^2/2].$$

The test is legitimate by smooth mollification and then compact cutoff: its first and second derivatives are bounded, it grows at most linearly, and its gradients converge in $L^2(p_t)$. Gaussian moments dominate the chain-rule terms. Weak derivatives at the strip endpoints do not create an additional term, since ϕ' is continuous. Weakening $3/4$ to $1/2$ gives (13). Solving this inequality for L under the stated threshold on η gives (14). □

E. A nonexplicit uniform gap for the broad source class

Proposition S6 (Positive residual gap at fixed Lipschitz bound)**.** *Fix a positive density $p \propto e^{-W}$ satisfying (10), an odd d, $\kappa > 0$, and a finite $L \geq 0$. Among all globally L-Lipschitz fields r with $\operatorname{div}(pr) = 0$, the quantity*

$$\mathcal{R}_\kappa(r) = \sup_{\substack{\phi \in C_c^\infty \\ \|\nabla\phi\|_{L^2(p)} \leq 1}} \left| \kappa \int x \cdot \nabla\phi \, \mathrm{d}p - \int r^\top D^2\phi \, r \, \mathrm{d}p \right|$$

attains a finite strictly positive minimum. For a generator of the scaling path at this time, $\mathcal{R}_\kappa(r) = \|\Pi_p \mathbf{a}_u\|$.

Proof. Cutoff tests in the weighted-divergence equation give $\mathbb{E}_p r = 0$. Consequently $\|r(0)\| \leq L\mathbb{E}_p\|X\|$, giving a uniform linear-growth bound for this class. Arzela–Ascoli and a diagonal extraction give a locally uniform subsequential limit of every sequence. The Gaussian tail bound and uniform linear growth upgrade this to strong $L^2(p)$ convergence. Both the Lipschitz bound and weighted divergence survive the limit, so the class is compact in $L^2(p)$. For each fixed test ϕ, its displayed quadratic integral is continuous in this topology. A supremum of continuous functions is lower semicontinuous. The minimum is attained, and is finite because the candidate $r = 0$ has value $\kappa\|x\|_{L^2(p)}$.

If the minimum were zero, its Lipschitz minimizer would satisfy (38) distributionally. The pointwise identity extends to this regularity almost everywhere. Here is a justification: $r \in W^{1,\infty}_{\mathrm{loc}}$ and $\operatorname{div} r = \nabla W \cdot r \in W^{1,\infty}_{\mathrm{loc}}$. Mollification, the product rule, and convergence in local Sobolev norms give

$$\operatorname{div}((Dr)r) = \operatorname{tr}((Dr)^2) + r \cdot \nabla(\operatorname{div} r)$$

as distributions. Adding the weight cancels the first-order score terms exactly as above. Since $\|Dr\| \leq L$ almost everywhere and $D^2W \preceq MI$ in the tails, this yields

$$\kappa(x \cdot \nabla W - d) \leq dL^2 + M\|r(x)\|^2$$

almost everywhere there. Both sides are continuous, so the inequality holds everywhere. Similarly $|\nabla W \cdot r| \leq dL$ holds everywhere. Applying the level-set hairy-ball argument to the continuous field r gives the same contradiction as in Theorem 6.

For an actual scaling generator, integrating its second-derivative identity against ϕ gives the formula for $\mathcal{R}_\kappa$. The cross term is zero because $\int r \cdot D^2\phi \, x \, \mathrm{d}p = \int r \cdot \nabla(x \cdot \nabla\phi - \phi) \, \mathrm{d}p = 0$. The dual projection formula then proves the asserted equality. □

F. A separate radial-score extension

For completeness, odd-dimensional affine-scale rigidity also holds for radial densities that need not meet the upper-Hessian assumption. Suppose $\rho > 0$ is smooth and

$$\nabla \log \rho(z) = -b(\|z\|)z, \qquad 0 < \lambda \leq b(r) \leq \Lambda < \infty.$$

These hypotheses give Gaussian tails. At a fixed scale g, let q_g be any one-dimensional marginal and define its scalar Stein kernel

$$\tau_g(y) = q_g(y)^{-1} \int_y^\infty s q_g(s) \, \mathrm{d}s.$$

The marginal is even and $-q_g'(y)/q_g(y) = y\mathbb{E}[b(\|X\|/g) \mid e \cdot X = y]/g^2$. Integration of these score bounds gives $g^2/\Lambda \leq \tau_g(y) \leq g^2/\lambda$.

For a scale path, the projected continuity equation first gives $\mathbb{E}[e \cdot r \mid e \cdot X = y] = 0$. Equation (38) then implies

$$\mathbb{E}[(e \cdot r)^2 \mid e \cdot X = y] = \kappa\tau_g(y).$$

Indeed, writing the conditional variance as Q_e, the equation is $(q_g Q_e)'' = -\kappa(yq_g)'$; integrability removes the affine integration constants, and $(q_g\tau_g)' = -yq_g$. On the other hand, weighted divergence gives $|x \cdot r(x)| \leq dg^2L/\lambda$. At a hairy-ball pole Re_R, tangency vanishes. For $X = Re_R + Z_\perp$ conditioned on $e_R \cdot X = R$, the same cancellation $Z_\perp \cdot r(Re_R) = 0$ bounds

$$|e_R \cdot r(X)| \leq \frac{dg^2L/\lambda + L\|Z_\perp\|^2}{R}.$$

Conditional integration by parts gives, uniformly in R, $\mathbb{E}\|Z_\perp\|^2 \leq (d-1)g^2/\lambda$ and $\mathbb{E}\|Z_\perp\|^4 \leq (d-1)(d+1)g^4/\lambda^2$. For the latter identities, test the transverse score equation with z and $\|z\|^2 z$ and use its coefficient lower bound λ/g^2; cutoff limits are justified by the uniform Gaussian tails. Thus the conditional variance tends to zero as $R \to \infty$, contradicting $\kappa\tau_g(R) \geq gg''/\Lambda > 0$ if $g'' > 0$. Equation (39) handles the other signs and completes the same rigidity conclusion. This argument uses no bound on derivatives of b.

APPENDIX III
CONSTRUCTING A NONRADIAL PROBABILITY PATH AND ITS CORRECTION

The acceleration condition can be imposed before solving a nonlinear Eulerian equation. Let ρ be a reference law, b a field preserving ρ, and F_t smooth diffeomorphisms. If

$$\begin{aligned} \partial_{tt} F_t(z) &= DF_t(z) b(z), \\ p_t &= (F_t)_{\#}\rho, \\ u_t(F_t z) &= \partial_t F_t(z). \end{aligned} \tag{47}$$

then $\mathbf{a}_t(F_t z) = DF_t(z)b(z)$. For every compactly supported smooth test ϕ, the change of variables gives $\int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t = \int \nabla(\phi \circ F_t) \cdot b \, \mathrm{d}\rho = 0$. Thus the linear equation for F_t in Eq. (47) constructs an admissible acceleration. A general nonlinear pushforward does not have this property: the displayed second-derivative identity is essential.

A. An explicit globally regular family

Identify $\mathbb{R}^2$ with $\mathbb{C}$, so $Jz = iz$. Take $\rho = \mathcal{N}(0, I_2)$, $a, T > 0$, and $\epsilon > 0$. Define

$$\begin{aligned} f(z) &= z^2 e^{-|z|^2}, \qquad A_t = e^{(1+i)at}, \qquad B_t = \epsilon e^{(1+i)a\sqrt{2}t}, \\ F_t(z) &= A_t z + B_t f(z), \qquad q = \frac{4\epsilon}{e} e^{a(\sqrt{2}-1)T} < 1. \end{aligned} \tag{48}$$

The source law is ρ; the initial target law is $p_0 = (F_0)_{\#}\rho$, which is already non-Gaussian. Writing $k_t(F_t z) = DF_t(z)Jz$, set

$$v_t(F_t z) = \partial_t F_t(z) - a k_t(F_t z), \qquad u_{\omega,t} = v_t + \omega k_t. \tag{49}$$

Here v is a base generator, not a claimed minimum-action field.

Theorem S7 (A constructive nonradial family)**.** *Under Eq. (48), each F_t is a global smooth bi-Lipschitz diffeomorphism, uniformly on $[0, T]$. The laws $p_t = (F_t)_{\#}\rho$ have smooth positive densities on $\mathbb{R}^2$ and finite moments of every order. For every fixed $\omega \in \mathbb{R}$, u_ω generates exactly this same path, with trajectories*

$$X_t = F_t\big(e^{i(\omega - a)t} Z\big), \qquad Z \sim \rho.$$

The choice $\omega = a$ satisfies the acceleration condition, and all smoothness, growth, global Lipschitz, and moment assumptions of Theorem 1. Consequently its Euler distributions have error $O(H^2)$ on every finite grid. For $\epsilon > 0$, no p_t is an invertible affine image of a radial law or of a product of two independent real laws.

The key identity is

$$\partial_{tt} F_t = 2a^2 \partial_\theta F_t = 2a^2 DF_t Jz.$$

It follows because the angular harmonics z and $f(z)$ have degrees one and two, and their time exponents have squares $2ia^2$ and $4ia^2$. Also $\|Df\|_\infty \le 4/e$, so $F_t^{-1}(x)$ is the unique limit of the contraction $z_{m+1} = A_t^{-1}(x - B_t f(z_m))$, with contraction factor at most q. Thus the density $p_t(x) = \rho(F_t^{-1}x)/\det DF_t(F_t^{-1}x)$ and the velocity are computable without a PDE solve. The inverse iteration adds real computational cost; the numerical comparison includes it.

This is a family, not just one isolated deformation. Replace $B_t f$ by $\sum_{n=2}^{m} \epsilon_n e^{(1+i)a\sqrt{n}t} f_n(z)$, where $f_n(e^{i\theta}z) = e^{in\theta} f_n(z)$ and the f_n are smooth rapidly decaying functions. The same construction works whenever

$$\sum_{n=2}^{m} |\epsilon_n| e^{a(\sqrt{n}-1)T} \|Df_n\|_\infty < 1.$$

Arbitrary smooth radial profiles times z^n give such harmonics. This constructs a specified class of paths. It does not solve the inverse problem for an arbitrary externally supplied density path. Appendix IV proves all assertions, including the nonproduct certificate.

Corollary S8 (A Gaussian starting law)**.** *Replace B_t by $\epsilon \sinh((1+i)a\sqrt{2}t)$ and assume $(4/e)\epsilon\sqrt{2}\sinh(a\sqrt{2}T) < 1$. Then F_0 is the identity, so $p_0 = \mathcal{N}(0, I_2)$, and the acceleration-preserving construction and its $O(H^2)$ bound still hold. For every $t > 0$ the target law is neither affine-radial nor affine-product.*

This follows from $B_t'' = 4ia^2 B_t$ and $|\sinh((1+i)s)|^2 = \sinh^2 s + \sin^2 s \le 2\sinh^2 s$. The numerical study below uses the exponential-coefficient family of Eq. (48); the Gaussian-start variant is an additional proved construction, not a separately reported empirical result.

B. Discovering the correction by a finite global search

For the two-mode family let $F_{1,t} = A_t z$, $F_{2,t} = B_t f(z)$, and

$$\zeta_n = a\sqrt{n} + i\{a\sqrt{n} + n(\omega - a)\}.$$

Then $\mathbf{a}_\omega(t, F_t z) = \sum_{n=1}^{2} \zeta_n^2 F_{n,t}(z)$. Every ck_t preserves p_t, so the following is an upper certificate for the projected acceleration residual:

$$\begin{aligned} R_t(\omega, c)^2 &= \|\mathbf{a}_\omega - ck_t\|^2_{L^2(p_t)} \\ &= \sum_{n=1}^{2} M_n(t)|\zeta_n^2 - inc|^2, \\ M_1(t) = 2e^{2at}, &\qquad M_2(t) = \frac{8}{125}\epsilon^2 e^{2a\sqrt{2}t}. \end{aligned} \tag{50}$$

Angular orthogonality makes this formula exact for the continuum law. It avoids numerical differentiation and a weighted Poisson solve.

Proposition S9 (Global correction recovery in the stated family)**.** *At any fixed t, minimizing Eq. (50) over $(\omega, c) \in \mathbb{R}^2$ has the unique zero $(a, 2a^2)$. For a fixed ω the unique minimizing coefficient is*

$$c_*(\omega) = \frac{\sum_{n=1}^{2} nM_n \operatorname{Im}(\zeta_n^2)}{\sum_{n=1}^{2} n^2 M_n}.$$

After substitution, $R_t(\omega, c_*(\omega))^2$ is a coercive quartic polynomial. Its global minimum is found by evaluating its real cubic stationary points; on a bounded interval include the endpoints. The same statement holds for a time-integrated squared residual after replacing $M_n(t)$ by their positive integrals.

The proof is elementary and useful operationally. A zero in mode one forces $\omega = \pm a$ and $c = 2a\omega$. Mode two excludes $\omega = -a$. The leading quartic coefficient is $\sum n^4 M_n > 0$, while c_* is affine in ω. A local optimizer can converge to a nonzero residual minimum; the finite global search resolves this failure within this one-parameter correction family. Floating-point root enumeration is validated against the exact sum-of-squares identity, not presented as an interval-arithmetic proof.

C. An explicit distribution coupling and a particle limit

Theorem S10 (Phase coupling for the constructed field)**.** *For u_a, a uniform grid $h = T/N$, and Euler initialized at $F_0(Z)$, there is a constant C independent of N such that*

$$W_2(\widehat{p}_N, p_T) \le \|\widehat{X}_N - F_T(e^{-ia^2Th}Z)\|_{L^2(\rho)} \le Ch^2. \tag{51}$$

Meanwhile the common-start particle error has the strictly positive limit

$$\lim_{h\downarrow 0} \frac{\|\widehat{X}_N - F_T(Z)\|_{L^2}}{h} = a^2 T \|DF_T(Z)JZ\|_{L^2(\rho)} > 0. \tag{52}$$

The comparison curve $F_t(e^{-ia^2ht}Z)$ solves the explicit modified field $u_a - (h/2)\mathbf{a}_{u_a}$, which generates the same marginals. A second-order Taylor expansion cancels Euler's quadratic local defect, giving the strong $O(h^2)$ comparison in Eq. (51); no infinite modified-equation series is used. This gives a computable coupling bound for a genuinely nonradial law. It is an upper bound on full W_2, not its exact value. The theorem does not claim a positive h^2 limit for that optimal distance.

APPENDIX IV
PROOFS AND ADDITIONAL DETAILS OF THE HARMONIC CONSTRUCTION

A. Reference coordinates and path preservation

Let ρ be a probability measure and let a globally defined smooth flow generated by b preserve ρ. For a proper smooth diffeomorphism F_t, the pushforward vector field $w_t(F_t z) = DF_t(z)b(z)$ has zero weighted divergence under $(F_t)_\#\rho$. Indeed,

$$\int \nabla\phi(F_t z) \cdot DF_t(z)b(z)\, \rho(\mathrm{d}z) = \int \nabla(\phi \circ F_t)(z) \cdot b(z)\, \rho(\mathrm{d}z) = 0.$$

Properness makes the pulled-back test compactly supported. The exact flow of w_t at frozen physical time is the conjugate of the flow of b by F_t. For $u_t(F_t z) = \partial_t F_t z$, material differentiation along a fixed reference label yields $\mathbf{a}_t(F_t z) = \partial_{tt} F_t z$ and $\mathbf{j}_t(F_t z) = \partial_{ttt} F_t z$. Equation (47) therefore suffices exactly.

More generally, for a time-dependent scalar $\gamma(t)$, the field $\partial_t F_t + \gamma(t) DF_t Jz$ has trajectories $F_t(R_{\int_0^t \gamma(s)\,\mathrm{d}s} z)$, where $R_\theta z = e^{i\theta} z$. For a radial reference law all these fields generate the same p_t. In Eq. (49), $\gamma = \omega - a$. Only the constant-ω family is used for the polynomial search.

B. Global diffeomorphism, density, and regularity

The differential of $f(z) = z^2 e^{-|z|^2}$ in any unit real direction ξ obeys

$$|Df(z)\xi| \le 2|z|(1+|z|^2)e^{-|z|^2} \le 4/e.$$

The last scalar function is maximal at $|z| = 1$, as follows by differentiating it; its derivative has the sign of $1 + r^2 - 2r^4$. Hence

$$|F_t(z) - F_t(y)| \ge e^{at}(1-q)|z-y|,$$
$$|F_t(z) - F_t(y)| \le e^{at}(1+q)|z-y|.$$

For every x the map $z \mapsto A_t^{-1}(x - B_t f(z))$ is a contraction of the complete space $\mathbb{R}^2$ with constant at most $q < 1$. Its unique fixed point proves surjectivity and gives the inverse. The differential is nonsingular everywhere; the inverse function theorem gives a smooth inverse. The singular values of DF_t are between $e^{at}(1-q)$ and $e^{at}(1+q)$. The determinant is positive, because the perturbation may be continuously reduced to zero without crossing a singular differential. The density formula in the main text follows by change of variables.

Every derivative of f is a polynomial times $e^{-|z|^2}$. It and its products with any fixed power of $|z|$ are bounded. On a neighborhood of the compact time interval, the strict inequality $q < 1$ persists after a small extension of the interval. The inverse maps have uniformly bounded first derivative, and bounded higher derivatives of each fixed order. Consequently each of u_a and $\mathbf{a}_{u_a}$ is a fixed linear field plus a smooth bounded perturbation with bounded first derivatives. They are globally Lipschitz, have at most linear growth, and the required time derivatives exist on this neighborhood. Explicitly,

$$\mathbf{a}_{u_a}(t, F_t z) = 2ia^2 A_t z + 4ia^2 B_t f(z), \qquad \mathbf{j}_{u_a}(t, F_t z) = ((1+i)a)^3 A_t z + ((1+i)a\sqrt{2})^3 B_t f(z).$$

Their $L^2(p_t)$ norms are uniformly finite. The same smoothness and growth conclusions hold for each fixed u_ω, and uniformly for ω in compact sets. Gaussian moments give all the integrability needed for the later Taylor and moment arguments. This proves the regularity part of Theorem S7.

For the multi-mode extension, the sum of Lipschitz perturbation bounds is strictly less than the leading linear singular value by hypothesis. The same contraction argument applies. Angular equivariance gives $\partial_\theta f_n = inf_n$, while $\partial_{tt}(e^{(1+i)a\sqrt{nt}} f_n) = 2ia^2 n e^{(1+i)a\sqrt{nt}} f_n$. Thus $F_{tt} = 2a^2 F_\theta$ persists. Rapid decay of the f_n and all their derivatives, and finiteness of the sum, give the same regularity bounds.

C. Why the example is neither affine-radial nor affine-product

For a standard planar Gaussian Z, angular orthogonality gives

$$\mathbb{E}F_t(Z) = 0, \qquad \mathbb{E}F_t(Z)^2 = 0, \qquad \mathbb{E}F_t(Z)^3 = 0, \tag{53}$$
$$\mathbb{E}\big[F_t(Z)^2 \overline{F_t(Z)}\big] = A_t^2 \overline{B}_t \, \mathbb{E}[|Z|^4 e^{-|Z|^2}] = \frac{8}{27} A_t^2 \overline{B}_t \neq 0.$$

Also $\mathbb{E}|F_t(Z)|^2 = 2|A_t|^2 + (8/125)|B_t|^2$, so the real covariance is a positive multiple of I_2. An affine image of a radial law is centrally symmetric about its mean, which would make the last cubic moment zero. Thus no such representation exists.

Suppose instead that the centered real vector $F_t(Z)$ were an invertible affine image of independent scalar variables. Finite third moments follow from those of $F_t(Z)$ and the inverse affine map. After centering and standardizing the independent variables, isotropic covariance forces the remaining mixing matrix to be a scalar multiple of an orthogonal matrix. For independent centered real U, V,

$$\mathbb{E}(U+iV)^3 = \mathbb{E}U^3 - i\mathbb{E}V^3, \qquad \mathbb{E}[(U+iV)^2(U-iV)] = \mathbb{E}U^3 + i\mathbb{E}V^3.$$

The two quantities therefore have equal moduli. Orthogonal rotations, reflections, and common scaling preserve this equality of moduli. Equation (53) violates it. This rules out an affine-product representation, proving the final assertion of the theorem. The small deformation is consequently not one of the earlier radial, Gaussian, affine-radial, or independent-product constructions.

For Corollary S8, the proposed bound is a uniform upper bound on $(4/e)|B_t/A_t|$, because $e^{-at} \le 1$ and $|\sinh((1+i)a\sqrt{2}t)| \le \sqrt{2}\sinh(a\sqrt{2}T)$. The diffeomorphism and regularity proofs therefore apply unchanged. The second-derivative identity still holds, $B_0 = 0$, and $B_t \neq 0$ for $t > 0$ since its real argument is positive. The moment argument above continues to apply with this coefficient. This proves the corollary.

D. Exact residual polynomial and correction recovery

Along the u_ω trajectories the reference label rotates at rate $\omega - a$. The material derivative on functions of (t, z) is $\partial_t + (\omega - a)\partial_\theta$. Applying it twice to each $F_{n,t}$ gives the coefficient ζ_n^2 in the main text. Angular orthogonality makes different harmonics orthogonal in complex $L^2(\rho)$, which is the real vector L^2 norm. Since $|Z|^2$ has density $\frac{1}{2}e^{-s/2}$ on $[0, \infty)$,

$$\mathbb{E}|Z|^2 = 2, \qquad \mathbb{E}[|Z|^4 e^{-2|Z|^2}] = \frac{1}{2}\int_0^\infty s^2 e^{-5s/2}\,\mathrm{d}s = \frac{8}{125}.$$

This proves Eq. (50). The first mode is $\zeta_1 = a + i\omega$. Its squared real part is $a^2 - \omega^2$, so a zero residual requires $\omega = \pm a$ and $c = 2a\omega$. If $\omega = -a$, the second mode has real part of its squared coefficient $2a^2 - a^2(\sqrt{2}-4)^2 \neq 0$. If $\omega = a$, both mode coefficients equal inc for $c = 2a^2$. As all M_n are positive, these alternatives prove uniqueness of the zero.

For fixed ω, differentiate the quadratic objective in c to obtain c_*. The imaginary part of ζ_n^2 is affine in ω, so eliminating c changes only terms of degree at most two in ω. The real squared coefficients contribute the positive quartic leading term $\sum n^4 M_n$. Coercivity and differentiation therefore give the finite global minimization procedure. The same calculation applies to time integrals, because the coefficients ζ_n are time-independent. This proves Proposition S9.

The upper certificate controls the canonical projected residual by orthogonal projection. It need not equal that residual away from the recovered zero, because a single solenoidal basis field need not be the best approximation among all solenoidal fields. The proof certifies the exact algebraic solution $\omega = a, c = 2a^2$; machine residuals near roundoff describe a numerical recovery of this solution.

E. A finite modified-flow argument for the explicit coupling

Let $u = u_a$ and $\mathbf{a} = \mathbf{a}_u$. For each $0 < h \le h_0$, define $u_h = u - (h/2)\mathbf{a}$. The curve

$$Y_h(t,z) = F_t(e^{-ia^2 ht} z)$$

solves $\dot{Y}_h = u_h(t, Y_h)$ and starts from $F_0(z)$. For $Z \sim \rho$, its law at each t is exactly p_t. On a fixed bounded range of h, u_h and its required derivatives satisfy uniform linear-growth and spatial Lipschitz bounds by the previous regularity calculation. Its material acceleration is

$$\partial_t u_h + (Du_h)u_h = \mathbf{a} - \frac{h}{2}\{\mathbf{j} + (Du)\mathbf{a}\} + \frac{h^2}{4}(D\mathbf{a})\mathbf{a},$$

All remainder terms have an L^2 bound proportional to $1 + |z|$ along Y_h, uniformly in h and t; the third derivatives needed for the Taylor remainder have the same bound. Taylor expansion over one step gives

$$\begin{aligned} Y_h(t+h,z) &= Y_h(t,z) + h[u(t,Y_h) - (h/2)\mathbf{a}(t,Y_h)] \\ &\quad + \frac{h^2}{2}\mathbf{a}(t,Y_h) + O_{L^2(\rho)}(h^3) \\ &= Y_h(t,z) + hu(t, Y_h(t,z)) + O_{L^2(\rho)}(h^3). \end{aligned}$$

Minkowski's inequality and Euler stability imply $e_{k+1} \le (1 + Lh)e_k + Ch^3$, where $e_k = \|\widehat{X}_k - Y_h(t_k, Z)\|_{L^2}$ and $e_0 = 0$. Summing yields $\max_k e_k \le CTe^{LT}h^2$. Because rotations preserve ρ, the pairing at $t = T$ proves Eq. (51).

Finally, Taylor expansion in the rotation angle gives in $L^2(\rho)$

$$F_T(e^{-ia^2 Th}Z) = F_T(Z) - a^2 Th\, DF_T(Z)JZ + O_{L^2}(h^2).$$

The preceding $O(h^2)$ coupling error therefore shows $h^{-1}(\widehat{X}_N - F_T(Z)) \to -a^2 TDF_T(Z)JZ$ in L^2. Taking norms proves Eq. (52). Its constant is positive since DF_T is nonsingular and $\rho(Z = 0) = 0$. For a uniform grid the same argument works at every grid point with T replaced by t_k. The law-level $O(H^2)$ assertion for arbitrary grids is supplied separately by Theorem 1.

APPENDIX V
CONSTRUCTING AND CERTIFYING APPROXIMATE CORRECTIONS

The useful residual is the component of acceleration that changes the distribution. This component has a canonical definition even when its orthogonal complement does not have a classical flow. For a probability measure p, write

$$\mathcal{G}_p = \overline{\{\nabla\phi : \phi \in C_c^\infty(\mathbb{R}^d)\}}^{L^2(p)}$$

and let Π_p be the orthogonal projection onto this space. Its orthogonal complement consists exactly of the fields with zero weighted divergence. This is the standard tangent space decomposition in Wasserstein geometry [4, Chapter 8]; the contribution below is its quantitative use in the Euler error bound. Define

$$\varepsilon_t = \|\Pi_{p_t}\mathbf{a}_t\|_{L^2(p_t)} = \sup_{\substack{\phi \in C_c^\infty \\ \|\nabla\phi\|_{L^2(p_t)} \le 1}} \left| \int \mathbf{a}_t \cdot \nabla\phi \, dp_t \right|. \tag{54}$$

Thus $\varepsilon_t = 0$ is equivalent to the acceleration condition. No density or Poincaré inequality is needed for this definition.

Here canonical refers to the projection for the chosen generator u, not an intrinsic acceleration of the probability path. For the expanding circular Gaussian, $u(x) = ax$ gives $\Pi_p\mathbf{a} = a^2 x$, whereas $u(x) = a(I + J)x$ gives $\Pi_p\mathbf{a} = 0$ on the same path. Intrinsic Wasserstein covariant acceleration instead first uses the minimum-energy gradient velocity $\Pi_p u$.

A. A residual theorem without a regular projected flow

Theorem S11 (Projected acceleration controls Euler error)**.** *Keep the smoothness, spatial Lipschitz, growth, and moment assumptions of Theorem 1, including the common Lipschitz constant L_a of $\mathbf{a}_t$, but do not impose the acceleration condition. Set*

$$E_H = \exp(L_a H^2/2), \qquad B_H = (1+LH)E_H,$$
$$D_H = \frac{C_j}{6} + E_H\left(\frac{LC_a}{2} + \frac{HL_aC_a}{8}\right). \tag{55}$$

Then on every finite increasing grid,

$$\max_k W_2(\widehat{p}_k, p_{t_k}) \le e^{LT}\left(D_H\sum_j h_j^3 + \frac{B_H}{2}\sum_j h_j^2\varepsilon_{t_j}\right) \tag{56}$$
$$\le e^{LT}T\left(D_H H^2 + \frac{B_H}{2}\varepsilon H\right).$$

The second line assumes $\sup_t \varepsilon_t \le \varepsilon$. In particular, the projected solenoidal field $\mathbf{a}_t - \Pi_{p_t}\mathbf{a}_t$ need not be continuous or Lipschitz.

Proposition S29 required a regular flow for the distribution-preserving part of acceleration. Theorem S11 instead follows the full acceleration field and bounds the change of its distribution by the projected residual. The price is the explicit factor $B_H = 1 + O(H)$ and a different cubic constant; no elliptic regularity of the projection is required. Appendix VI proves the result. A certificate $\varepsilon_{t_j} \le \kappa H$ at every Euler node gives an $O(H^2)$ bound when the other constants are uniform. The certificate may belong to one fixed field or to a family of fields indexed by H; these are different assertions. A small value only at a training grid does not control a different evaluation grid without an interpolation bound.

Corollary S12 (Exact local Wasserstein coefficient)**.** *Under Theorem S11, at every fixed $t < T$,*

$$\lim_{h\downarrow 0}\frac{W_2((E_{t,h})_{\#}p_t, p_{t+h})}{h^2} = \tfrac{1}{2}\|\Pi_{p_t}\mathbf{a}_t\|_{L^2(p_t)}. \tag{57}$$

The projected residual is therefore the exact leading local error coefficient. This strengthens the local necessity result, but does not rule out cancellation between different steps at the final time.

B. Upper and lower certificates for the residual

A finite test space detects only part of the gradient projection. Its norm is a lower bound, so reducing it alone cannot certify a small residual. The following identity supplies both sides using a gradient approximation and an independently admissible solenoidal field.

Proposition S13 (Primal and dual residual certificates)**.** *Fix p and $a \in L^2(p;\mathbb{R}^d)$. For any $q \in \mathcal{G}_p$ and $w \in \mathcal{G}_p^{\perp}$, put*

$$\mathcal{L} = 2\langle a, q\rangle_{L^2(p)} - \|q\|^2_{L^2(p)},$$
$$\mathcal{U} = \|a - w\|^2_{L^2(p)}.$$

Then

$$\max(0,\mathcal{L}) \le \|\Pi_p a\|^2_{L^2(p)} \le \mathcal{U},$$
$$\mathcal{U} - \mathcal{L} = \|a - w - q\|^2_{L^2(p)}. \tag{58}$$

For an exact Galerkin solution $q_N = \sum_i c_i\nabla\phi_i$, where $A_{ij} = \langle\nabla\phi_i, \nabla\phi_j\rangle$ and $b_i = \langle a, \nabla\phi_i\rangle$, $Ac = b$ gives $\mathcal{L} = \|q_N\|^2 = b^{\top}A^{-1}b$ when A is positive definite. The general certificate remains valid for inexact coefficients.

This Hilbert-space identity turns a numerical projection into an upper certificate for Eq. (56). For fixed acceleration, gradient and solenoidal linear bases give two quadratic least-squares problems; the gap checks their resolution. An explicit admissible w already gives an upper bound, so a Poisson solve is optional.

A weighted Poisson formulation, an a posteriori Poincaré estimate, and explicit bounds for quadrature and acceleration approximation errors appear in Appendix VI. A raw divergence penalty or Monte Carlo standard error alone is not a deterministic residual certificate.

C. A finite-dimensional construction with explicit scope

In two dimensions let J denote the quarter-turn matrix. For a known positive density, stream potentials give the established gauge parameterization [6]

$$r_\theta(t,x) = p_t(x)^{-1} J \nabla_x \psi_\theta(t,x),$$
$$\psi_\theta = \sum_{\ell=1}^{m} \theta_\ell \psi_\ell, \qquad \nabla \cdot (p_t r_\theta) = 0. \tag{59}$$

Compactly supported potentials avoid dividing by arbitrarily small density in the tails. The proof appendix gives the antisymmetric-tensor version in higher dimensions. A finite family need not represent every admissible field.

For a base field satisfying the theorem's bounds, a positive C^4 density, uniformly compactly supported C^4 potentials and a compact parameter set, the corrected fields preserve the exact path and have uniform regularity bounds. Their objective $\mathcal{E}(\theta) = \sup_t \|\Pi_{p_t} \mathbf{a}_{u_\theta,t}\|$ is Lipschitz and attains a minimum. Proposition S16 in the proof appendix gives the assumptions and a certified finite-net search guarantee: parameter mesh δ and objective certificate width η achieve residual at most $\min_\Theta \mathcal{E} + K_\Theta \delta + \eta$.

This finite-dimensional guarantee does not imply efficiency or convergence of a local nonlinear optimizer. A positive minimum in one family does not prove path infeasibility. The appendix also certifies the continuous time supremum from a time grid.

D. Separating path error from integration error

Let p_t now be a target path generated by v, and let u be the field actually integrated. Its exact path need not equal p_t. Besides the acceleration residual under the target, define the path defect

$$\beta_t = \|\Pi_{p_t}(u_t - v_t)\|_{L^2(p_t)}. \tag{60}$$

This defect ignores velocity differences that preserve the target continuity equation. Both diagnostics in the next theorem are evaluated under the target p_t; no density substitution for an unknown learned path is made.

Theorem S14 (Target-path, residual, and integration errors)**.** *Let p_t be generated by a smooth, uniformly spatially Lipschitz v from $p_0 \in \mathcal{P}_2(\mathbb{R}^d)$, with $\sup_t \|v_t\|_{L^2(p_t)} < \infty$. Let u satisfy the smoothness, growth, and Lipschitz assumptions of Theorem S11. Write $F^u_{t,s}$ for its exact flow and assume*

$$C_a = \sup_t \|\mathbf{a}_{u,t}\|_{L^2(p_t)} < \infty,$$
$$C_j^\diamond = \sup_{0 \le t \le s \le T} \|\mathbf{j}_u(s, F^u_{t,s}(\cdot))\|_{L^2(p_t)} < \infty.$$

Suppose β_t is integrable. Let $\widehat{q}_k$ be Euler's laws for u from any $q_0 \in \mathcal{P}_2$. With $\varepsilon_t = \|\Pi_{p_t}\mathbf{a}_{u,t}\|$ and with D_H in Eq. (55) using $C_j^\diamond$ in place of C_j,

$$\max_k W_2(\widehat{q}_k, p_{t_k}) \le e^{LT}\Big[W_2(q_0,p_0) + D_H \sum_j h_j^3 + \frac{B_H}{2}\sum_j h_j^2 \varepsilon_{t_j} + \int_0^T \beta_t \, \mathrm{d}t\Big]. \tag{61}$$

The mixed jerk assumption concerns exact u-trajectories started from the target at any time. It is automatic from a uniform linear-growth bound on $\mathbf{j}_u$ and bounded target second moments, but does not follow from a training loss alone. If u generates p_t, then $C_j^\diamond = C_j$ and $\beta_t = 0$, recovering Theorem S11.

For conditional flow matching with $v_t(x) = \mathbb{E}[Y_t \mid X_t = x]$ and $\mathrm{Law}(X_t) = p_t$, the population excess squared regression loss is $\mathcal{R}_{\mathrm{excess}} = \int_0^T \|u_t - v_t\|^2_{L^2(p_t)} \, \mathrm{d}t$. Projection contraction gives $\int_0^T \beta_t \, \mathrm{d}t \le \sqrt{T\mathcal{R}_{\mathrm{excess}}}$. These population and exact-flow facts are standard [1, 14]; Eq. (61) combines them with the Euler residual. There is no finite-sample learning guarantee. An exactly target-solenoidal correction leaves β_t unchanged, but can change the other constants.

For a stream correction formed with an approximate density, the resulting path defect can be bounded using the score discrepancy and a Poincaré constant; Eq. (65) gives the precise bound.

APPENDIX VI
PROOFS FOR RESIDUAL CERTIFICATION AND TARGET-PATH STABILITY

A. Projection and the weighted weak problem

The space $L^2(p;\mathbb{R}^d)$ is Hilbert, and $\mathcal{G}_p$ is closed by definition. The Hilbert projection theorem therefore gives a unique $g = \Pi_p a$. Its complement $w = a - g$ satisfies $\int w \cdot \nabla\phi \, \mathrm{d}p = 0$ for every smooth compactly supported ϕ. Conversely this

identity places w in $\mathcal{G}_p^\perp$. Taking a supremum over a dense set of unit gradients proves Eq. (54). This argument makes no regularity claim about g or w.

For a smooth positive density with a Poincaré constant C_P, the projection has a potential, unique modulo constants, satisfying

$$\int \nabla\varphi \cdot \nabla\eta\, p \,\mathrm{d}x = \int a \cdot \nabla\eta\, p \,\mathrm{d}x, \qquad -\nabla\cdot(p\nabla\varphi) = -\nabla\cdot(pa). \tag{62}$$

The potential is understood in the completion of centered smooth test functions in weighted H^1. On $\mathbb{R}^d$ this is a finite-energy weak problem; no unmentioned boundary condition at a numerical truncation boundary is imposed. If an exact Galerkin q_N has strong residual $d_N = p^{-1}\nabla\cdot[p(a-q_N)] \in L^2(p)$ with zero mean, then

$$\|q_N\|_{L^2(p)}^2 \le \|\Pi_p a\|_{L^2(p)}^2 \le \|q_N\|_{L^2(p)}^2 + C_P\|d_N\|_{L^2(p)}^2. \tag{63}$$

This explains precisely when a divergence penalty gives an upper bound: the weighted norm, the Poincaré constant, and the weak residual identity must all be controlled.

Numerical integration error must also be enclosed. If computed values $\widehat{\mathcal{L}},\widehat{\mathcal{U}}$ have deterministic error bounds η_L,η_U, and the acceleration supplied to those computations differs from the true acceleration by at most δ_a in $L^2(p)$, then

$$\max\left\{0, \sqrt{\max(0,\widehat{\mathcal{L}}-\eta_L)} - \delta_a\right\} \le \|\Pi_p a\| \le \sqrt{\max(0,\widehat{\mathcal{U}}+\eta_U)} + \delta_a. \tag{64}$$

Here the primal and dual fields must be admissible for the same exact p. Sample standard errors alone are not deterministic quadrature certificates; confidence bounds give statements with their specified coverage.

For the potential formulation, assume p is a smooth positive density with $\|\eta - \mathbb{E}_p\eta\|_{L^2(p)}^2 \le C_P\|\nabla\eta\|_{L^2(p)}^2$. Complete the centered functions $f - \mathbb{E}_p f$, with $f \in C_c^\infty$, in the weighted H^1 norm. The gradient norm is equivalent to that norm on this centered space. The form $\int \nabla\varphi\cdot\nabla\eta\, p\,\mathrm{d}x$ is continuous and coercive, and the functional $\int a\cdot\nabla\eta\, p\,\mathrm{d}x$ is bounded. The Hilbert-space representation theorem yields a unique centered φ with Eq. (62). Its gradient is the projection. Other representatives differ by a constant within the uncentered completion.

For an exact Galerkin solution, $g - q_N$ is orthogonal to the finite gradient space and hence to q_N. If the stated strong residual exists, then for centered tests

$$\langle g - q_N, \nabla\eta\rangle = -\int d_N\eta \,\mathrm{d}p \le \sqrt{C_P}\|d_N\|_{L^2(p)}\|\nabla\eta\|_{L^2(p)}.$$

Density gives $\|g - q_N\| \le \sqrt{C_P}\|d_N\|$. Pythagoras proves Eq. (63). The same argument applied to $p^{-1}\nabla\cdot(pr)$ proves Eq. (65); compact support removes boundary terms and implies zero mean of this divergence.

Finally, using an approximate density has a measurable cost. If $r = \widetilde{p}^{-1}J\nabla\psi$ has compact support, then

$$p^{-1}\nabla\cdot(pr) = r\cdot(\nabla\log p - \nabla\log\widetilde{p}).$$

When p has Poincaré constant C_P, integration by parts gives

$$\|\Pi_p r\|_{L^2(p)} \le \sqrt{C_P}\,\|r\cdot(\nabla\log p - \nabla\log\widetilde{p})\|_{L^2(p)}. \tag{65}$$

Thus a score approximation controls a path-defect term under explicit norm and Poincaré assumptions; it does not preserve the exact target path automatically.

Proof of Proposition S13. Write $a = g + z$ with $g = \Pi_p a$ and $z \in \mathcal{G}_p^\perp$. Orthogonality gives

$$\|g\|^2 - \mathcal{L} = \|g - q\|^2, \qquad \mathcal{U} - \|g\|^2 = \|z - w\|^2.$$

Their sum is $\|g - q + z - w\|^2 = \|a - w - q\|^2$. For an exact Galerkin solution, $\langle a, q_N\rangle = \|q_N\|^2$, giving the stated matrix formula. For Eq. (64), first enclose the certificate for the approximate acceleration. Orthogonal projection is contractive, so changing that acceleration by δ_a changes its projected norm by at most δ_a. □

B. A quantitative flow lemma requiring no solenoidal flow

We use the standard continuity-equation estimate

$$W_2(\mu_b,\mu_a) \le \int_a^b \|V_s\|_{L^2(\mu_s)}\,\mathrm{d}s \quad \text{if} \quad \partial_s\mu_s + \nabla\cdot(\mu_s V_s) = 0, \tag{66}$$

for narrowly continuous finite-second-moment curves and integrable L^2 velocities [4, Theorem 8.3.1]. The estimate requires neither uniqueness nor classical trajectories for V_s. All curves constructed below are continuous pushforwards by globally Lipschitz diffeomorphisms, so their continuity and moment conditions hold.

Lemma S15 (Frozen flow displacement controlled by projection)**.** *Let $p \in \mathcal{P}_2(\mathbb{R}^d)$ and let $a \in C^1(\mathbb{R}^d;\mathbb{R}^d)$ be globally L_a-Lipschitz, with $a \in L^2(p)$. Write Φ^s for its autonomous flow and set*

$$Q_a(\tau) = \int_0^\tau e^{L_a s}\,\mathrm{d}s = \begin{cases} (e^{L_a\tau}-1)/L_a, & L_a > 0, \\ \tau, & L_a = 0. \end{cases}$$

For every $\tau \geq 0$,

$$W_2((\Phi^\tau)_\# p, p) \leq Q_a(\tau)\|\Pi_p a\|_{L^2(p)}. \tag{67}$$

Proof. Let $g = \Pi_p a$ and $w = a - g$. On $0 \leq s \leq \tau$, define $\mu_s = (\Phi^{\tau-s})_\# p$ and, with $y = \Phi^{\tau-s}(x)$,

$$V_s(y) = -D\Phi^{\tau-s}(x)g(x).$$

For a compactly supported smooth test f, differentiating its integral under μ_s initially gives the same expression with a instead of g. The omitted term is $\int \nabla(f\circ\Phi^{\tau-s})(x)\cdot w(x)\,p(\,\mathrm{d}x) = 0$. The composition is C^1 and compactly supported; convolution extends the defining orthogonality from smooth tests to these tests. Here $D\Phi^r(x)a(x) = a(\Phi^r(x))$, the autonomous flow identity, gives the differentiated expression. Thus (μ_s, V_s) satisfies the continuity equation. Moreover

$$\|V_s\|_{L^2(\mu_s)} \leq e^{L_a(\tau-s)}\|g\|_{L^2(p)}.$$

Apply Eq. (66) between $\mu_0 = (\Phi^\tau)_\# p$ and $\mu_\tau = p$. The proof never constructs a flow for w. □

Proof of Theorem S11. Fix t and an admissible step h, write $X \sim p_t$, and set $\tau = h^2/2$. For the frozen acceleration flow, the elementary Gronwall estimate $\|\mathbf{a}_t(\Phi^s X)\|_{L^2} \leq e^{L_a s}\|\mathbf{a}_t(X)\|_{L^2}$ gives

$$\|\Phi^\tau X - X\|_{L^2} \leq Q_a(\tau)C_a,$$
$$\|\Phi^\tau X - X - \tau\mathbf{a}_t(X)\|_{L^2} \leq \tfrac{1}{2}L_a C_a \tau^2 e^{L_a\tau}.$$

For completeness, the first pointwise estimate follows by writing $a(\Phi^s x) = a(x) + \int_0^s Da(\Phi^r x)a(\Phi^r x)\,\mathrm{d}r$ and using $\|Da\|_{\mathrm{op}} \leq L_a$. The second bound follows by integrating $\|a(\Phi^s X) - a(X)\| \leq L_a\|\Phi^s X - X\|$.

The exact physical flow has expansion

$$F_{t,t+h}X = X + hu_t(X) + \tfrac{1}{2}h^2\mathbf{a}_t(X) + R, \qquad \|R\|_{L^2} \leq C_j h^3/6.$$

The actual marginal at intermediate physical time is p_s, which justifies the C_j bound under the integral Taylor remainder. Consequently, for $E_{t,h}(x) = x + hu_t(x)$,

$$\begin{aligned}\|E_{t,h}(\Phi^\tau X) - F_{t,t+h}X\|_{L^2} &\leq C_j h^3/6 + hLC_a Q_a(\tau) + \tfrac{1}{2}L_a C_a\tau^2 e^{L_a\tau} \\ &\leq D_H h^3.\end{aligned}$$

The remaining change of input distribution is bounded by Lemma S15 and the Euler Lipschitz constant:

$$W_2((E_{t,h})_\# p_t, (E_{t,h})_\#(\Phi^\tau)_\# p_t) \leq (1+Lh)Q_a(\tau)\varepsilon_t \leq \tfrac{1}{2}B_H h^2\varepsilon_t.$$

The triangle inequality gives the local bound. The recurrence $e_{k+1} \leq (1+Lh_k)e_k + D_H h_k^3 + B_H h_k^2\varepsilon_{t_k}/2$, with $e_0 = 0$, proves Eq. (56) after multiplication by the later stability factors and use of $\prod_k(1+Lh_k) \leq e^{LT}$. □

Proof of Corollary S12. Fix $t < T$ and let $\mu_h = (E_{t,h})_\# p_t$, $\nu_h = p_{t+h}$, and $d_h = W_2(\mu_h, \nu_h)$. The single-step estimate proved above gives $\limsup_{h\downarrow 0} d_h/h^2 \leq \varepsilon_t/2$. In particular $d_h = O(h^2)$, and $\mu_h \to p_t$ in W_2 because $\|E_{t,h}X - X\|_{L^2} = h\|u_t\|_{L^2(p_t)}$.

For $\phi \in C_c^\infty$, Taylor expansion in an optimal coupling of μ_h and ν_h and Cauchy–Schwarz give

$$\left|\int \phi\,\mathrm{d}\nu_h - \int \phi\,\mathrm{d}\mu_h\right| \leq \|\nabla\phi\|_{L^2(\mu_h)}d_h + \tfrac{1}{2}\|D^2\phi\|_\infty d_h^2.$$

The same-start Taylor expansion used for local necessity gives

$$\int \phi\,\mathrm{d}\nu_h - \int \phi\,\mathrm{d}\mu_h = \tfrac{1}{2}h^2\langle\mathbf{a}_t, \nabla\phi\rangle_{L^2(p_t)} + o(h^2).$$

This expansion needs only the stated L^2 acceleration and jerk bounds: replace the exact-minus-Euler displacement by $h^2\mathbf{a}_t(X)/2$ with an L^2 remainder $O(h^3)$, and use bounded first and second derivatives of ϕ. Since $\|\nabla\phi\|_{L^2(\mu_h)} \to \|\nabla\phi\|_{L^2(p_t)}$, division by h^2 gives, whenever the latter norm is positive,

$$\liminf_{h\downarrow 0}\frac{d_h}{h^2} \geq \frac{|\langle\mathbf{a}_t, \nabla\phi\rangle_{L^2(p_t)}|}{2\|\nabla\phi\|_{L^2(p_t)}}.$$

The supremum over these tests is $\varepsilon_t/2$ by Eq. (54). If $\varepsilon_t = 0$, the upper bound and nonnegativity suffice. This proves the limit, including singular p_t, without requiring an optimal transport map. □

C. Regularity and optimization of compact gauges

Proposition S16 (Compact gauge search)**.** *Suppose p_t is a strictly positive C^4 density and is generated by a field v satisfying the regularity and moment bounds of Theorem S11. Let the C^4 potentials in Eq. (59) have spatial support in one compact set K, uniformly in time, and let θ range over a compact set $\Theta \subset \mathbb{R}^m$. Then $u_\theta = v + r_\theta$ has a global flow generating exactly p_t. The regularity and moment constants in Theorem S11 can be chosen uniformly on Θ. The objective*

$$\mathcal{E}(\theta) = \sup_{0\le t\le T} \|\Pi_{p_t}[\partial_t u_\theta + (D_x u_\theta)u_\theta]\|_{L^2(p_t)} \tag{68}$$

is Lipschitz on Θ and attains its minimum. If its Lipschitz constant is K_Θ, a finite δ-net of Θ, evaluated with certified upper and lower objective bounds of width at most η, has a candidate $\widehat{\theta}$ chosen by its smallest upper bound satisfying

$$\mathcal{E}(\widehat{\theta}) \le \min_{\theta\in\Theta} \mathcal{E}(\theta) + K_\Theta\delta + \eta.$$

Proof of Proposition S16. Positivity and continuity of p give a positive minimum on $[0,T]\times K$. Therefore r_θ and its derivatives through the orders used below are bounded and have support in K, uniformly for $\theta \in \Theta$. The identity $\nabla\cdot(J\nabla\psi) = 0$ proves exact weighted divergence freedom. In higher dimensions, let $A_{ij} = -A_{ji}$ and set $r_i = p^{-1}\sum_j \partial_j A_{ij}$. For this antisymmetric tensor, mixed partial derivatives cancel in pairs in $\sum_{ij}\partial_i\partial_j A_{ij}$.

The field u_θ is globally Lipschitz and has linear growth, uniformly over Θ. Its global flow is therefore defined on the whole interval. The prescribed p_t satisfies its continuity equation, and uniqueness for a globally Lipschitz field shows that this is its actual path. The acceleration difference

$$\mathbf{a}_{u_\theta} - \mathbf{a}_v = \sum_i \theta_i b_i + \sum_{ij}\theta_i\theta_j c_{ij}, \quad b_i = \partial_t r_i + (D_x v)r_i + (D_x r_i)v, \quad c_{ij} = (D_x r_i)r_j, \tag{69}$$

is supported in K with uniformly bounded first derivatives. Thus the acceleration Lipschitz and L^2 bounds are uniform. The jerk difference is also compactly supported and bounded, because it uses only the displayed regular derivatives and the locally bounded derivatives of v. This proves all uniform moment assertions.

Let $R = \sup_{\theta\in\Theta}\|\theta\|_2$, $B = (\sum_i \sup_t \|b_i\|^2_{L^2(p_t)})^{1/2}$, and $C = (\sum_{ij}\sup_t\|c_{ij}\|^2_{L^2(p_t)})^{1/2}$. Equation (69) and $\|\theta\otimes\theta - \eta\otimes\eta\|_2 \le 2R\|\theta-\eta\|_2$ imply

$$\sup_t \|\mathbf{a}_{u_\theta,t} - \mathbf{a}_{u_\eta,t}\|_{L^2(p_t)} \le (B + 2RC)\|\theta-\eta\|_2.$$

Projection is contractive. The reverse triangle inequality and then the supremum over time prove Lipschitz continuity of $\mathcal{E}$ with $K_\Theta = B + 2RC$. Compactness gives a minimizer θ_*. Choose a net point θ_i within δ of θ_*. Its upper certificate is at most $\mathcal{E}(\theta_*) + K_\Theta\delta + \eta$; selecting the smallest upper certificate proves the claimed bound. This proof requires certificates of the continuous-time supremum, not unqualified values at sampled times.

For the time envelope, choose a computable C^1 field w_t with zero weighted divergence under the target and define $R_t = \mathbf{a}_t - w_t$. Assume $K_R = \sup_t\|\partial_t R_t + (D_x R_t)v_t\|_{L^2(p_t)} < \infty$. Then $\varepsilon_t \le \|R_t\|$. This regularity is imposed on the chosen upper-certificate field, not on the orthogonal projection. Couple p_s and p_t using the exact target flow. The chain rule and Minkowski's inequality give

$$\big|\|R_t\|_{L^2(p_t)} - \|R_s\|_{L^2(p_s)}\big| \le \int_s^t \|\partial_r R_r + (D_x R_r)v_r\|_{L^2(p_r)}\,\mathrm{d}r.$$

The bound $K_R|t-s|$ and the covering property of the time net give the envelope $\sup_t\|R_t\| \le \max_i\|R_{s_i}\| + K_R\Delta$, where Δ is the covering radius of the time net. A lower bound for $\sup_t\varepsilon_t$ is the maximum of the lower certificates at those nodes. Achieving uniformly vanishing bracket widths further requires control of the primal–dual gaps; it is not a consequence of time-grid refinement alone. □

D. A target-measure stability estimate

Lemma S17 (Exact-flow comparison using the projected path defect)**.** *Under the path and field assumptions of Theorem S14, for $s < t$,*

$$W_2((F^u_{s,t})_\# p_s, p_t) \le \int_s^t e^{L(t-r)}\beta_r\,\mathrm{d}r. \tag{70}$$

Proof. Set $b_r = \Pi_{p_r}(u_r - v_r)$ and $\mu_r = (F^u_{r,t})_\# p_r$. Differentiating a smooth compactly supported test integral and using the target continuity equation gives

$$\frac{\mathrm{d}}{\mathrm{d}r}\int f\,\mathrm{d}\mu_r = \int \nabla f(F^u_{r,t}x)\cdot D_x F^u_{r,t}(x)(v_r(x) - u_r(x))\,p_r(\mathrm{d}x).$$

The term $u_r - v_r - b_r$ has zero weighted divergence under p_r. Testing it against $f\circ F^u_{r,t}$ replaces $v_r - u_r$ by $-b_r$. This gives a continuity-equation velocity of μ_r with norm at most $e^{L(t-r)}\beta_r$. A Borel version of the projections may be obtained by

projecting onto an increasing countable family of smooth gradient tests and taking the L^2 limit; the resulting weak identities are sufficient here. Equation (66) proves the claim. □

Proof of Theorem S14. Start a single exact u-step from $X \sim p_{t_k}$. The integral Taylor remainder is at most $C_j^{\diamond} h_k^3/6$ by assumption. All frozen-flow estimates in the proof of Theorem S11 therefore apply under this target measure, giving

$$W_2((E^u_{t_k,h_k})_{\#}p_{t_k}, (F^u_{t_k,t_{k+1}})_{\#}p_{t_k}) \le D_H h_k^3 + \tfrac{1}{2} B_H h_k^2 \varepsilon_{t_k}.$$

Lemma S17 bounds the difference between the second law and $p_{t_{k+1}}$. Consequently, with $e_k = W_2(\widehat{q}_k, p_{t_k})$,

$$e_{k+1} \le (1 + Lh_k) e_k + D_H h_k^3 + \tfrac{1}{2} B_H h_k^2 \varepsilon_{t_k} + \int_{t_k}^{t_{k+1}} e^{L(t_{k+1}-s)} \beta_s \, \mathrm{d}s.$$

Iteration proves Eq. (61). In fact the path term can retain the smaller factor $\int_0^{t_k} e^{L(t_k-s)}\beta_s \, \mathrm{d}s$, since later Euler stability factors are bounded by $e^{L(t_k - t_{j+1})}$.

To check the stated sufficient jerk condition, suppose

$$\|\mathbf{j}_u(s,x)\| \le J_0 + J_1\|x\|, \qquad \sup_t \|u(t,0)\| \le U_0, \qquad \sup_t \mathbb{E}_{p_t}\|X\|^2 \le M.$$

The growth estimate $\|F^u_{t,s}(x)\| \le e^{LT}(\|x\| + TU_0)$ yields $C_j^{\diamond} \le J_0 + J_1 e^{LT}(\sqrt{M} + TU_0)$. Finally, conditional expectation makes the cross term in the squared regression loss zero. Projection contraction and Cauchy–Schwarz in time give $\int \beta_t \, \mathrm{d}t \le \sqrt{T\mathcal{R}_{\mathrm{excess}}}$. Linearity of Π_{p_t} proves that a target-solenoidal correction leaves β_t unchanged. □

APPENDIX VII
FEASIBILITY CERTIFICATES BEYOND THE SECOND MOMENT

The scalar second-moment test is one member of a larger family of path-level restrictions. These restrictions can reject a path before a nonlinear correction search and give positive lower bounds on the best possible acceleration residual. The underlying second-derivative chain rule is standard transport calculus; here it yields certificates that strictly strengthen the second-moment restriction.

For fixed t, define the dual norm

$$\varepsilon_t^{\mathrm{acc}} = \sup_{\eta \in C_c^\infty,\, \|\nabla\eta\|_{L^2(p_t)} > 0} \frac{\left| \int \nabla\eta \cdot \mathbf{a}_t \, \mathrm{d}p_t \right|}{\|\nabla\eta\|_{L^2(p_t)}}. \tag{71}$$

It equals the norm of the weighted-gradient projection of $\mathbf{a}_t$ and is no larger than $\|\mathbf{a}_t - w\|_{L^2(p_t)}$ for every w of zero weighted divergence. This identity alone asserts no regularity of that projection.

Theorem S18 (Convex-observable and weighted-energy certificates)**.** *Assume the regularity and moment hypotheses of Theorem 1, except for the acceleration condition. For $\phi \in C^2(\mathbb{R}^d)$ with bounded Hessian, set $F_\phi(t) = \int \phi \, \mathrm{d}p_t$. Then*

$$F_\phi''(t) = \int u_t^\top D^2\phi \, u_t \, \mathrm{d}p_t + \int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t. \tag{72}$$

If the acceleration condition holds and ϕ is convex, F_ϕ is convex in time. More generally, let $K_\phi(t)$ be the infimum of $\int v^\top D^2\phi \, v \, \mathrm{d}p_t$ over finite-energy instantaneous velocities satisfying $\partial_t p_t + \nabla\cdot(p_t v) = 0$. For convex ϕ and $\|\nabla\phi\|_{L^2(p_t)} > 0$,

$$\varepsilon_t^{\mathrm{acc}} \ge \frac{[K_\phi(t) - F_\phi''(t)]_+}{\|\nabla\phi\|_{L^2(p_t)}} \ge \frac{[-F_\phi''(t)]_+}{\|\nabla\phi\|_{L^2(p_t)}}. \tag{73}$$

Unbounded Hessians are permitted when the chain rule is dominated and $\nabla\phi$ belongs to the weighted-gradient closure, including polynomial tests under the corresponding finite moments.

Thus a feasible path satisfies $p_{(1-\lambda)s+\lambda t} \preceq_{\mathrm{cx}} (1-\lambda)p_s + \lambda p_t$: each convex observable at an intermediate time is bounded by its endpoint mixture average. This is necessary, not sufficient. Every straight stochastic interpolation $\mathrm{Law}((1-t)X_0 + tX_1)$ already passes this test. Under weighted Poincare and uniform ellipticity assumptions, K_ϕ is obtained by a linear weighted elliptic problem. Appendix VIII states these hypotheses and gives the characterization and all proofs.

Proposition S19 (Covariance constraints)**.** *Under the acceleration condition and the preceding hypotheses, let $m(t) = m(0) + t\xi$ and $C(t) = \mathrm{Cov}(X_t) > 0$. There exists B_t such that*

$$C' = B_t + B_t^\top, \qquad \begin{pmatrix} C & B_t^\top \\ B_t & C''/2 \end{pmatrix} \succeq 0. \tag{74}$$

In particular $C'' \succeq 0$, and every directional standard deviation $\sqrt{e^\top C(t) e}$ is convex.

The block-matrix condition is a semidefinite necessary test using only the prescribed covariance and its first two derivatives. It can fail even when the scalar second-moment test passes. The following example passes both tests but is excluded by a fourth-order convex observable.

Proposition S20 (A residual floor beyond covariance tests)**.** *Let $Z \sim \mathcal{N}(0, I_2)$ and $0 \le t \le 1/2$. Define*

$$\begin{aligned} \kappa(r) &= \frac{-1 + 5e^{-r^2/2}}{4}, \qquad b(z) = \kappa(\|z\|)z, \\ G_t(z) &= z + \frac{t^2}{2} b(z), \qquad p_t = (G_t)_{\#} \operatorname{Law}(Z). \end{aligned} \tag{75}$$

This path has smooth positive densities and a smooth globally Lipschitz minimum-action generator $v_t(G_t z) = t b(z)$. It passes the scalar second-moment and covariance tests throughout the interval. Nevertheless, no generator with the stated regularity can satisfy the acceleration condition. Every such generator obeys

$$\varepsilon_0^{\mathrm{acc}} \ge \frac{\sqrt{3}}{16}. \tag{76}$$

The separation is explicit: $M'' - 2K_{\min} = 1/4 + 23t^2/144 > 0$, and $C(t) = (1 + t^2/16 + 23t^4/1152)I_2$ satisfies (74). However, $(\mathbb{E}\|X_t\|^4)'' < 0$ on the entire interval, with value -3 at zero. Since $\|\nabla\|x\|^4\|_{L^2(p_0)} = 16\sqrt{3}$, (73) gives the displayed floor. The obstruction applies to every admissible generator, independently of its parameterization or optimization method. In particular no correction family on this path can achieve $\sup_t \varepsilon_t^{\mathrm{acc}} = O(H)$ as $H \downarrow 0$. By Corollary S12, the same obstruction yields an actual local error floor for every such generator, with $E^u_{0,h}(x) = x + hu(0, x)$:

$$\lim_{h \downarrow 0} h^{-2} W_2((E^u_{0,h})_{\#} p_0, p_h) \ge \frac{\sqrt{3}}{32}.$$

This local conclusion does not exclude accidental cancellation of terminal integration error.

Appendix VIII
Proofs of the path-level feasibility certificates

A. Convex observables and weighted kinetic energy

Proof. Along an exact trajectory, two derivatives of $\phi(X_t)$ give $u_t(X_t)^\top D^2\phi(X_t)u_t(X_t) + \nabla\phi(X_t)\cdot \mathbf{a}_t(X_t)$. The global growth bounds and the finite second moment dominate these terms when $D^2\phi$ is bounded. Smooth compact cutoffs of ϕ have gradients converging to $\nabla\phi$ in $L^2(p_t)$: both the gradient and the cutoff error are bounded by $C(1 + \|x\|)$, with the error supported in the tails. Thus the acceleration condition removes the final integral in (72). A convex Hessian makes the first integral nonnegative. Without that condition, the first integral is at least $K_\phi(t)$, and the definition of $\varepsilon_t^{\mathrm{acc}}$ bounds the absolute value of the final integral by $\varepsilon_t^{\mathrm{acc}}\|\nabla\phi\|_{L^2(p_t)}$. This proves (73). The additional hypotheses stated for unbounded tests justify the identical argument. □

The theorem implies the convex-order condition

$$p_{(1-\lambda)s+\lambda t} \preceq_{\mathrm{cx}} (1-\lambda)p_s + \lambda p_t, \qquad 0 \le s < t \le T, \quad 0 \le \lambda \le 1. \tag{77}$$

Here $\preceq_{\mathrm{cx}}$ compares expectations of convex functions, and the right side is a mixture of endpoint measures. To extend the proved inequality beyond smooth tests, first mollify convex Lipschitz functions and then approximate an arbitrary convex function from below by maxima of finitely many supporting affine functions. Subtracting one supporting affine function permits monotone convergence; the mean is affine by Corollary S26. This is a necessary condition, not an existence theorem. In particular, every straight stochastic interpolation $\operatorname{Law}((1-t)X_0 + tX_1)$ already satisfies the convex-observable test, regardless of whether it has an acceleration-preserving deterministic generator.

For a smooth positive density, the weighted quantity K_ϕ has a linear variational characterization. If $B = D^2\phi$ satisfies $b_- I \preceq B \preceq b_+ I$ for positive finite constants, a weighted Poincare inequality and one finite-energy generating field give a solution χ, unique modulo constants in the weighted Sobolev completion of smooth functions, of

$$\int \nabla\eta^\top B^{-1}\nabla\chi \, \mathrm{d}p_t = \int \eta \, \partial_t p_t \, \mathrm{d}x.$$

Its velocity $v_\phi = B^{-1}\nabla\chi$ attains K_ϕ. Indeed, every other admissible velocity differs by a zero-weighted-divergence field r, and $\int r \cdot \nabla\chi \, \mathrm{d}p_t = 0$. Expansion of the quadratic energy proves minimality. This permits a linear elliptic feasibility diagnostic without asserting that the nonlinear correction problem is solvable.

B. Covariance constraints and their limits

Proof. Applying (72) to quadratic coordinate products, or polarizing its quadratic-form identity, gives $\frac{\mathrm{d}^2}{\mathrm{d}t^2}\mathbb{E}[X_tX_t^\top] = 2\mathbb{E}[u_tu_t^\top]$. Subtracting $m(t)m(t)^\top$ yields $C'' = 2\mathbb{E}[(u_t-\xi)(u_t-\xi)^\top]$. The displayed block matrix is the covariance of $(X_t, u_t(X_t))$. For $Z = e\cdot(X_t - m(t))$ and $V = e\cdot(u_t(X_t)-\xi)$, $q = \sigma_e^2$ satisfies $q' = 2\mathbb{E}[ZV]$ and $q'' = 2\mathbb{E}[V^2]$. Hence

$$\sigma_e'' = \frac{\mathbb{E}[Z^2]\mathbb{E}[V^2] - (\mathbb{E}[ZV])^2}{(\mathbb{E}[Z^2])^{3/2}} \geq 0$$

by Cauchy–Schwarz. □

These tests form a strict hierarchy. For example, a diagonal Gaussian path with standard deviations $s_1(t) = 1+t-t^2/10$ and $s_2(t) = 1+t+t^2$ on $[0,1/4]$ has positive C'' and passes the scalar minimum-energy test, because $M''/2 - K_{\min} = s_1s_1'' + s_2s_2'' > 0$. It is nevertheless infeasible because $s_1'' = -1/5 < 0$. The next example also passes the full covariance test.

C. The quartic obstruction and its residual floor

Proof. The bounds $|\kappa| \leq 1$ and $r|\kappa'(r)| \leq 5/(2e) < 1$ give $\|Db\|_{\mathrm{op}} < 2$. Thus G_t is a global smooth bi-Lipschitz map, with lower Lipschitz constant at least $1-t^2 \geq 3/4$. Its nonlinear terms have bounded derivatives, and its inverse has uniformly bounded first derivatives on the interval. The generator and its material acceleration $b\circ G_t^{-1}$ are globally Lipschitz. The source is Gaussian and the maps have linear growth, so every required moment is finite. The generator is radial, hence a gradient, with a potential of at most quadratic growth; the weighted-gradient orthogonality argument establishes its minimum kinetic energy.

For $R = \|Z\|$, $R^2/2$ is exponential with mean one, and

$$\mathbb{E}[R^{2\ell}e^{-jR^2/2}] = \frac{2^\ell \ell!}{(1+j)^{\ell+1}}.$$

It follows exactly that

$$M(t) = 2 + \frac{t^2}{8} + \frac{23t^4}{576}, \qquad K_{\min}(t) = \frac{23t^2}{144}, \qquad M'' - 2K_{\min} = \frac{1}{4} + \frac{23t^2}{144} > 0.$$

The mean is zero and $C(t) = q(t)I_2$, where $q(t) = 1 + t^2/16 + 23t^4/1152$. For $q = 1+\alpha t^2 + \beta t^4$ with $\alpha, \beta > 0$,

$$qq'' - \tfrac{1}{2}(q')^2 = 2\alpha + 12\beta t^2 + 6\alpha\beta t^4 + 4\beta^2 t^6 > 0.$$

Thus $\sqrt{q}$ is convex and the choice $B_t = (q'/2)I_2$ satisfies (74).

For the convex observable $\phi(x) = \|x\|^4$, direct expansion instead gives

$$F_\phi(t) = 8 - \frac{3}{2}t^2 + \frac{73}{144}t^4 + \frac{29}{9216}t^6 + \frac{179}{73728}t^8,$$
$$F_\phi''(t) = -3 + \frac{73}{12}t^2 + \frac{145}{1536}t^4 + \frac{1253}{9216}t^6 < 0 \qquad (0 \leq t \leq 1/2).$$

For the last inequality, the expression is increasing in t^2 and its value at $t = 1/2$ is $-867715/589824 < 0$. The polynomial test is justified by the Gaussian moments and global linear growth. Theorem S18 therefore excludes every acceleration-preserving generator of this path, not only the displayed base field or a selected correction family. Finally, $F_\phi''(0) = -3$ and $\|\nabla\phi\|_{L^2(p_0)} = 4\sqrt{\mathbb{E}R^6} = 16\sqrt{3}$. Equation (73) gives (76). Corollary S12 then gives the stated local Euler coefficient lower bound. For the displayed radial base field itself, $\mathbf{a}_t(G_tz) = b(z)$ is a gradient with linear growth, so its projection equals itself and $\varepsilon_t^{\mathrm{acc}} = \|b\|_{L^2(\rho)} = \sqrt{23}/12$ throughout the interval; this is larger than the universal floor. □

APPENDIX IX
NONRADIAL EXPERIMENT: IMPLEMENTATION AND VALIDATION

A. Exact references and inverse-map cost

We identify $\mathbb{R}^2$ with $\mathbb{C}$ and use a standard planar Gaussian label Z. Write $F_1(t,z) = e^{(1+i)at}z$ and $F_2(t,z) = be^{(1+i)a\sqrt{2}t}z^2e^{-|z|^2}$, with $a = 0.4$, $b = 0.18$, and $F_t = F_1 + F_2$. For the gauge parameter ω, the exact trajectory from $F_0(z)$ is

$$X_t = F_t(e^{i(\omega-a)t}z).$$

All tested gauge parameters therefore preserve the same continuous probability path. We evaluate the field in Cartesian coordinates: the algorithm receives x, recovers $z = F_t^{-1}(x)$ by contraction, and evaluates the velocity there. It does not evolve hidden labels instead of applying the stated numerical solver.

Starting with $z_0 = e^{-(1+i)at}x$, the inverse iteration is

$$z_{j+1} = e^{-(1+i)at}x - be^{(1+i)a(\sqrt{2}-1)t}z_j^2e^{-|z_j|^2}.$$

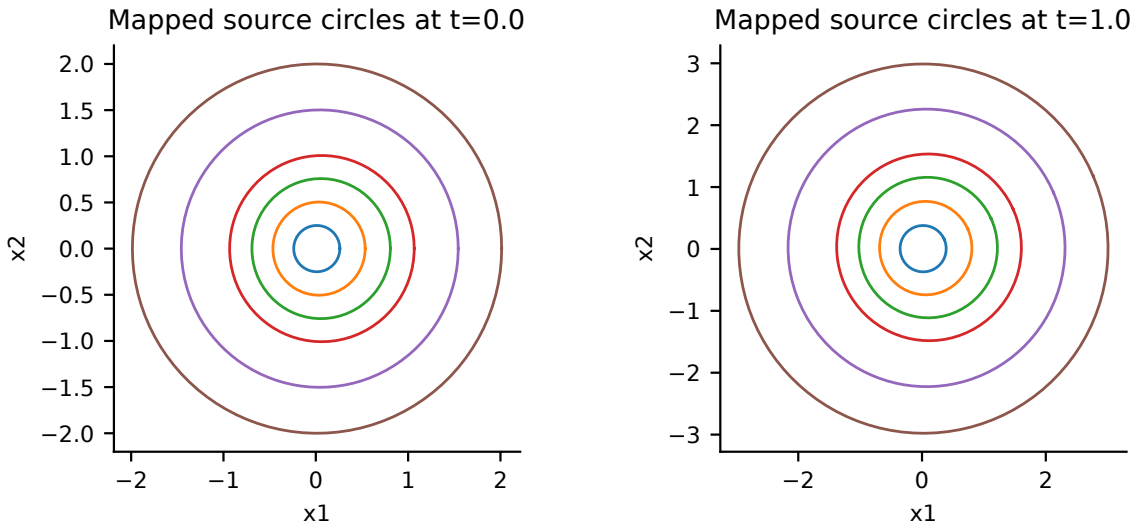


Fig. 3. Images of concentric source circles under F_0 and F_1, showing the asymmetric deformation. These are mapped source contours, not claimed level sets of the target density.

Its declared contraction factor is $\rho_t = (4/e)be^{a(\sqrt{2}-1)t} < 0.313$. We stop when $\rho_t\|z_{j+1} - z_j\|_\infty/(1-\rho_t) \le 2 \times 10^{-14}$. This is the exact-arithmetic contraction error bound evaluated in floating point, not a roundoff-inclusive enclosure. A refinement uses 2×10^{-15}. The main $N = 512$ corrected-Euler run uses 8,319 batch inverse iterations for 512 field evaluations, about 16.25 iterations per evaluation. Every solver stage, including every Heun and RK3 stage, pays this cost. No neural network, numerical score, numerical Jacobian, or Poisson solve is used in this example.

B. An exact residual objective with an observable bad local minimum

Set $n \in \{1, 2\}$,

$$\zeta_n = a\sqrt{n} + i\{a\sqrt{n} + n(\omega - a)\}, \qquad k_t(F_t z) = DF_t(z)Jz = \sum_{n=1}^{2} inF_n(t, z).$$

The acceleration along this gauge is $\sum_n \zeta_n^2 F_n$. The field $w_t = ck_t$ preserves p_t, so the following is an upper certificate for the weighted-gradient acceleration residual, not generally an equality with that projection norm:

$$R_t(\omega, c)^2 = \|\mathbf{a}_\omega - ck_t\|^2_{L^2(p_t)} = \sum_{n=1}^{2} M_n(t)|\zeta_n^2 - inc|^2,$$

where

$$M_1(t) = 2e^{2at}, \qquad M_2(t) = \frac{8}{125}b^2 e^{2a\sqrt{2}t}.$$

The second identity follows from angular orthogonality and $\mathbb{E}[|Z|^4 e^{-2|Z|^2}] = 8/125$. Both masses increase, so $\sup_{0\le t\le 1} R_t = R_1$ for fixed (ω, c). Eliminating c gives

$$c_*(\omega) = \frac{\sum_n nM_n(1)\,\mathrm{Im}(\zeta_n^2)}{\sum_n n^2 M_n(1)}.$$

The remaining objective is a coercive quartic. The code enumerates all real roots of its cubic derivative, then compares objective values using the original sum of squares. Its three stationary points have ω approximately -0.392394, 0.000331763, and 0.4. Their residuals are approximately 0.0724104, 0.3388934, and floating-point zero. The first and third are local minima; the middle is a local maximum. Local least squares from $(\omega, c) = (-0.3, 0)$ terminates at the inferior minimum. Starts $(0.07, 0.03)$ and $(0.8, -0.2)$ reach the zero. The optimizer is therefore not treated as a global feasibility certificate; the finite stationary-point enumeration and the exact coefficient argument supply that distinction.

C. Distribution bounds and quadrature refinement

For a numerical endpoint $\widehat{X}_N(Z)$, the lower bound is

$$\left|\sqrt{\mathbb{E}|\widehat{X}_N|^2} - \sqrt{M_1(T) + M_2(T)}\right| \le W_2(\widehat{p}_N, p_T).$$

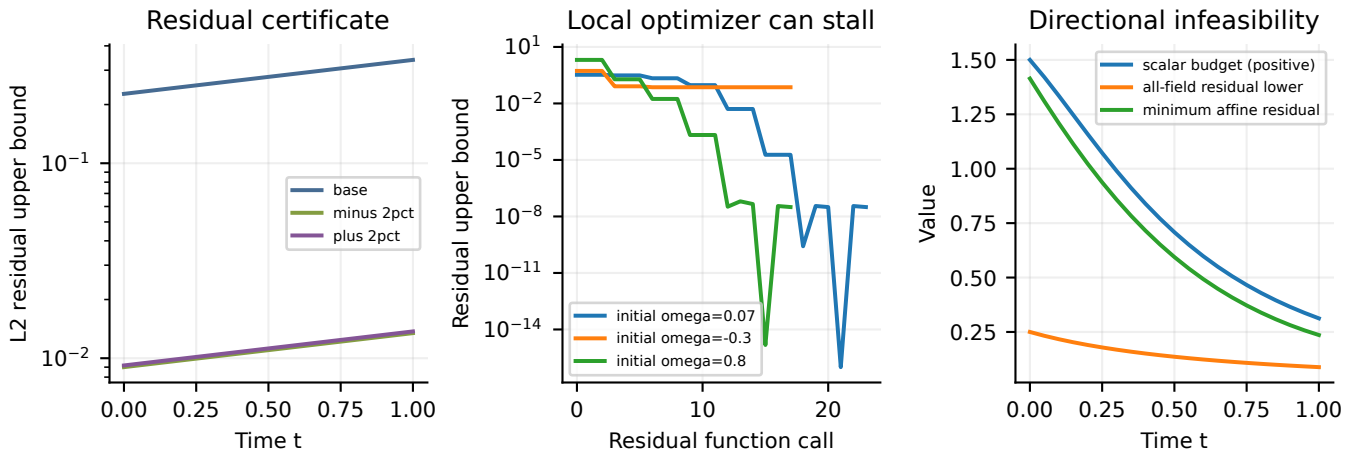


Fig. 4. Left: exact formulas for residual upper certificates. The discovered solution is omitted from this logarithmic panel because its exact residual is zero. Center: local optimization from three initial values; the negative start stagnates at a positive residual. Right: the directional obstruction for a Gaussian path whose scalar energy budget is positive. The affine residual curve uses the global minimum of its even quartic objective.

For corrected Euler, the upper bound uses the valid target coupling $F_T(e^{-ia^2Th}Z)$, with $h = T/N$, as proved in Theorem S10. For a perturbed ω, we use $F_T(e^{i\{(\omega-a)T-c_*(\omega)Th/2\}}Z)$. Every such fixed rotation preserves the Gaussian source, so the pairing remains a valid upper bound even when no second-order guarantee is claimed. For base Euler, Heun, and RK3, the same-start coupling is used. We do not minimize these couplings or identify their RMS with full W_2.

The radial rule uses $s = |Z|^2/2 \sim \mathrm{Exp}(1)$, with positive Gauss–Laguerre weights; the angular rule uses equally spaced points on $[0, 2\pi)$. The main rule has $64 \times 64 = 4{,}096$ nodes. Refining to $96 \times 128 = 12{,}288$ nodes changes the $N = 512$ corrected upper estimate by 7.8×10^{-17} and the base upper estimate by less than 7×10^{-15}. The corrected lower estimate is unchanged at printed double precision. Tightening inverse tolerance by a factor ten changes the corrected upper estimate by less than 4×10^{-18}. A coarser 32×32 rule changes the corrected lower bound by about 3.1×10^{-11}; the fine rule is used throughout. This is a refinement study, not a rigorous quadrature error enclosure.

An independent DOP853 solve uses relative tolerance 2×10^{-12} and absolute tolerance 2×10^{-13} on five fixed labels, for the base, discovered, and -2%-perturbed fields. Four observation times $1/4, 1/2, 3/4, 1$ give a maximum discrepancy of 6.5×10^{-12} from the analytic trajectories. Tests also compare acceleration with a second time difference of the exact nonlinear trajectory, so omission of the nonlinear-coordinate Hessian term would be detected.

D. A path that passes the scalar budget but is infeasible

Let $\Sigma_t = \mathrm{diag}(1+t, 1+t+2t^2)$ and $p_t = \mathcal{N}(0, \Sigma_t)$. The gradient field is $v_t = \frac{1}{2}\dot{\Sigma}_t\Sigma_t^{-1}x$. The scalar budget $\frac{1}{2}(\mathrm{tr}\,\Sigma_t)'' - \mathbb{E}|v_t|^2$ remains at least 0.3125 over $[0, 1]$. Nevertheless, $m_1(t) = \mathbb{E}X_1^2 = 1 + t$ has zero second derivative and nonzero first derivative. Every admissible velocity satisfies

$$\mathbb{E}[u_1^2] \geq \frac{1}{4(1+t)}, \qquad \mathbb{E}[X_1 a_1] = -\mathbb{E}[u_1^2], \qquad \|P_{p_t}^{\nabla}\mathbf{a}_t\|_{L^2(p_t)} \geq \frac{1}{4(1+t)^{3/2}}.$$

This is an all-field obstruction under the stated moment and integration-by-parts assumptions, not a failure of one optimizer. As a diagnostic comparison, the code minimizes over all two-dimensional affine Gaussian gauges $A = v + \omega\Sigma^{1/2}J\Sigma^{-1/2}$. The acceleration's weighted-gradient projection is Sx, where

$$S\Sigma + \Sigma S = \ddot{\Sigma} - 2A\Sigma A^{\top}, \qquad \varepsilon^2 = \mathrm{tr}(S\Sigma S).$$

This objective is an even quartic in ω, so its global minimum follows from a quadratic minimization in $\omega^2 \geq 0$. The affine curve in the figure does not assert that every admissible non-affine field has the same minimum.

E. A learned coefficient and a continuous-path error floor

To separate learned-path error from integration error, define the restricted regression family

$$u_{\delta,\omega}(t, F_t z) = \partial_t F_t z + \delta DF_t(z)z + (\omega - a)DF_t(z)Jz.$$

Its exact trajectory is $F_t(e^{\delta t}e^{i(\omega-a)t}z)$. We observe 512 independent pairs (t, z), with t uniform on $[0, 1]$ and z standard planar Gaussian, and noisy base velocities $v(t, F_t z) + \xi$, where the two coordinates of ξ are independent $\mathcal{N}(0, 0.08^2)$. Ordinary one-parameter least squares fits the coefficient of $DF_t(z)z$ around the specified base field. The true coefficient is zero; it is

not supplied to the fit. Seeds $7, 19, 42$ give fitted coefficients approximately -0.00060835, -0.00031947, and 0.00253611, respectively. Their known-noise standard errors are about 0.0020. This experiment tests one identifiable coefficient with an exact model, not general neural velocity learning.

After adding the gauge $\omega = a$, the learned path is $p_t^\delta = (F_t \circ e^{\delta t})_\# \rho$. Its second moment is known exactly:

$$\mathbb{E}_{p_t^\delta}|X|^2 = 2e^{2at}s^2 + b^2 e^{2a\sqrt{2}t}\frac{8s^4}{(1+4s^2)^3}, \qquad s = e^{\delta t}.$$

We bound $W_2(p_T^\delta, p_T)$ below by the second-moment discrepancy and above by the common-label RMS between $F_T(e^{\delta T}Z)$ and $F_T(Z)$. These bounds give path-error intervals approximately $[0.0012814, 0.0012850]$, $[0.0006730, 0.0006749]$, and $[0.0053504, 0.0053654]$ for the three seeds. At $N = 512$, numerical-to-learned-path coupling upper bounds are between 8.9×10^{-7} and 6.0×10^{-6}, while numerical-to-target error remains near these much larger path-error floors. The approximate fields are not claimed to satisfy exact acceleration preservation.

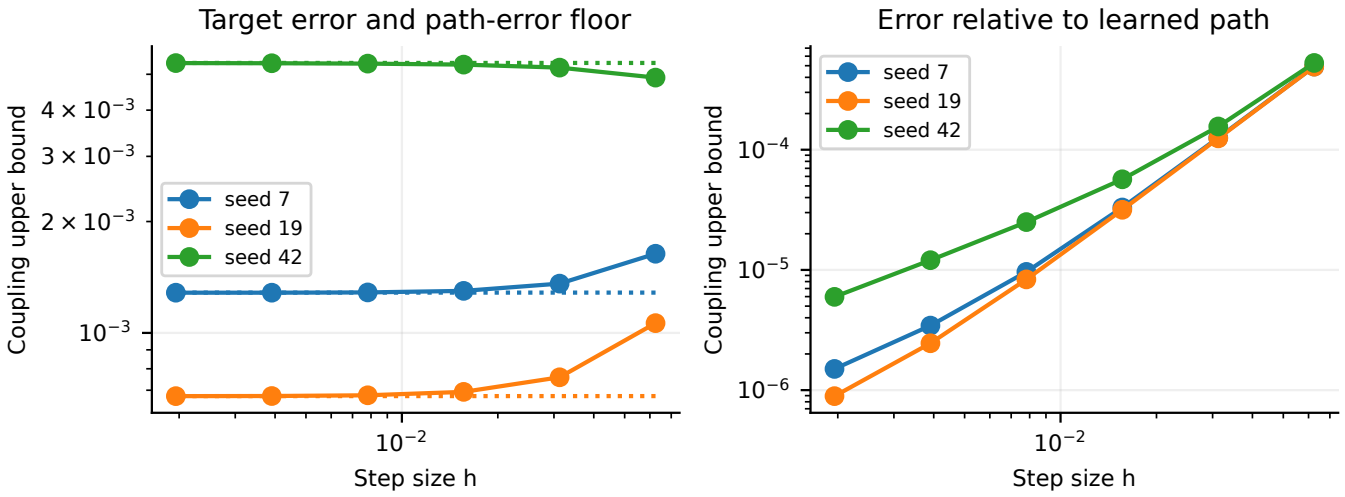


Fig. 5. Restricted learned-field experiment. Dotted lines are continuous-path coupling upper bounds. Integration refinement reduces error relative to the learned path but cannot remove error in the learned continuous distribution. Curves and floors are quadrature evaluations of valid upper couplings; they are not exact W_2 evaluations.

F. Runtime, reproducibility, and limitations

Equal-cost runs use budgets $48, 96, 192, 384$ field evaluations; Euler uses N steps, Heun $N/2$, and RK3 $N/3$. Each batch contains $4{,}096$ deterministic quadrature labels. Timings are the median of three repetitions, with minima and maxima retained in `equal_cost.csv`. At 384 evaluations, batch times are of the same order (a few tenths of a second) for all three methods. The correction search, including unsuccessful local searches, takes about 0.01 seconds. The one-coefficient regression solve takes about 10^{-5} seconds after generating the data. These are CPU implementation measurements; they neither predict neural-network evaluation costs nor establish an end-to-end generation speedup. The observed Heun advantage is small, so we retain bounds and numerical refinement data rather than interpreting wall-clock differences of this size as significant. RK3's accuracy advantage is much larger.

The recorded environment is Linux x86_64, Python 3.12.14, NumPy 2.3.5, and SciPy 1.17.0. The CPU reports Intel Xeon Platinum 8573C. We set `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS`, and `MKL_NUM_THREADS` to one. The code does not claim a measured peak-memory advantage; it uses vectorized arrays of batch size $4{,}096$, and the exact small optimization eliminates any neural-training storage in this example. Run

```
OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 \
MKL_NUM_THREADS=1 python3 -B scripts/run_revision.py
python3 -B -m unittest discover -s tests -v
```

from the repository root after installing the declared dependencies. The first command regenerates raw CSV files, the machine-readable summary, the table, and five figures under `results/revision/`. The original reproduction command regenerates the pre-existing Gaussian, nonlinear radial, and solver-comparison results separately. All 63 original tests and 9 new independent tests pass. The original reproduction regenerated 81 baseline runs and 33 nonlinear runs with $16{,}983$ and $24{,}600$ node comparisons, respectively; the original manuscript's rounded numerical macros agree exactly. No image-generation benchmark, high-dimensional learned model, or universal correction algorithm is claimed.

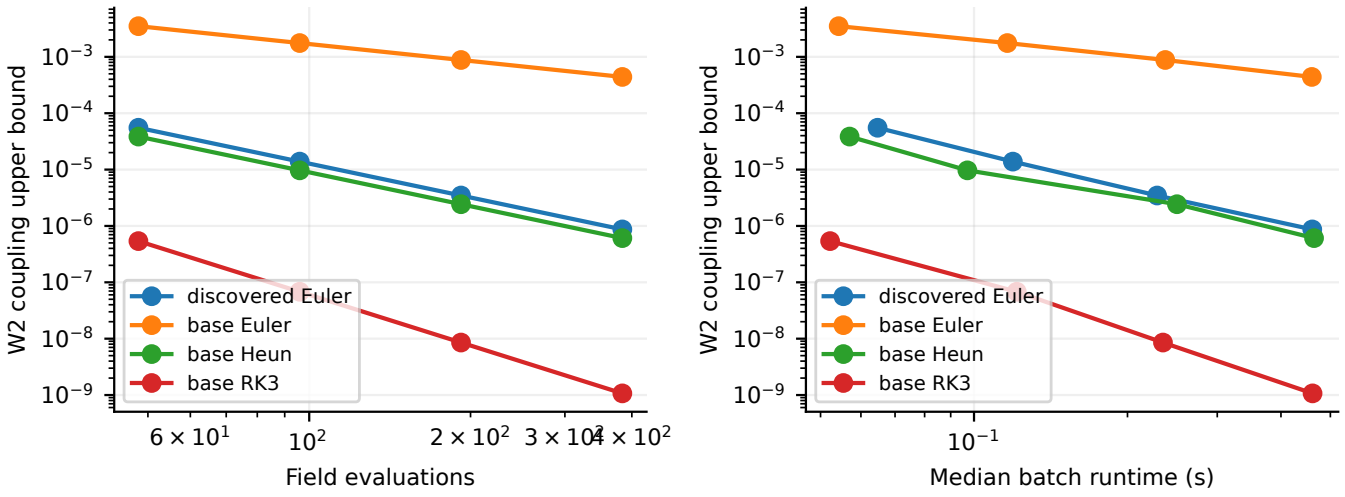


Fig. 6. Coupling upper bounds versus field-evaluation count and measured median batch runtime. The different methods use different couplings, so upper-bound curves alone do not order exact Wasserstein errors; Table II also reports lower bounds.

APPENDIX X
ERROR MEASURES AND THE ACCELERATION CONDITION

Term or symbol	Meaning	Definition
Probability path p_t	The distribution specified at each time t	Section X-A
Local and global error	Error from one Euler step started from the exact distribution, and error accumulated through repeated steps	Section X-A
Particle error	Root mean squared distance using the same starting point	Section X-B
Coupling and W_2	A joint distribution with given marginals, and the least root mean squared distance over all such pairings	Eq. (79)
Material acceleration $\mathbf{a}_t$	Velocity change along an exact particle	Eq. (80)
Acceleration condition	$\int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t = 0$ for every smooth compactly supported ϕ and every t	Definition S21
Kinetic action $\mathcal{A}_T$	Time integral of kinetic energy $\mathbb{E}_{p_t}\|u_t\|^2$	Eq. (82)
Second moment $M(t)$	$\mathbb{E}_{p_t}\|X\|^2$	Section XIV

A. The probability path and Euler's method

A velocity field u generates the probability path $(p_t)_{0\le t\le T}$ when the solution of $\dot{X}_t = u(t, X_t)$, with $X_0 \sim p_0$, has distribution p_t at each time. Each p_t, called a *marginal distribution*, describes positions at one time without recording the particles' starting points. Different particle motions can therefore give the same probability path, as the outward and rotating Gaussian fields do. We assume finite second moments and write $\mathbb{E}$ for expectation, Law for distribution, and $\|\cdot\|$ for the Euclidean norm. Single bars $|\cdot|$ denote scalar absolute value or complex modulus, while $\|\cdot\|_{\mathrm{op}}$ denotes the induced operator norm of a linear or multilinear map.

On a grid $0 = t_0 < \cdots < t_N = T$, with step lengths $h_k = t_{k+1} - t_k$ and largest step $H = \max_k h_k$, Euler's method is

$$\widehat{X}_{k+1} = \widehat{X}_k + h_k u(t_k, \widehat{X}_k), \qquad \widehat{X}_0 \sim p_0, \qquad \widehat{p}_k = \mathrm{Law}(\widehat{X}_k). \tag{78}$$

For $u(x) = x$, a particle starting at one reaches 1.1 after an Euler step of length 0.1 and reaches 1.21 after the next step. The second step starts from the first numerical position, so it includes the error from the first step. We call the error after repeated steps the *global error*. The *local error* is the error from one step that starts from the exact distribution p_t. We compare fields on the same grid. When studying convergence, we decrease the step sizes while keeping both fields and the final time fixed.

B. Particle error and distribution error

A rotation shows why particle displacement and distribution change can differ. Rotating the uniform unit circle by 30° moves each point by $2\sin(15^\circ) \approx 0.518$, although the distribution stays the same. We can pair each rotated point with an original point at the same location, giving zero distance between the distributions. Such a pairing is a *coupling*, a joint distribution with the two given marginal distributions. The Wasserstein distance is the least root mean squared distance over these couplings,

$$W_2(\mu,\nu) = \left(\inf_{\gamma\in\Gamma(\mu,\nu)} \int \|x-y\|^2 \, \mathrm{d}\gamma(x,y) \right)^{1/2}, \tag{79}$$

where $\Gamma(\mu,\nu)$ is the set of couplings of μ and ν. Particle error uses the numerical and exact particles from the same starting point. Its root mean squared value is $\|\widehat{X}_N - X_T\|_{L^2} = (\mathbb{E}\|\widehat{X}_N - X_T\|^2)^{1/2}$. This pairing is one of the couplings in Eq. (79), so distribution error cannot exceed particle error. Practical sampling metrics also compare distributions. For example, Fréchet Inception Distance uses the squared Wasserstein distance between Gaussian fits to feature distributions [15]. Our examples use W_2 between the full distributions.

We calculate the errors from Gaussian formulas or deterministic quadrature, so they have no sampling uncertainty. Smaller values mean better accuracy in all figures and tables. An $O(H^2)$ bound means that the error is at most a constant times H^2 for small H, but the error could decrease faster. A positive limit of error divided by h^2 shows that second order is the actual rate. In that case, halving a sufficiently small uniform step reduces the error by approximately a factor of four.

C. The acceleration condition

For a density p_t, the continuity equation $\partial_t p_t + \nabla\cdot(p_t u_t) = 0$ describes how the velocity field changes the distribution. Adding a field r_t with $\nabla\cdot(p_t r_t) = 0$ leaves this equation unchanged, although it can change how individual particles move. When the probability measure has no density, we express the same condition using expectations of smooth functions. We apply it to material acceleration and use *jerk* for the change in acceleration along an exact trajectory,

$$\mathbf{a}_t = \partial_t u_t + (D_x u_t)u_t, \qquad \mathbf{j}_t = \partial_t \mathbf{a}_t + (D_x \mathbf{a}_t)u_t. \tag{80}$$

Here $\mathbf{a}_t(X_t)$ and $\mathbf{j}_t(X_t)$ are the second and third time derivatives of the exact particle position. The term $(D_x u_t)u_t$ accounts for the change in velocity caused by the particle's motion through space.

Definition S21 (Distribution-preserving fields and the acceleration condition)**.** A field w has *zero weighted divergence under* a probability measure p if

$$\int \nabla\phi(x)\cdot w(x)\, p(\,\mathrm{d}x) = 0 \quad \text{for every } \phi \in C_c^\infty(\mathbb{R}^d), \tag{81}$$

where the tests are smooth and compactly supported. A velocity field u satisfies the *acceleration condition* along (p_t) if Eq. (81) holds with $w = \mathbf{a}_t$ and $p = p_t$ at every time.

Zero weighted divergence refers to the probability measure p, while ordinary zero divergence refers to volume. For example, rotation preserves a circularly symmetric distribution because it keeps the angle uniform. A constant rightward field also has zero ordinary divergence, but it moves a Gaussian to the right and changes its distribution. Under the regularity assumptions of Proposition S31, Eq. (81) means that the flow of w preserves p exactly. For acceleration, this means that motion along $\mathbf{a}_t$ preserves p_t while physical time t is held fixed.

D. Kinetic action as a comparison objective

Penalties on speed or velocity derivatives are used to make numerical integration easier [16–18]. We compare Euler's distribution error with kinetic action, the time integral of average squared speed,

$$\mathcal{A}_T(u) = \int_0^T \mathbb{E}_{p_t}\|u(t,X)\|^2\,\mathrm{d}t, \tag{82}$$

where the integrand is kinetic energy, defined here without the factor $1/2$. The base fields in Sections XI and XIII minimize this action among finite-action fields that satisfy the same weak continuity equation. Their Euler distribution errors are nevertheless larger than those of the corrected fields. Section XIV proves this action comparison and shows which restrictions the acceleration condition places on the path.

Appendix XI
Rotation separates the two error orders for a Gaussian

For a Gaussian path, we can find the correction by comparing how Euler and the exact flow change the radius. Let J be a quarter-turn in two dimensions, so Jx is perpendicular to x and has the same length. The field $ax + bJx$ combines expansion at rate $a > 0$ with rotation at rate b. An Euler step sends x to $(1+ah)x + bhJx$ and multiplies the squared radius by $(1+ah)^2 + b^2h^2$. The exact expansion gives $e^{2ah} = 1 + 2ah + 2a^2h^2 + O(h^3)$, while Euler without rotation misses one a^2h^2 term. Choosing $b = a$ supplies this term, and the same choice works as the step size changes.

For the probability path

$$p_t = \mathcal{N}(0, e^{2at}I_2), \qquad a > 0, \tag{83}$$

the base and corrected fields are therefore

$$v_0(x) = ax, \qquad v_1(x) = a(I+J)x. \tag{84}$$

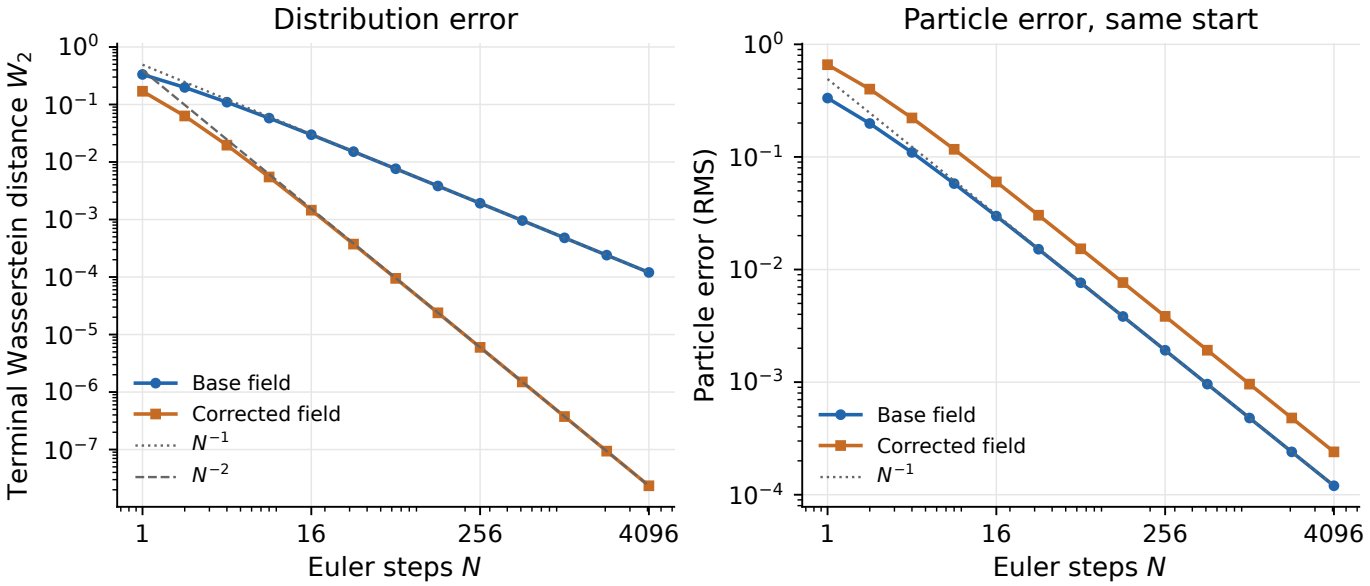


Fig. 7. Adding rotation gives second-order Gaussian distribution error, although particle error remains first order with twice the base field's leading coefficient. The left panel shows W_2 between numerical and exact distributions. The right panel shows particle error from the same starting point. The fields stay fixed as the uniform step count N increases, and the reference lines show the N^{-1} and N^{-2} rates. Parameters are $a = 0.6$, $T = 1$, and $\Sigma_0 = \left(\begin{smallmatrix} 1.5 & 0.25 \\ 0.25 & 0.75 \end{smallmatrix}\right)$.

These fields are part of the known Gaussian rotation family [19]. We use them to compare action, distribution error, and particle error. A linear change of coordinates gives the same construction for every positive definite covariance in even dimension. Euler also reproduces an added affine mean exactly. Appendix XVI gives these extensions, and Figure 7 shows an elliptical Gaussian example.

Theorem S22 (Gaussian distribution and particle errors)**.** *For the fields in Eq. (84) on $[0, T]$ with $T > 0$, the base field v_0 minimizes kinetic action among finite-action fields satisfying the same weak continuity equation. The corrected field adds $a(e^{2aT} - 1) > 0$ to that action. Euler's numerical covariance is $q_{i,k} I_2$, where*

$$q_{0,k} = \prod_{j<k} (1 + ah_j)^2, \qquad q_{1,k} = \prod_{j<k} (1 + 2ah_j + 2a^2 h_j^2). \tag{85}$$

At every grid point after the start,

$$0 < W_2(\widehat{p}_k^1, p_{t_k}) < W_2(\widehat{p}_k^0, p_{t_k}), \qquad W_2(\widehat{p}_k^i, p_{t_k}) = \sqrt{2}\,[e^{at_k} - \sqrt{q_{i,k}}]. \tag{86}$$

For uniform $h = T/N \to 0$, write $C_\Sigma = \sqrt{2} e^{aT}$. Then

$$W_2(\widehat{p}_N^0, p_T) = C_\Sigma \tfrac{1}{2} a^2 T h + O(h^2), \tag{87}$$

$$W_2(\widehat{p}_N^1, p_T) = C_\Sigma \tfrac{2}{3} a^3 T h^2 + O(h^3), \tag{88}$$

$$\|\widehat{X}_N^1 - X_T^1\|_{L^2} = C_\Sigma\, a^2 T h + O(h^2). \tag{89}$$

For $z = ah > 0$, the covariance factors satisfy $(1 + z)^2 < 1 + 2z + 2z^2 < e^{2z}$. Both numerical clouds are therefore too small, but the corrected cloud is closer to the exact size. Multiplying these factors keeps the same strict ordering on every increasing grid. To find the error orders, we take logarithms and add the errors over the steps. The one-step log-scale errors are $z - \log(1 + z) = z^2/2 + O(z^3)$ for the base field and $z - \frac{1}{2}\log(1 + 2z + 2z^2) = 2z^3/3 + O(z^4)$ for the corrected field. On a uniform grid, the sum over T/h steps gives first-order and second-order errors, respectively.

An optimal coupling pairs Gaussian points at the same angle, so the distribution error depends only on scale. Particle error also includes the angle error between particles from the same starting point. The base field has no angle error. Eqs. (87) and (89) show that adding the correction doubles the leading particle error coefficient, although distribution error becomes second order. The elliptical Gaussian in Figure 7 has the same behavior. Increasing the uniform step count from 32 to 64 reduces base distribution error from 1.51×10^{-2} to 7.63×10^{-3}, a factor of 1.98. Corrected distribution error decreases from 3.74×10^{-4} to 9.47×10^{-5}, a factor of 3.94. Appendix XVI gives the proof and bounds for unequal steps.

This improvement on every grid requires expansion. During contraction, the added rotation can make the numerical radius too large. For $a = -0.6$ and one step of length one, the base scale 0.4 is closer to the exact scale $e^{-0.6} \approx 0.549$ than the corrected scale $\sqrt{0.52} \approx 0.721$.

For the expanding circular Gaussian, the corrected acceleration is $2a^2 Jx$. Motion along this field follows circles and preserves the distribution. We use the same property below to bound the error without assuming a Gaussian path.

APPENDIX XII
THE ACCELERATION CONDITION BOUNDS GLOBAL DISTRIBUTION ERROR

A velocity field can preserve a distribution even when its acceleration does not. For steady rotation $u(x) = Jx$ on the uniform unit circle, the velocity preserves the distribution, but the acceleration $\mathbf{a}(x) = J^2x = -x$ points inward. Motion along this acceleration shrinks the circle. The corrected expanding Gaussian has tangential acceleration, so motion along its acceleration preserves the distribution. This is the property we use to bound Euler's distribution error.

A. A coupling that reduces the local error

Let $E_{t,h}(x) = x + hu_t(x)$ be the Euler map and $F_{t,t+h}$ the exact flow over the same interval. Hold physical time t fixed, and let Ψ_t^τ be the flow that follows $\mathbf{a}_t$ for duration τ. Under the acceleration condition and the regularity assumptions below, $X \sim p_t$ and $\Psi_t^\tau X$ have the same distribution. Applying Euler to either starting point therefore gives the same output distribution. We can use the shifted start for Euler and the original start for exact motion to form a closer pairing.

The choice $\tau = h^2/2$ supplies the leading displacement that Euler omits. We write $O_{L^2}(h^q)$ for a remainder whose root mean squared norm is at most Ch^q, with C independent of h as $h \downarrow 0$. The shifted start satisfies $\Psi_t^{h^2/2}X = X + h^2\mathbf{a}_t(X)/2 + O_{L^2}(h^4)$. Evaluating velocity at this point changes the Euler displacement by only $O_{L^2}(h^3)$. Thus the root mean squared distance between $E_{t,h}(\Psi_t^{h^2/2}X)$ and $F_{t,t+h}(X)$ is $O(h^3)$, as Figure 8 shows. For the circular Gaussian with $a = 0.6$, the shift rotates the start by $2a^2\tau = 0.36h^2$.

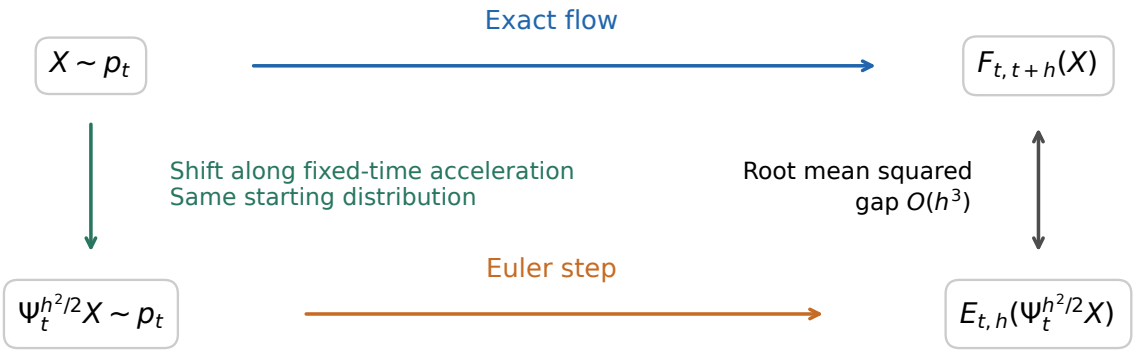


Fig. 8. Shifting the start along the acceleration field gives a closer pairing and keeps Euler's output distribution unchanged. During the shift, physical time is held fixed and the distribution remains p_t. The shift supplies the leading displacement that Euler omits. We pair the Euler output from the shifted start with exact motion from the original start. Under Theorem 1, their root mean squared distance is $O(h^3)$.

To obtain the global bound, we also need to control how later steps increase this local error. Two continuous derivatives of velocity give the trajectory expansion. Global Lipschitz bounds control how far apart the outputs can be when the inputs differ, and mean square bounds allow us to integrate over the distribution. The theorem states these assumptions.

The exact theorem statement is Theorem 1 in the main paper.

Each step adds an error bounded by a constant times h_j^3. The sum $\sum_j h_j^3 \le TH^2$ gives the total contribution of these errors, and e^{LT} bounds how much later Euler steps can increase them. The error is therefore at most a constant times H^2 at every grid point, without any Gaussian or symmetry assumption. The theorem gives a test for a particular field. It does not give a suitable field for every path, and Section XIV shows that some paths cannot satisfy the condition. Appendix XVII proves the bound using the local coupling and the Euler stability estimate $1 + Lh$.

B. Local necessity and weak error

The acceleration condition is also necessary if one Euler step from the exact distribution has error $o(h^2)$. Here $o(h^2)$ means that the error divided by h^2 tends to zero. For this rate to hold, the acceleration must preserve the distribution in the sense of Eq. (81).

Proposition S23 (Local converse)**.** *Let u and p_t satisfy the regularity, growth, and moment assumptions of Theorem 1, without imposing the acceleration condition. At a fixed $t < T$, suppose*

$$W_2((E_{t,h})_\# p_t, p_{t+h}) = o(h^2) \quad \text{as } h \downarrow 0.$$

Then $\mathbf{a}_t$ satisfies Eq. (81) under p_t.

Here $(E_{t,h})_{\#}p_t$ is the distribution obtained by applying Euler to a point drawn from p_t. The result concerns one step from the exact distribution. Errors across several steps can cancel at the final time, even when the acceleration condition fails. For example, the field $u(t,x) = \sin(2\pi t)e_1$ on $[0,1]$ with a Gaussian source has zero exact displacement and zero total Euler displacement on every uniform grid. Euler therefore gives the exact terminal distribution, although the acceleration changes the current distribution whenever it is nonzero. Appendix XVIII gives the proof and this example.

For a smooth compactly supported test function ϕ, the difference between expectations after one Euler step and exact motion is

$$\mathbb{E}_{p_t}\phi(E_{t,h}X) - \mathbb{E}_{p_{t+h}}\phi = -\tfrac{1}{2}h^2 \int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t + o(h^2). \tag{90}$$

Under the proposition's assumptions, the absolute difference in expectations is at most the Lipschitz constant of ϕ times W_2. If $W_2 = o(h^2)$, the leading integral in Eq. (90) must therefore be zero. This gives the acceleration condition. Appendix XVIII justifies the remainder, and Proposition S32 in Appendix XVIII-A gives an explicit cubic bound under a stronger moment assumption. Earlier global weak expansions follow expectation errors through the later exact flow [11, 20]. The coupling in Theorem 1 bounds the distribution distance directly.

APPENDIX XIII
A NONLINEAR PATH HAS DIFFERENT ERROR ORDERS

The same difference in error orders is possible when the distribution changes shape. We use a radial distribution whose outer radii expand by larger relative factors. Unlike uniform expansion, this motion does not multiply all radii by the same number. Radial symmetry still gives an optimal pairing by radius, so we can compare errors when a covariance calculation is no longer enough.

We choose the rotation to cancel the radial part of acceleration. For the outward field $\alpha(r)x$, this part is $[\alpha(r)^2 + \alpha(r)r\alpha'(r)]x$. Adding the tangential velocity $\beta(r)Jx$ contributes $-\beta(r)^2 x$ to the radial acceleration. Choosing $\beta(r)^2 = \alpha(r)^2 + \alpha(r)r\alpha'(r)$ cancels the radial acceleration. The remaining tangential acceleration preserves a radial distribution. We choose an increasing α so that the relative expansion depends on radius. We also keep α bounded to control growth at large radii. With $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $a, \eta > 0$, define

$$\alpha(r) = a + \eta\frac{r^2}{1+r^2}, \qquad P(r) = \alpha(r)^2 + \alpha(r)r\alpha'(r), \qquad \beta(r) = \sqrt{P(r)}. \tag{91}$$

The positive function P gives a real rotation rate, and the resulting base and corrected fields are

$$u_0(x) = \alpha(\|x\|)x, \qquad u_1(x) = \alpha(\|x\|)x + \beta(\|x\|)Jx. \tag{92}$$

For both fields, the radius satisfies $\dot{R} = \alpha(R)R$. We write $R(t,r)$ for its solution from radius r, with $R(0,r) = r$. When the source is rotationally invariant, the added rotation keeps the angle uniform at each positive radius and leaves the origin fixed. The two fields therefore give the same distribution at every time, although their particles follow different trajectories. Since α increases with radius, the distribution changes shape whenever the source support contains two distinct positive radii.

For example, $a = 0.6$ and $\eta = 0.3$ give relative expansion rates $\alpha(1) = 0.75$ and $\alpha(2) = 0.84$. The outer circle therefore grows faster relative to its own radius. Figure 9 shows this shape change by dividing radius quantiles by the median radius. These ratios change along the nonlinear path. They would stay constant if every radius grew by the same factor.

The corrected field also satisfies the acceleration condition. Differentiation gives

$$\mathbf{a}_{u_1}(x) = \beta_a(\|x\|)Jx, \qquad \beta_a(r) = 2\alpha(r)\beta(r) + \alpha(r)r\beta'(r) \geq 2a^2. \tag{93}$$

This acceleration is tangent to each circle, so motion along it at fixed physical time preserves every radial distribution. The acceleration is nonzero away from the origin, so these trajectories are not straight with constant velocity as in Proposition S27. The next theorem shows that the correction gives smaller distribution error on every grid. It also gives the exact error orders on uniform grids. Appendix XIX contains the regularity checks and proofs.

Theorem S24 (Nonlinear error separation on every grid)**.** *Let $a, \eta, T > 0$, and let p_0 be rotationally invariant on $\mathbb{R}^2$, with $0 < \mathbb{E}\|X_0\|^2 < \infty$. The fields in Eq. (92) are smooth and globally Lipschitz. They generate the same probability path. The base field minimizes kinetic action among finite-action fields satisfying the same weak continuity equation, and the correction strictly increases it. On every finite increasing time grid,*

$$0 < W_2(\widehat{p}^1_k, p_{t_k}) < W_2(\widehat{p}^0_k, p_{t_k}) \qquad (k \geq 1). \tag{94}$$

On uniform grids $h = T/N \to 0$, there are finite, strictly positive constants $\kappa_0, \kappa_1, \kappa_{\mathrm{p}}$ such that

$$\frac{W_2(\widehat{p}^0_N, p_T)}{h} \to \kappa_0, \qquad \frac{W_2(\widehat{p}^1_N, p_T)}{h^2} \to \kappa_1, \qquad \frac{\|\widehat{X}^1_N - X^1_T\|_{L^2}}{h} \to \kappa_{\mathrm{p}}. \tag{95}$$

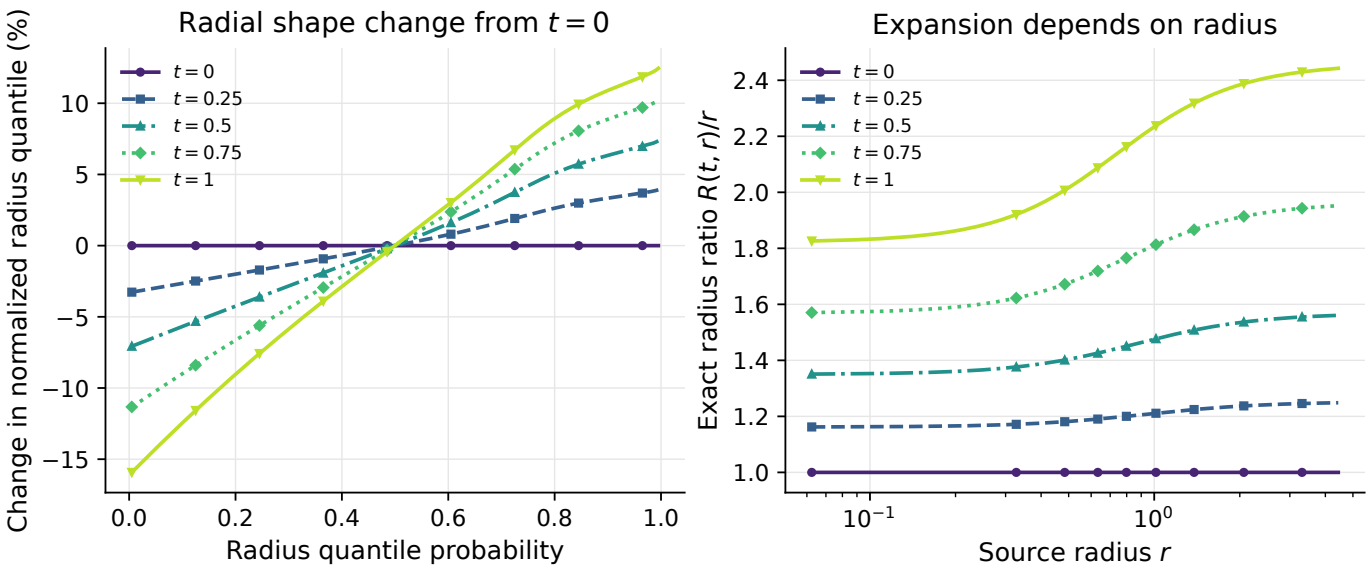


Fig. 9. The nonlinear distribution changes shape because outer radii expand by larger relative factors. The left panel shows the percentage change, from $t = 0$, in each radius quantile divided by the median radius. Uniform scaling would leave these ratios unchanged. The right panel shows the expansion factor as a function of starting radius, with the parameters of Figure 10.

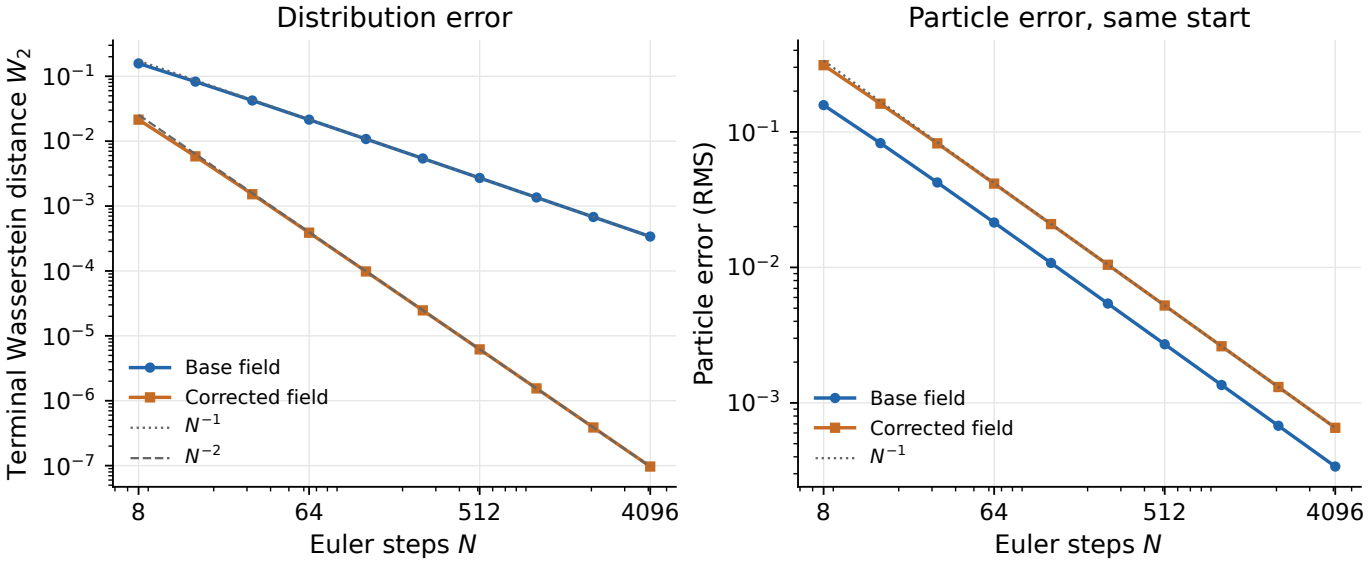


Fig. 10. The corrected nonlinear field gives second-order distribution error, although particle error remains first order with a larger leading coefficient. The left panel uses Eq. (96) to measure distribution error, while the right panel compares particles from the same start. Both fields remain fixed as the uniform step size decreases, and the reference slopes give the rates in Eq. (95). The source is $0.6\mathcal{N}(0, 0.5^2 I_2) + 0.4\mathcal{N}(0, 1.5^2 I_2)$, with $a = 0.6$, $\eta = 0.3$, and $T = 1$.

The error comparison follows from the one-step radius maps

$$G_{0,h}(r) = r(1 + h\alpha(r)), \qquad G_{1,h}(r) = r\sqrt{(1 + h\alpha(r))^2 + h^2 P(r)}.$$

For every $h, r > 0$, both maps increase with the starting radius and satisfy $G_{0,h}(r) < G_{1,h}(r) < R(h, r)$. The corrected radius is below the quadratic Taylor approximation of the exact radius. That approximation is below the exact radius because the exact radius has a positive third time derivative. Since the maps are increasing, the same ordering holds after repeated steps. Let $R_{i,k}(r)$ be the numerical radius after k steps under u_i. An optimal coupling pairs points at the same angle and matches their radii in increasing order. Since all radius maps are increasing, this pairs the radii that came from the same source radius. Thus

$$W_2(\widehat{p}^{\,i}_k, p_{t_k})^2 = \mathbb{E}\Big[\big(R_{i,k}(R_0) - R(t_k, R_0)\big)^2\Big], \qquad R_0 = \|X_0\|. \tag{96}$$

Equation (96) shows how the radius ordering gives the stated distribution error ordering. In one step, the radius falls short by positive leading terms of orders h^2 and h^3 for the base and corrected fields. After repeated steps, these give positive leading

terms of orders h and h^2. The error bounds are proportional to the source radius, so a finite second moment is enough to average the squared errors.

Particle error remains first order because each corrected Euler step rotates too little by a term of order h^2. Over repeated steps, this gives a first-order angle error that is larger than the second-order radius error for sufficiently small steps. The base field keeps each angle fixed, so its particle and distribution errors are equal. For the mixture in Figure 10, the correction increases the leading particle error coefficient from 1.39 to 2.68, a factor of about 1.93. Appendix XIX derives these coefficients.

A fixed invertible affine transformation keeps the same error orders, and independent products give examples in higher dimensions. These extensions need not keep the same exact distances or strict error comparison between fields. Appendix XIX-E states which properties hold.

a) Numerical verification.: For the mixture in Figure 10, increasing the step count from 2,048 to 4,096 gives observed orders 0.9997, 1.9995, and 0.9998 for base distribution, corrected distribution, and corrected particle error, respectively. These values agree with the limiting orders in the theorem. We check numerical accuracy using Cartesian updates, an independent reference calculation, and finer quadrature. Appendix XXVII and Table VI give the results. The supplement includes the summary outputs and reproduction commands.

APPENDIX XIV
WHICH PROBABILITY PATHS CAN SATISFY THE CONDITION?

A. Mean, second moment, and kinetic action

The acceleration condition rules out some probability paths before we choose a velocity field. To see why, let $M(t) = \mathbb{E}\|X_t\|^2$ be the second moment. Its second derivative equals twice the kinetic energy plus $2\mathbb{E}[X_t \cdot \mathbf{a}_t(X_t)]$. Under the assumptions below, the acceleration condition makes the last term zero. The path therefore determines the kinetic energy of every field that satisfies the condition.

Proposition S25 (Second-moment restriction)**.** *Suppose exact trajectories and their first two time derivatives are continuous, and*

$$\mathbb{E} \sup_{0\le t\le T} \left[\|X_t\|^2 + \|u_t(X_t)\|^2 + \|\mathbf{a}_t(X_t)\|^2\right] < \infty.$$

If $\mathbf{a}_t$ satisfies Eq. (81), then

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\mathbb{E}\|X_t\|^2 = 2\mathbb{E}\|u_t(X_t)\|^2 \ge 2K_{\min}(t), \tag{97}$$

where $K_{\min}(t)$ is the infimum of kinetic energy among fields satisfying the weak continuity equation at time t for the given probability path.

When the second moment is constant, Eq. (97) forces velocity to be zero almost everywhere. This rules out a changing distribution with constant second moment. For example, a centered noncircular Gaussian can rotate without changing its second moment. No field that generates this path can satisfy both the acceleration condition and the proposition's assumptions. A circular Gaussian stays unchanged under rotation, so its path does satisfy the condition with the zero field.

The condition also requires an affine mean, which moves at constant velocity. To prove this, we apply smooth cutoffs to the coordinate functions and let their supports expand. The acceleration condition gives zero second derivative of the mean. Integrating the second-moment identity also determines the action.

Corollary S26 (Affine mean and action fixed by the path)**.** *Under the assumptions and acceleration condition of Proposition S25, let $m(t) = \mathbb{E}X_t$, $M(t) = \mathbb{E}\|X_t\|^2$, and $\xi = \mathbb{E}u_0(X_0)$. Then*

$$m(t) = m(0) + t\xi, \qquad \mathcal{A}_T(u) = \tfrac{1}{2}\big[M'(T) - M'(0)\big]. \tag{98}$$

Thus any two fields that generate the same probability path and satisfy these assumptions have the same kinetic action.

The assumptions of Theorem 1 imply those of Proposition S25, as verified in Appendix XXII. The energy inequality also excludes any interval with $g'' < 0$ on a scaling path $p_t = \mathrm{Law}(g(t)X_0)$, provided $g > 0$ and $\mathbb{E}\|X_0\|^2 > 0$.

The action required by the acceleration condition need not be the minimum action on the path. Suppose a finite-action gradient field v^∇ generates the path and lies in the $L^2(p_t)$ closure of compactly supported smooth gradients for almost every t. Every finite-action correction r_t that preserves the weak continuity equation is orthogonal to this closure. The cross term in the squared velocity is therefore zero, which gives

$$\mathcal{A}_T(v^\nabla + r) = \mathcal{A}_T(v^\nabla) + \int_0^T \mathbb{E}_{p_t}\|r_t\|^2 \,\mathrm{d}t. \tag{99}$$

The added term is nonnegative. It is positive whenever the correction is nonzero in space-time L^2, so v^∇ minimizes action within this class [4, 5, 10]. Here the distribution at every time is fixed. We are not minimizing over different paths with the same endpoints. Appendix XXIII checks the closure assumption for the Gaussian and nonlinear examples.

B. Tests for generative probability paths

We next apply these moment restrictions to paths used in generative modeling. Let X_0 be standard Gaussian noise independent of data X_1 with finite second moment, and write $m_1 = \mathbb{E}X_1$ and $M_1 = \mathbb{E}\|X_1\|^2$. We test whether a field that generates each path can satisfy both the acceleration condition and the assumptions of Proposition S25.

a) Variance-preserving interpolation.: Independence gives the following mean and second moment for $X_t = \rho_t X_1 + \sqrt{1-\rho_t^2}X_0$:

$$m(t) = \rho_t m_1, \qquad M(t) = d + (M_1 - d)\rho_t^2.$$

When $M_1 = d$, the second moment is constant, so only a stationary path can satisfy the stated condition and assumptions. This rules out a path from Gaussian noise to a different data distribution with the same second moment. When $M_1 \neq d$, the cosine schedule $\rho_t = \sin(\pi t/(2T))$ is also ruled out. In this case, $(\rho_t^2)''$ changes sign, so M'' is negative somewhere and contradicts Eq. (97). The schedule also has an affine mean only if $m_1 = 0$.

b) Independent linear interpolation.: For $X_t = (1-t)X_0 + tX_1$ on $[0,1]$, each noise-data pair moves along a straight segment. The conditional-mean field averages the velocities of segments that pass through each position. Its trajectories can therefore curve even though the original segments are straight. Conditional averaging cannot increase mean squared speed, so

$$\tfrac{1}{2}M''(t) = d + M_1 = \mathbb{E}\|X_1 - X_0\|^2 \geq \mathbb{E}\|\mathbb{E}[X_1 - X_0 \mid X_t]\|^2 \geq K_{\min}(t).$$

The last inequality uses $v_t(x) = \mathbb{E}[X_1 - X_0 \mid X_t = x]$ as a candidate for $K_{\min}(t)$. This assumes enough regularity for the field to generate the path. With this assumption, independent linear interpolation passes the necessary moment test. Passing this test does not prove that a field satisfying the acceleration condition exists. Any such field must have constant kinetic energy $d + M_1$, and its action over the unit interval must also equal $d + M_1$.

C. Gradient fields force straight motion

The moment test does not decide whether a suitable field exists for independent linear interpolation. For global gradient fields, we can say more. Under all assumptions of the main theorem, the acceleration condition forces the acceleration to be zero.

Proposition S27 (Gradient fields have straight trajectories)**.** *Under all assumptions of Theorem 1, suppose each u_t is a global spatial gradient on $\mathbb{R}^d$. Then $\mathbf{a}_t = 0$ p_t-almost everywhere for every $t \in [0,T]$. For p_0-almost every start,*

$$X_t = X_0 + tu(0, X_0) \qquad (0 \leq t \leq T).$$

Euler is exact on every finite increasing grid for this same full-probability set of starts. Both particle error and distribution error are zero.

A gradient velocity has a symmetric Jacobian, which makes its material acceleration a gradient as well. The global assumptions put this acceleration in the weighted gradient closure. The acceleration condition makes it orthogonal to that closure, so the acceleration must be zero. Continuity along trajectories then gives straight motion for one full-probability set of starts. Appendix XXIII-A gives the proof.

For independent linear interpolation from standard Gaussian noise, the exact conditional-mean field is a gradient when the intermediate densities are smooth and positive [1]:

$$v_t(x) = \frac{x}{t} + \frac{1-t}{t}\nabla \log p_t(x), \qquad 0 < t < 1.$$

Both terms are spatial gradients. Under the theorem's global assumptions, the acceleration condition can therefore hold only when Euler is exact for almost every start. Exact probability-flow fields are also gradients when their forward drift is isotropic and linear [3]. A learned field need not equal these exact fields, so the assumptions must be checked for the learned field separately.

Regular displacement interpolation gives one class of straight trajectories. It pairs endpoints optimally and moves each pair at constant velocity. Appendix XXIV states sufficient regularity conditions. Another class follows from Liu's straight rotated coupling between standard Gaussians [5, Example 3.5]. We scale the target and use the quarter-turn pairing $X_1 = \sigma J X_0$. In even dimension, the resulting straight segments have the same distribution at every time as independent interpolation to $\mathcal{N}(0, \sigma^2 I)$. Appendix XXV proves this identity.

APPENDIX XV
SOLVER CHOICE AND IMPERFECT CORRECTIONS

A. The useful correction depends on the solver

A correction selected for Euler need not improve another solver. We compare Euler with Heun's method, which averages the initial velocity with the velocity at an Euler prediction:

$$y = x + hu(t,x), \qquad x^+ = x + \tfrac{h}{2}\{u(t,x) + u(t+h,y)\}.$$

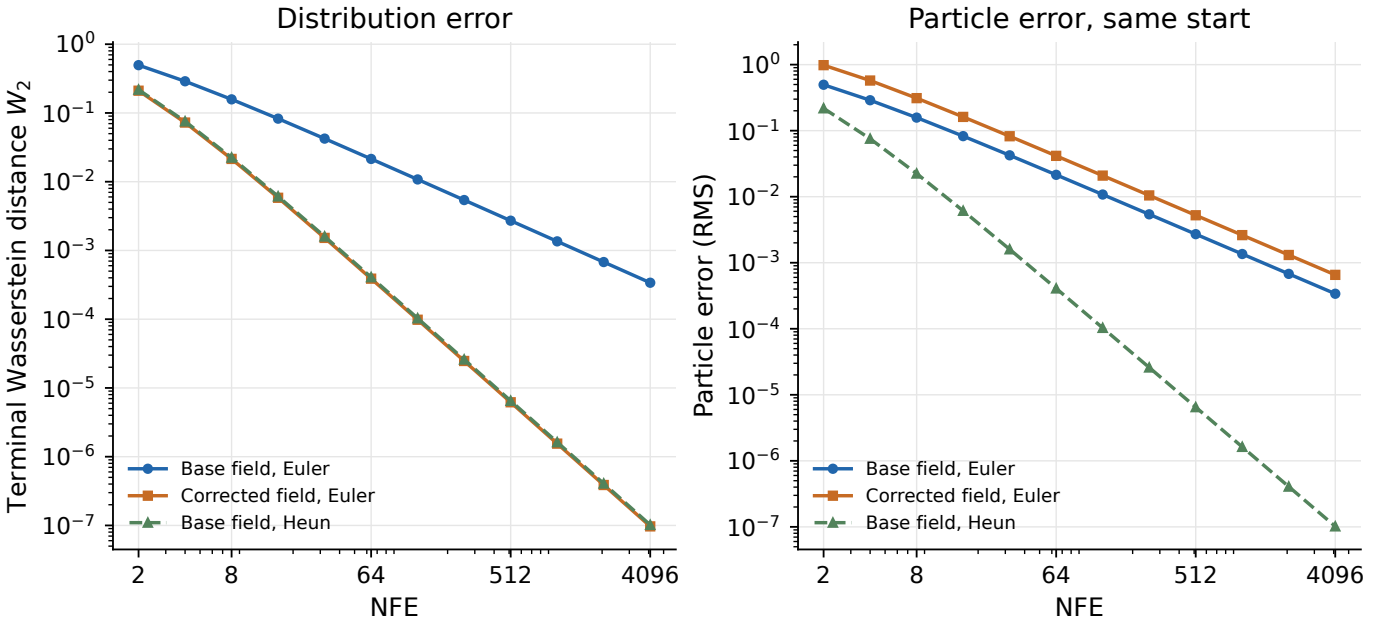


Fig. 11. Corrected Euler closely matches base-field Heun in nonlinear distribution error at equal NFE, although it has larger particle error. The source and parameters are those of Figure 10, and each even N gives N uniform Euler steps or $N/2$ Heun steps of twice the length. The left and right panels compare distribution and particle errors, respectively. Base-field Heun has particle error equal to W_2.

Heun uses two field evaluations per step and is also used in generative sampling [21]. For the circular Gaussian field $(aI+bJ)x$, its one-step complex multiplier is $1+h(a+ib)+\frac{1}{2}h^2(a+ib)^2$. The magnitude of this multiplier gives the scale change, and its argument gives the rotation. Adding the log-scale errors over a uniform grid gives distribution error

$$C_\Sigma \frac{T|a^3-3ab^2|}{6}h^2+O(h^3), \qquad C_\Sigma=\sqrt{2}e^{aT}. \tag{100}$$

The Euler choice $b=a$ doubles the coefficient in Eq. (100) compared with $b=0$. Choosing $b=a/\sqrt{3}$ instead cancels this coefficient for Heun. Thus the useful correction depends on the solver. Appendix XXI derives the expansion and extends it to the Gaussian family of Appendix XVI.

a) Comparison at equal NFE.: Heun uses twice as many evaluations per step as Euler. To compare them at equal NFE, we therefore use two corrected Euler steps of length h for each base-field Heun step of length $2h$. Both methods then give the same length multiplier,

$$\left(\sqrt{1+2ah+2a^2h^2}\right)^2=1+a(2h)+\tfrac{1}{2}a^2(2h)^2.$$

Proposition S28 (Gaussian equality at equal NFE)**.** *For the circular Gaussian fields in Theorem S22, let N be even and $h=T/N$. Compare corrected-field Euler with N steps of length h and base-field Heun with $N/2$ steps of length $2h$. Their distributions agree at every shared grid point. Each method uses N field evaluations over $[0,T]$.*

The identity extends to the Gaussian construction in Appendix XVI and to nonuniform grids with equal lengths within each consecutive pair of Euler steps. Appendix XXI-A proves both statements.

The two methods have the same distributions at shared grid points, although their particle errors differ. Corrected Euler still has a first-order angle error, while base-field Heun has second-order particle error.

For the nonlinear mixture in Figure 11, corrected Euler has 3.3–4.7% smaller distribution error than base-field Heun over 2 to 4,096 NFE. Its particle error is much larger. Base-field Heun keeps each angle fixed and has an increasing radius map, so its particle error equals its distribution error. Table III gives the low-NFE values and ratios.

With the rotation $b=a/\sqrt{3}$, Heun has third-order distribution error while particle error remains second order. We compare this corrected Heun method with Kutta's third-order Runge–Kutta method, denoted RK3, on the base field at equal NFE. Corrected Heun has 18.0–20.9% smaller distribution error at 12, 48, 192, and 384 NFE, although its particle error is larger. Appendix Table V gives the values. Appendix XXI-B derives the coefficients, and Appendix Figure 13 shows convergence.

B. A residual acceleration adds a first-order term to the bound

An imperfect correction can keep the probability path unchanged without fully satisfying the acceleration condition. Write $\mathbf{a}_t=w_t+g_t$, where w_t satisfies Eq. (81) under p_t and g_t is the residual. Shifting the starting points along w_t supplies the part of Euler's missing displacement that preserves the distribution. The contribution from g_t remains.

TABLE III
AT THESE LOW NFE VALUES, CORRECTED EULER HAS SLIGHTLY SMALLER DISTRIBUTION ERROR AND LARGER PARTICLE ERROR THAN BASE-FIELD HEUN. THE SETUP IS THAT OF FIGURE 10, AND THE DISPLAYED W_2 EQUALS PARTICLE ERROR FOR BOTH BASE-FIELD METHODS. THE FINAL COLUMN GIVES THE RATIO OF CORRECTED EULER TO BASE-FIELD HEUN DISTRIBUTION ERROR, CALCULATED FROM UNROUNDED VALUES.

NFE	Base field Euler W_2	Corrected field Euler W_2	Base field Heun W_2	Corrected field Euler particle error	Corrected Euler / base-field Heun W_2
2	4.96×10^{-1}	2.11×10^{-1}	2.18×10^{-1}	9.82×10^{-1}	0.9671
4	2.89×10^{-1}	7.25×10^{-2}	7.54×10^{-2}	5.74×10^{-1}	0.9609
8	1.58×10^{-1}	2.14×10^{-2}	2.24×10^{-2}	3.11×10^{-1}	0.9570
16	8.26×10^{-2}	5.84×10^{-3}	6.11×10^{-3}	1.62×10^{-1}	0.9548

Proposition S29 (Residual acceleration)**.** *Keep the assumptions on u and C_j from Theorem 1. Assume that w_t is C^1, globally Lipschitz with common constant L_w, and satisfies Eq. (81) under p_t. If $C_{w,2} = \sup_t \|w_t\|_{L^2(p_t)} < \infty$ and $\sup_t \|g_t\|_{L^2(p_t)} \le \varepsilon$, then*

$$\max_k W_2(\widehat{p}_k, p_{t_k}) \le e^{LT} T \left(C_w H^2 + \tfrac{1}{2}\varepsilon H \right), \qquad C_w = \frac{C_j}{6} + \frac{L C_{w,2}}{2} + \frac{T L_w C_{w,2}}{8}. \tag{101}$$

In each step, the residual adds at most $\varepsilon h^2/2$ to the root mean squared distance between the paired points. Adding these contributions over the steps gives the εH term. If the other constants stay bounded, $\varepsilon = O(H)$ is enough for second-order accuracy. For example, halving both H and ε divides both terms in $H^2 + \varepsilon H$ by four. Halving only H divides the residual term by two. A fixed residual leaves a first-order term in the bound, but this does not mean that the actual error is always first order. Appendix XX proves the bound and gives the sum over individual steps.

In the nonlinear example, a fixed mismatch in rotation gives first-order error. Replacing β by $\lambda\beta$, with $\lambda \ge 0$ fixed as the grid is refined, keeps the probability path unchanged but gives acceleration

$$\mathbf{a}_{u_\lambda}(x) = (1 - \lambda^2) P(r) x + \lambda \beta_a(r) J x.$$

For $\lambda \ne 1$, the remaining radial acceleration leads to $W_2(\widehat{p}_N^{\lambda}, p_T)/h \to |1 - \lambda^2| \kappa_0 > 0$, as proved in Appendix XIX-D. Even a small fixed mismatch therefore gives first-order convergence. It can still reduce error compared with the base field over a finite range of step sizes.

We can estimate where the mismatch starts to matter by equating its leading contribution $|1 - \lambda^2|\kappa_0 h$ with the exact correction's contribution $\kappa_1 h^2$. This gives

$$h_* \approx |1 - \lambda^2| \frac{\kappa_0}{\kappa_1}.$$

For the Gaussian path, the corresponding coefficients are $c_0^{\mathrm{G}} = C_\Sigma a^2 T/2$ and $c_1^{\mathrm{G}} = 2 C_\Sigma a^3 T/3$, where $C_\Sigma = e^{aT}\sqrt{\operatorname{tr} \Sigma_0}$. They give $h_*^{\mathrm{G}} \approx |1 - \lambda^2| c_0^{\mathrm{G}}/c_1^{\mathrm{G}} = 3|1 - \lambda^2|/(4a)$. At $a = 0.6$ on $[0, 1]$, the 2% mismatches $\lambda = 0.98$ and 1.02 give about 20 steps, as shown in Appendix Figure 12. The first-order error dominates at sufficiently smaller step sizes. When $\lambda > 1$, the two contributions can have opposite signs and cancel near this estimate. Thus the estimate gives a range where both terms matter, rather than an exact threshold between orders.

The appendices give the proofs and numerical details for the paper. In the Gaussian example, distribution error depends on scale, while particle error also depends on angle. The nonlinear example shows the same difference when the distribution changes shape. The general proof shows how the acceleration condition improves distribution accuracy in both examples. The table below gives the location of each result.

Main claim or comparison	Support
Finite-time W_2 bound and its coupling	Appendix XVII
Minimum action and the gradient-field consequence	Appendix XXIII
Gaussian distribution and particle errors	Appendix XVI
Nonlinear rates, coefficient error, and extensions	Appendix XIX
Residual acceleration bound	Appendix XX
Local converse and terminal cancellation	Appendix XVIII
Solver dependence and equal NFE	Appendix XXI
Mean, second-moment, and action restrictions	Appendix XXII
Scaled rotated Gaussian coupling	Appendix XXV
Weak expansion and regular displacement interpolation	Appendices XVIII-A, XXIV
Arbitrarily small action cost on one grid	Appendix XXVI
Population calculations and refinement results	Appendix XXVII

For a map F, the pushforward $F_{\#}\mu$ is the distribution of $F(X)$ when $X \sim \mu$. We write $\mathcal{P}_2(\mathbb{R}^d)$ for probability measures with finite second moment, and define the root mean squared size of a vector field w by

$$\|w\|_{L^2(p)} = \left(\int \|w(x)\|^2 p(\,\mathrm{d}x) \right)^{1/2}.$$

For a random vector Z, $\|Z\|_{L^2}$ is its root mean squared norm. Table IV lists the main symbols.

TABLE IV
NOTATION USED IN THE PROOFS. MATRICES AND COEFFICIENTS SPECIFIC TO AN EXAMPLE ARE DEFINED WITH THAT EXAMPLE.

Symbol	Meaning
p_t, $\widehat{p}_k$	Exact distribution at time t and Euler distribution at node k
u_t, r_t	Velocity field and a correction added to that field
$\mathbf{a}_t$, $\mathbf{j}_t$	Second and third time derivatives along an exact particle
T, h_k, H	Final time, step length, and largest step length
W_2	Distribution distance from the best root mean squared pairing
$\mathcal{A}_T$	Kinetic action over the time interval $[0, T]$
$F_{s,t}$	Exact particle flow from time s to time t
Ψ_t^τ	Auxiliary acceleration flow with physical time t held fixed
L, L_a	Common spatial Lipschitz bounds for velocity and acceleration
C_a, C_j	Uniform $L^2(p_t)$ bounds for acceleration and its material derivative
Σ_0, Ω	Initial Gaussian covariance and its weighted rotation matrix
C_0, K_t	Gaussian covariance root and skew matrix in the rotation representation
$C_{w,2}$, C_w	L^2 bound for w_t and constant in the residual bound
α, β, β_a	Nonlinear radial rate, rotation rate, and tangential acceleration coefficient
$\mathcal{C}_i$, Θ_1	Leading radius-error and angle-error coefficient functions
$\mathcal{T}$, G, d_{aff}	Affine map, its matrix, and its shift
$R(t,r)$, $R_{i,k}(r)$	Exact and numerical radii from starting radius r

APPENDIX XVI
FIXED GAUSSIAN CORRECTION AND RECURSIVE ERRORS

The Gaussian result also holds when the mean moves at constant velocity and the covariance is any positive definite matrix in even dimension. Assume $T, a > 0$ and $\Sigma_0 = C_0 C_0^\top$, with $J^\top = -J$, $J^\top J = I$, and $\Omega = C_0 J C_0^{-1}$. For $m_t = m_0 + t\xi$, the fields $v_0(t,x) = \xi + a(x - m_t)$ and $v_1(t,x) = \xi + a(I + \Omega)(x - m_t)$ both generate $p_t = \mathcal{N}(m_t, e^{2at}\Sigma_0)$. These parameters stay fixed as the grid changes. The base field v_0 minimizes action among finite-action fields satisfying the same weak continuity equation. The correction increases action by

$$\mathcal{A}_T(v_1) - \mathcal{A}_T(v_0) = a^2 \operatorname{tr}(\Sigma_0) \frac{e^{2aT} - 1}{2a} > 0.$$

Euler reproduces m_t exactly and has covariance $q_{i,k}\Sigma_0$, where

$$q_{0,k} = \prod_{j<k} (1 + ah_j)^2, \qquad q_{1,k} = \prod_{j<k} (1 + 2ah_j + 2a^2 h_j^2).$$

Euler's distribution error is $\sqrt{\operatorname{tr}(\Sigma_0)}[e^{at_k} - \sqrt{q_{i,k}}]$. The corrected error is positive and strictly smaller for every $k \geq 1$. Equations (87)–(89) hold with $C_\Sigma = \sqrt{\operatorname{tr}(\Sigma_0)} e^{aT}$. Setting $m_0 = \xi = 0$ and $\Sigma_0 = I_2$ gives Theorem S22.

Write

$$I_a(T) = \int_0^T e^{2at}\,\mathrm{d}t = \frac{e^{2aT} - 1}{2a}.$$

Subtracting the mean and applying the inverse covariance factor gives a Gaussian source with identity covariance. These coordinates are called whitened coordinates. Each corrected Euler step changes scale and angle, but rotation leaves the Gaussian distribution unchanged. Distribution error therefore depends only on scale. Particles from the same starting point also have an angle error. The scale error has a positive second-order coefficient, while the angle error has a positive first-order coefficient.

The correction belongs to the Gaussian family of Lim [19, Proposition 5.5]. For a single Gaussian with $\Sigma_t = e^{2at} C_0 C_0^\top$, the skew matrix $K_t = be^{-2at} C_0^{-\top} J C_0^{-1}$ gives $\Sigma_t K_t (x - m_t) = b\Omega(x - m_t)$. This form lets us calculate kinetic action and Euler error over the full time interval.

Proof of Theorem S22 and its Gaussian extension. **Common path and action.** The assumptions on J imply $J^2 = -I$. Direct matrix multiplication verifies

$$\Omega\Sigma_t + \Sigma_t \Omega^\top = 0, \qquad \Omega\Sigma_t\Omega^\top = \Sigma_t, \qquad \Sigma_t = e^{2at}\Sigma_0.$$

The centered flows $e^{at} I$ and $e^{at} C_0 e^{atJ} C_0^{-1}$ both give covariance Σ_t and mean m_t. Both flows are global and affine, so the distributions remain Gaussian. Equal means and covariances therefore give the same distribution at every intermediate time.

For $Y \sim \mathcal{N}(0, \Sigma_t)$, we have $\mathbb{E}\|\Omega Y\|^2 = \operatorname{tr}(\Sigma_t)$ and $\mathbb{E}[Y^\top \Omega Y] = \operatorname{tr}(\Omega\Sigma_t) = 0$, since $\Omega\Sigma_t$ is skew. The cross terms with ξ also have zero expectation because Y is centered. Expanding the squared speed and integrating over time therefore gives the action identities.

The base field is the gradient of $\xi^\top x + (a/2)\|x - m_t\|^2$. This potential has quadratic growth, so the projection argument in Appendix XXIII applies. The base field uniquely minimizes action up to almost-everywhere equality. For the continuous fields in the class $\mathcal{V}(p)$ defined there, positive Gaussian density gives equality at every time and position. In particular,

$$\mathcal{A}_T(v_0) = T\|\xi\|^2 + a^2 \operatorname{tr}(\Sigma_0) I_a(T).$$

Recursive covariance and strict improvement. Let $r_{i,k} = \sqrt{q_{i,k}}$ and $\widehat{p}^i_k = \widehat{p}^{v_i}_k$ for $i = 0, 1$. Euler reproduces the affine mean exactly, while its centered whitened matrices are $(1 + ah_j)I$ and $(1 + ah_j)I + ah_j J$. The second matrix multiplied by its transpose is $(1 + 2ah_j + 2a^2h_j^2)I$, so repeated covariance updates give the product formulas. For equal-mean Gaussians with covariances $r^2\Sigma_0$ and $s^2\Sigma_0$, W_2 equals $|r - s|\sqrt{\operatorname{tr}(\Sigma_0)}$.

For $z > 0$,

$$(1 + z)^2 < 1 + 2z + 2z^2 < e^{2z}.$$

The first difference is z^2, while the second inequality follows from the positive remaining terms in the exponential series. Taking square roots and multiplying over the steps gives $r_{0,k} < r_{1,k} < e^{at_k}$ for every $k \geq 1$. Both numerical distributions have too small a scale, but the corrected scale is closer to the exact value. The corrected error is strictly smaller at every grid point, so its maximum and sum are also smaller.

Accumulated error and convergence rates. To find the convergence orders, we calculate the scale error over all steps. Taking logarithms changes the product of step factors into a sum. The positive error terms for each step are

$$f_0(z) = z - \log(1 + z), \qquad f_1(z) = z - \tfrac{1}{2}\log(1 + 2z + 2z^2).$$

Their derivatives are

$$f_0'(z) = \frac{z}{1 + z}, \qquad f_1'(z) = \frac{2z^2}{1 + 2z + 2z^2}.$$

Integrate from zero to obtain $0 < f_0(z) \leq z^2/2$ and $0 < f_1(z) \leq 2z^3/3$. Put $\Delta^{\log}_{i,k} = \sum_{j<k} f_i(ah_j)$. Then $r_{i,k} = e^{at_k} e^{-\Delta^{\log}_{i,k}}$ and

$$W_2(\widehat{p}^i_k, p_{t_k}) = \sqrt{\operatorname{tr}(\Sigma_0)} e^{at_k}(1 - e^{-\Delta^{\log}_{i,k}}).$$

Use $1 - e^{-\Delta^{\log}} \leq \Delta^{\log}$, $\sum_{j<k} h_j^2 \leq t_k H$, and $\sum_{j<k} h_j^3 \leq t_k H^2$ to obtain the explicit unequal-step bounds

$$\max_k W_2(\widehat{p}^0_k, p_{t_k}) \leq \sqrt{\operatorname{tr}\Sigma_0} e^{aT} \frac{a^2 T}{2} H, \qquad \max_k W_2(\widehat{p}^1_k, p_{t_k}) \leq \sqrt{\operatorname{tr}\Sigma_0} e^{aT} \frac{2a^3 T}{3} H^2.$$

Near zero,

$$f_0(z) = \tfrac{1}{2}z^2 - \tfrac{1}{3}z^3 + O(z^4), \qquad f_1(z) = \tfrac{2}{3}z^3 - z^4 + O(z^5).$$

On the uniform grid,

$$\Delta^{\log}_{0,N} = \frac{a^2 T}{2} h + O(h^2), \qquad \Delta^{\log}_{1,N} = \frac{2a^3 T}{3} h^2 + O(h^3).$$

Substitute these expansions into $1 - e^{-\Delta^{\log}} = \Delta^{\log} + O((\Delta^{\log})^2)$ to obtain the terminal expansions. □

Proposition S30 (Particle error for the fixed correction)**.** *Pair the exact and numerical particles for v_1 through the same $X_0 \sim p_0$. On a uniform grid, set*

$$r_N = (1 + 2ah + 2a^2h^2)^{N/2}, \qquad \theta_N = N \arctan \frac{ah}{1 + ah}.$$

Their terminal mean squared difference is

$$\mathbb{E}\|\widehat{X}_N - X_T\|^2 = \operatorname{tr}(\Sigma_0) \left[(r_N - e^{aT})^2 + 4 r_N e^{aT} \sin^2 \frac{\theta_N - aT}{2} \right].$$

Consequently, their root mean squared difference satisfies

$$\left(\mathbb{E}\|\widehat{X}_N - X_T\|^2 \right)^{1/2} = \sqrt{\operatorname{tr}(\Sigma_0)} e^{aT} a^2 T\, h + O(h^2).$$

Proof. Particle error compares the exact and numerical maps from the same starting vector, so it includes their angle difference. In whitened coordinates these maps are $e^{aT} e^{aTJ}$ and $r_N e^{\theta_N J}$. The difference multiplied by its transpose is a scalar matrix with factor $r_N^2 + e^{2aT} - 2r_N e^{aT} \cos(\theta_N - aT)$. Multiplication by C_0 and $C_0^\top$ makes its trace equal to this factor times the initial covariance trace, proving the formula. The numerical mean is exact.

Expanding the numerical angle shows that

$$\theta_N - aT = -a^2 T h + O(h^2).$$

The radial scale difference is $O(h^2)$ by the theorem. The angle difference is first order and gives the leading particle error through the sine term. An optimal distribution pairing matches angles, so the angle difference does not contribute to W_2. □

APPENDIX XVII
A DISTRIBUTION BOUND OVER A FIXED TIME INTERVAL

We bound distribution error by moving Euler's starting point along the acceleration field while holding physical time fixed. If this motion preserves the starting distribution, Euler gives the same output distribution. Its output can then be paired more closely with exact motion from the original starting point. Zero weighted divergence ensures that the starting distribution stays exactly the same. A suitable shift supplies the leading displacement that Euler omits. Preserving selected averages, or preserving the distribution only to first order, would not be enough for this coupling.

Let $F_{s,t}$ be the exact flow of u, and let $p_t = (F_{0,t})_{\#} p_0$. Define

$$\mathbf{a}_t = \partial_t u_t + (D_x u_t) u_t, \qquad \mathbf{j}_t = \partial_t \mathbf{a}_t + (D_x \mathbf{a}_t) u_t.$$

These fields are the second and third time derivatives along an exact particle. A field w has *zero weighted divergence under* a measure p if

$$\int w \cdot \nabla \phi \, \mathrm{d}p = 0 \quad \text{for every } \phi \in C_c^\infty(\mathbb{R}^d). \tag{102}$$

The tests in C_c^∞ are smooth and vanish outside a bounded set. The definition therefore applies even when the measure has no density. If a density exists, the condition is the weak form of $\nabla \cdot (pw) = 0$. The auxiliary flow is autonomous, which means that its field stays fixed as auxiliary time changes.

Proposition S31 (Preservation with physical time fixed)**.** *Let p be a probability measure and let $w \in C^1(\mathbb{R}^d; \mathbb{R}^d)$ be globally Lipschitz. If w has zero weighted divergence under p, its autonomous flow Ψ^τ satisfies $(\Psi^\tau)_{\#} p = p$ for every $\tau \in \mathbb{R}$.*

Proof. Global Lipschitz continuity gives a flow at all positive and negative times, with Ψ^τ a C^1 diffeomorphism and inverse $\Psi^{-\tau}$. For a smooth compactly supported test ϕ, the support of $\phi \circ \Psi^\tau$ lies in the image of the support of ϕ under this continuous inverse. The composition therefore has compact support and is C^1, although it need not be infinitely differentiable. Convolution extends Eq. (102) to C_c^1 tests, so it applies to the composition.

The autonomous flow satisfies $D\Psi^\tau(x) w(x) = w(\Psi^\tau(x))$. For $\phi \in C_c^\infty$,

$$\frac{\mathrm{d}}{\mathrm{d}\tau} \int \phi(\Psi^\tau(x)) p(\mathrm{d}x) = \int w(x) \cdot \nabla(\phi \circ \Psi^\tau)(x) p(\mathrm{d}x) = 0.$$

On each bounded time interval, one compact set contains the supports of these integrands, where both the integrands and their derivatives are bounded. Differentiation under the integral is therefore valid. The derivative is zero, so every test expectation stays constant and the measures are equal. □

Because the auxiliary flow preserves the distribution exactly, we can form a coupling with local error bounded by a constant times h^3. Euler's Lipschitz bound then controls how much later steps increase this error. We restate Theorem 1 with the assumptions needed for these two estimates.

Theorem 1, restated. Let $T > 0$ and $p_0 \in \mathcal{P}_2(\mathbb{R}^d)$. Suppose u is C^2 on a neighborhood of $[0,T] \times \mathbb{R}^d$, globally Lipschitz in space with a common constant L, and $\sup_{t \in [0,T]} \|u(t,0)\| < \infty$. Suppose each $\mathbf{a}_t$ is globally Lipschitz with a common constant L_a and has zero weighted divergence under p_t. Assume

$$C_a = \sup_{0 \le t \le T} \|\mathbf{a}_t\|_{L^2(p_t)} < \infty, \qquad C_j = \sup_{0 \le t \le T} \|\mathbf{j}_t\|_{L^2(p_t)} < \infty.$$

For every finite increasing grid $0 = t_0 < \cdots < t_N = T$, put $h_k = t_{k+1} - t_k$, $H = \max_k h_k$, and $E_{t,h}(x) = x + h u(t,x)$. Euler's method means

$$\widehat{p}_0 = p_0, \qquad \widehat{p}_{k+1} = (E_{t_k, h_k})_{\#} \widehat{p}_k.$$

Then

$$\begin{aligned} \max_{0 \le k \le N} W_2(\widehat{p}_k, p_{t_k}) &\le e^{LT} \left(\frac{C_j}{6} + \frac{L C_a}{2} + \frac{T L_a C_a}{8} \right) \sum_{j=0}^{N-1} h_j^3 \\ &\le e^{LT} \left(\frac{C_j}{6} + \frac{L C_a}{2} + \frac{T L_a C_a}{8} \right) T H^2. \end{aligned} \tag{103}$$

Proof of Theorem 1. The growth and Lipschitz assumptions give an exact flow on the full interval with finite second moments. For $X \sim p_t$, Taylor's formula shows the displacement that Euler omits:

$$F_{t,t+h}(X) = X + hu_t(X) + \frac{h^2}{2}\mathbf{a}_t(X) + \rho_{t,h}.$$

The remainder integrates the third trajectory derivative, which has the stated L^2 bound because the intermediate distribution is p_{t+s}. Minkowski's inequality gives

$$\|\rho_{t,h}\|_{L^2} \le \frac{1}{2}\int_0^h (h-s)^2 \|\mathbf{j}_{t+s}(X_{t+s})\|_{L^2}\,\mathrm{d}s \le \frac{C_j}{6}h^3. \tag{104}$$

Let Ψ_t^τ be the flow of $\mathbf{a}_t$ with physical time t fixed. By Proposition S31, $\Psi_t^\tau X \sim p_t$ for every auxiliary time τ, so the bound C_a applies throughout the motion. Integrating acceleration bounds the displacement, while integrating its change from the initial value bounds the remainder. For $\tau \ge 0$,

$$\|\Psi_t^\tau X - X\|_{L^2} \le C_a\tau, \qquad \|\Psi_t^\tau X - X - \tau\mathbf{a}_t(X)\|_{L^2} \le \int_0^\tau L_a C_a v\,\mathrm{d}v = \frac{L_a C_a}{2}\tau^2.$$

Choose $\tau = h^2/2$. Euler's output from the shifted start still has distribution $(E_{t,h})_\# p_t$. Pairing this output with $F_{t,t+h}(X)$ gives the Wasserstein bound

$$\begin{aligned}
\Psi_t^{h^2/2}X &= X + \tfrac{1}{2}h^2\mathbf{a}_t(X) + O_{L^2}(h^4),\\
W_2((E_{t,h})_\# p_t, p_{t+h}) &\le \|E_{t,h}(\Psi_t^{h^2/2}X) - F_{t,t+h}(X)\|_{L^2}\\
&\le \frac{L_a C_a}{8}h^4 + \frac{LC_a}{2}h^3 + \frac{C_j}{6}h^3.
\end{aligned} \tag{105}$$

The shift supplies $\tau\mathbf{a}_t(X)$, which is the second-order displacement in the exact expansion. It also changes the point at which Euler evaluates velocity, producing the difference $h[u_t(\Psi_t^\tau X) - u_t(X)]$. Since the shift is $O(h^2)$ in root mean square, the spatial Lipschitz bound makes this additional displacement $O(h^3)$.

This local estimate starts from the exact distribution. In a sequence of Euler steps, we must also bound error from earlier steps. For a Lipschitz map E and measures $\mu, \nu \in \mathcal{P}_2$, apply E to both members of a coupling and take the infimum over couplings. This gives $W_2(E_\#\mu, E_\#\nu) \le \mathrm{Lip}(E) W_2(\mu,\nu)$. For Euler, $\mathrm{Lip}(E_{t,h}) \le 1 + Lh$. With $e_k = W_2(\widehat{p}_k, p_{t_k})$ and $C_* = C_j/6 + LC_a/2 + TL_aC_a/8$, the triangle inequality separates the earlier error from the new local error. Since $h_k \le T$,

$$\begin{aligned}
e_{k+1} &\le W_2((E_{t_k,h_k})_\#\widehat{p}_k, (E_{t_k,h_k})_\# p_{t_k})\\
&\quad + W_2((E_{t_k,h_k})_\# p_{t_k}, p_{t_{k+1}})\\
&\le (1 + Lh_k)e_k + C_* h_k^3.
\end{aligned}$$

Starting from $e_0 = 0$, we apply this inequality at each step. Later steps increase each local error by at most $\prod_j (1+Lh_j) \le e^{LT}$. The sum satisfies $\sum_j h_j^3 \le H^2 \sum_j h_j = TH^2$, which gives the claimed order. □

If the initial numerical distribution differs from p_0, the same proof adds $e^{LT} W_2(\widehat{p}_0, p_0)$ to the right side.

a) The modified equation.: The modified equation produces the same cancellation by changing the velocity field. For a fixed step parameter h, let Y_s solve $\dot{Y}_s = u(t+s, Y_s) - (h/2)\mathbf{a}(t+s, Y_s)$ from $Y_0 = x$ over $0 \le s \le h$. Physical time evolves while h remains fixed. On compact sets, $u \in C^2$ and $\mathbf{a} \in C^1$ give $Y_s = x + su(t,x) + O(hs + s^2)$, and substitution yields

$$u(t+s, Y_s) = u(t,x) + s\mathbf{a}(t,x) + O(hs+s^2), \qquad \mathbf{a}(t+s, Y_s) = \mathbf{a}(t,x) + O(s).$$

After integration, the terms $h^2\mathbf{a}(t,x)/2$ cancel, giving $Y_h = x + hu(t,x) + O(h^3)$. For sufficiently small h, the remainders are uniform on compact sets. This is the first field correction in backward error analysis [22, Chapter IX]. Under the theorem's global assumptions, $u - (h/2)\mathbf{a}$ has a unique global flow. The acceleration condition leaves the weak continuity equation unchanged, so uniqueness gives the same distribution as u at every time. The expansion on compact sets alone does not bound mean squared error for an unbounded source. The coupling proof gives the integrable estimates needed for the finite-time bound.

APPENDIX XVIII
A NECESSARY LOCAL CONDITION AND TERMINAL CANCELLATION

If one Euler step from the exact distribution has error $o(h^2)$, the field must satisfy the acceleration condition at that time. Terminal accuracy on a family of grids does not imply this local condition, because errors from different times can cancel. We prove the local result and then give an example of terminal cancellation.

Proposition S23, restated. Let u and p_t satisfy the smoothness, growth, Lipschitz, and moment assumptions of Theorem 1, without requiring zero weighted divergence of acceleration under the current distribution. At a fixed $t < T$, suppose

$$W_2((E_{t,h})_\# p_t, p_{t+h}) = o(h^2) \quad \text{as } h \downarrow 0.$$

Then $\mathbf{a}_t$ has zero weighted divergence under p_t.

Proof of Proposition S23. For $X \sim p_t$, compare one Euler step $Y = E_{t,h}X$ with exact motion $Y + \Delta X$, where $\Delta X = F_{t,t+h}X - Y = h^2\mathbf{a}_t(X)/2 + \rho_{t,h}$. Equation (104) gives $\|\rho_{t,h}\|_{L^2} = O(h^3)$ and $\|\Delta X\|_{L^2} = O(h^2)$. Since a smooth compactly supported test ϕ has bounded first and second derivatives, Taylor expansion gives

$$\mathbb{E}[\phi(Y) - \phi(Y + \Delta X)] = -\mathbb{E}[\nabla\phi(Y) \cdot \Delta X] + O(\mathbb{E}\|\Delta X\|^2).$$

Replacing ΔX by $h^2\mathbf{a}_t(X)/2$ changes the expectation by $O(h^3)$. Replacing $\nabla\phi(Y)$ by $\nabla\phi(X)$ changes it by at most $Ch^3\|u_t\|_{L^2(p_t)}\|\mathbf{a}_t\|_{L^2(p_t)}$. Hence

$$\mathbb{E}\phi(E_{t,h}X) - \mathbb{E}\phi(F_{t,t+h}X) = -\frac{h^2}{2}\int \nabla\phi \cdot \mathbf{a}_t \, \mathrm{d}p_t + o(h^2). \tag{106}$$

The expectation difference has absolute value at most $\mathrm{Lip}(\phi)W_2((E_{t,h})_{\#}p_t, p_{t+h})$, which tends to zero after division by h^2. The acceleration integral is therefore zero for every smooth compactly supported test, which proves zero weighted divergence. The moment requirements are the finite second moment and the stated L^2 bounds on acceleration and jerk, with C_j controlling the trajectory remainder through Eq. (104). □

Conversely, under all assumptions and common bounds in Appendix XVII, zero weighted divergence of acceleration gives local $O(h^3)$ error. Both directions concern one step started from the exact distribution.

For a terminal counterexample, let $T = 1$, $p_0 = \mathcal{N}(0, I_d)$, and

$$u(t, x) = \sin(2\pi t)e_1.$$

Here e_1 is the first coordinate vector. The particles translate in this direction during the first half of the interval and return during the second half, giving zero exact terminal displacement. Euler also gives zero displacement on every uniform grid with $N \geq 1$ steps, since

$$\frac{1}{N}\sum_{k=0}^{N-1} \sin(2\pi k/N)e_1 = 0.$$

For $N > 1$, summing the Nth roots of unity and taking imaginary parts gives this identity, while $N = 1$ follows directly. However, $\mathbf{a}_t = 2\pi\cos(2\pi t)e_1$ has nonzero weighted divergence whenever it is nonzero. The current distribution is a translated Gaussian, whose density has a nonzero derivative along any nonzero constant direction. Thus terminal exactness does not imply the local condition.

A. Weak expansion with an explicit remainder

Weak error compares expectations of test functions. We calculate its leading term and give an explicit cubic remainder for C_b^3 tests. These tests have continuous bounded values and derivatives through order three. This larger test class requires a stronger moment assumption than the local converse. The expansion alone does not bound W_2. The coupling in Appendix XVII gives that bound under the assumptions stated there.

Proposition S32 (Local weak error of Euler)**.** *Fix t and $h_0 > 0$. Suppose u is C^2 on a neighborhood of $[t, t + h_0] \times \mathbb{R}^d$, and one constant C satisfies*

$$\|u(s, x)\| + \|\partial_s u(s, x)\| + \|\partial_s^2 u(s, x)\| \leq C(1 + \|x\|),$$
$$\|D_x u(s, x)\|_{\mathrm{op}} + \|D_x\partial_s u(s, x)\|_{\mathrm{op}} + \|D_x^2 u(s, x)\|_{\mathrm{op}} \leq C.$$

Let $X \sim p_t$ have finite third moment, let p_{t+h} be its exact evolved law, and let $\phi \in C_b^3(\mathbb{R}^d)$. Then

$$\mathbb{E}\phi(X + hu_t(X)) - \mathbb{E}_{p_{t+h}}\phi = -\frac{h^2}{2}\mathbb{E}_{p_t}[\nabla\phi \cdot \mathbf{a}_t] + R_h, \qquad |R_h| \leq C_{\mathrm{weak}}h^3\mathbb{E}(1 + \|X\|^3).$$

The constant C_{weak} depends on C, h_0, and the supremum norms of $\nabla\phi$, $D^2\phi$, and $D^3\phi$, but not on h.

Proof. The numerical and exact test expansions agree at first order. Gronwall's inequality bounds the exact position by a constant times one plus the source radius, while acceleration has at most linear growth. Its derivative along the exact flow is

$$\mathbf{j} = \partial_t^2 u + 2(D_x\partial_t u)u + D_x^2 u[u, u] + (D_x u)\partial_t u + (D_x u)^2 u.$$

This derivative has at most quadratic growth. Taylor's formula along the straight Euler step is

$$\phi(x + hu_t(x)) = \phi(x) + h\nabla\phi \cdot u_t + \frac{h^2}{2}D^2\phi[u_t, u_t] + O(h^3(1 + \|x\|^3)).$$

Along the exact flow the second derivative of the test is $D^2\phi[u,u]+\nabla\phi\cdot\mathbf{a}$, and its third derivative is $D^3\phi[u,u,u]+3D^2\phi[u,\mathbf{a}]+\nabla\phi\cdot\mathbf{j}$. The derivative and flow bounds make this third derivative at most $C_{\mathrm{weak}}(1+\|x\|^3)$ in absolute value. Subtracting the expansions cancels the terms that depend only on velocity and leaves the stated acceleration term. The finite third moment makes the remainder integrable. □

For smooth tests with compact support, integration by parts expresses the leading difference between the two distributions as

$$(\mathrm{Id}+hu_t)_{\#}p_t - p_{t+h} = \frac{h^2}{2}\nabla\cdot(p_t\mathbf{a}_t) + O_{\mathrm{weak}}(h^3).$$

The equality holds after integration against smooth compactly supported tests, up to the stated weak remainder. Integration by parts changes the sign of the acceleration integral and gives the positive divergence coefficient. When acceleration has zero weighted divergence, cutoff approximation extends this cancellation to the bounded tests in the proposition because acceleration is integrable.

For quadratic tests, the Euler expansion stops at order two, while the third derivative along the exact flow grows at most cubically. The third-moment assumption therefore makes the same formula valid. For affine fields $u(s,x)=B(s)x+d_{\mathrm{aff}}(s)$ with bounded coefficient derivatives through order two, this growth is only quadratic, so a finite second moment suffices for quadratic tests, including mean and covariance entries.

For the fixed Gaussian correction, write $y=x-m_t$ and $B=a(I+\Omega)$. The time derivative $-B\xi$ cancels the constant part $B\xi$ in $(D_xu)u$. Thus

$$\mathbf{a}_{v_1} = B^2y = 2a^2\Omega y.$$

The matrix identities imply $\operatorname{tr}\Omega=0$ and skew-symmetry of $\Sigma_t^{-1}\Omega$. Hence the Gaussian density satisfies

$$\nabla\cdot(p_t\Omega y) = p_t[\operatorname{tr}\Omega - y^\top\Sigma_t^{-1}\Omega y] = 0.$$

Acceleration is nonzero away from the mean, but its weighted divergence is zero. The leading weak error therefore cancels. The Gaussian covariance recursion gives the convergence rate directly, while the coupling theorem bounds distribution error for more general distributions. Proposition S23 shows that the local condition is necessary under the stated second-moment, acceleration, and jerk bounds.

APPENDIX XIX
A NONLINEAR RADIAL PATH AND ITS EXACT ERROR COEFFICIENTS

For a nonlinear radial path, covariance is not enough to determine the distribution. We therefore calculate distribution error from the full distribution of radii. Radial symmetry gives an optimal coupling that matches angles and pairs radii in increasing order. We show that the numerical radius maps keep this order and calculate their error over all steps. Bounds valid at every source radius then let us take expectations of the error limits. To calculate particle error, we also include the angle difference between particles from the same starting point.

For Theorem S24, assume $a,\eta,T>0$. Let p_0 be rotationally invariant on $\mathbb{R}^2$, with $m_{2,0}=\int\|x\|^2p_0(\,\mathrm{d}x)\in(0,\infty)$. A density is not required. Define

$$J=\begin{pmatrix}0&-1\\1&0\end{pmatrix}, \qquad \alpha(r)=a+\eta\frac{r^2}{1+r^2}, \qquad \mathcal{D}=r\frac{\mathrm{d}}{\mathrm{d}r},$$
$$q_\alpha=\mathcal{D}\alpha, \qquad P=\alpha^2+\alpha q_\alpha, \qquad \beta=\sqrt{P},$$

using the positive square root, and put

$$u_0(x)=\alpha(\|x\|)x, \qquad u_1(x)=\alpha(\|x\|)x+\beta(\|x\|)Jx. \tag{107}$$

Let $R(t,r)$ solve $\partial_tR=f(R)$, with $f(r)=\alpha(r)r$ and $R(0,r)=r$. Write $R_0=\|X_0\|$ for $X_0\sim p_0$.

Proof of Theorem S24. Both fields generate the same probability path. The correction increases action, but it also makes acceleration preserve the current distribution. We first compare the numerical and exact radii to show that the correction improves distribution accuracy on every grid. We then calculate the radius and angle errors over all steps. Their positive leading coefficients give the stated convergence orders, with mean square bounds that justify taking the limits.

A. Regularity, path preservation, and action

a) Velocity bounds.: For positive radius, $\mathcal{D}$ differentiates with respect to log radius. This gives the radius-weighted derivatives needed to differentiate the fields in space. With $a_+ = a + \eta$ and $s = r^2$, direct differentiation gives

$$q_\alpha = \frac{2\eta s}{(1+s)^2}, \quad \mathcal{D}q_\alpha = \frac{4\eta s(1-s)}{(1+s)^3}, \quad \mathcal{D}^2 q_\alpha = \frac{8\eta s(1-4s+s^2)}{(1+s)^4}.$$

Thus

$$a \le \alpha \le a_+, \quad 0 \le q_\alpha \le \eta/2, \quad |\mathcal{D}q_\alpha| \le \eta, \quad |\mathcal{D}^2 q_\alpha| \le 2\eta, \quad \mathcal{D}q_\alpha + 2q_\alpha = \frac{8\eta s}{(1+s)^3} \ge 0.$$

We have $P \ge a^2$ and $\mathcal{D}P = \alpha(2q_\alpha + \mathcal{D}q_\alpha) + q_\alpha^2 \ge 0$, with strict positivity for $r > 0$. Thus β is nondecreasing and strictly increases at positive radii. The coefficients are smooth functions of r^2, so the Cartesian fields are smooth at the origin as well.

We next bound β and its derivatives using P and its first two $\mathcal{D}$ derivatives. These estimates will give spatial Lipschitz bounds valid at every radius. Define

$$P_* = a_+^2 + a_+\eta/2, \qquad P_1 = 2a_+\eta + \eta^2/4, \qquad P_2 = 4a_+\eta + 2\eta^2,$$
$$b_* = \sqrt{P_*}, \qquad b_1 = P_1/(2a), \qquad b_2 = P_2/(2a) + P_1^2/(4a^3).$$

The identity $\mathcal{D}^2 P = 2q_\alpha^2 + 2\alpha\mathcal{D}q_\alpha + 3q_\alpha\mathcal{D}q_\alpha + \alpha\mathcal{D}^2 q_\alpha$ implies $|\mathcal{D}P| \le P_1$ and $|\mathcal{D}^2 P| \le P_2$. Hence

$$\beta \le b_*, \qquad |\mathcal{D}\beta| \le b_1, \qquad |\mathcal{D}^2\beta| \le b_2.$$

The spatial derivative of either $k(r)x$ or $k(r)Jx$ has operator norm at most $|k| + |\mathcal{D}k|$. Thus $a_+ + \eta/2$ and $a_+ + \eta/2 + b_* + b_1$ are global Lipschitz constants for u_0 and u_1, respectively, and both fields have linear growth and global flows.

b) The common path and its changing shape.: Both fields have radial velocity $f(r)$. The correction adds the angle change $\Theta(t,r) = \int_0^t \beta(R(v,r))\,\mathrm{d}v$, which depends on the starting radius. At each positive source radius, rotation preserves the uniform distribution of angles. Both flows also keep the origin fixed, including any mass there. The two fields therefore generate the same p_t at every time. Their common radius flow satisfies

$$e^{at} r \le R(t,r) \le e^{a_+ t} r, \qquad \partial_r R(t,r) = \exp\left(\int_0^t f'(R(v,r))\,\mathrm{d}v\right) > 0. \tag{108}$$

The distributions have finite second moments.

To show that the distribution changes shape, the source support must contain at least two distinct positive radii. For $r_2 > r_1 > 0$, scalar-flow ordering and strict increase of α imply

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \frac{R(t,r_2)/r_2}{R(t,r_1)/r_1} = \alpha(R(t,r_2)) - \alpha(R(t,r_1)) > 0.$$

Distinct positive source radii therefore expand by different ratios for $t > 0$. Increasing transport between radius distributions is unique up to zero source probability, so no single scale factor can give the same distribution. The path thus changes shape when the source support contains at least two distinct positive radii. A source concentrated on one positive radius does not give this conclusion.

c) Acceleration preservation and moment bounds.: The angular correction preserves the path. Choosing $\beta^2 = P$ also cancels the radial component of vector acceleration, even though the scalar radius can have a nonzero second derivative. For example, $u(x) = (I+J)x$ gives $X_t = e^t(\cos t, \sin t)$ from $(1,0)$, with $R'' = R$ but vector acceleration $2JX_t$. In polar coordinates its radial component is $R'' - R(\dot\theta)^2 = 0$. For the nonlinear field,

$$\mathbf{a}_{u_1}(x) = (\alpha^2 + \alpha q_\alpha - \beta^2)x + (2\alpha\beta + \alpha\mathcal{D}\beta)Jx = \beta_a(r)Jx, \quad \beta_a = \alpha(2\beta + \mathcal{D}\beta) \ge 2a^2. \tag{109}$$

With physical time fixed, the acceleration flow keeps radius r fixed and changes the angle by $v\beta_a(r)$. This rotation preserves every rotationally invariant measure, including measures without a density. The expectation of each smooth test stays constant, so differentiation proves zero weighted divergence.

To verify the general theorem's remaining assumptions, put

$$C_{\beta_a,0} = a_+(2b_* + b_1), \qquad C_{\beta_a,1} = \eta b_* + (2a_+ + \eta/2)b_1 + a_+ b_2.$$

Then $|\beta_a| \le C_{\beta_a,0}$, $|\mathcal{D}\beta_a| \le C_{\beta_a,1}$, and $L_a = C_{\beta_a,0} + C_{\beta_a,1}$ is a global Lipschitz bound for acceleration. The derivative of acceleration along the exact flow is

$$\mathbf{j}_{u_1}(x) = -\beta_a\beta x + \alpha(\beta_a + \mathcal{D}\beta_a)Jx.$$

The moment bounds required by the theorem are

$$C_a \le C_{\beta_a,0} e^{a_+ T}\sqrt{m_{2,0}}, \qquad C_j \le [b_* C_{\beta_a,0} + a_+(C_{\beta_a,0} + C_{\beta_a,1})]e^{a_+ T}\sqrt{m_{2,0}}.$$

These require only the source second moment.

d) Action comparison.: The base field has potential

$$\Phi(x) = \frac{a+\eta}{2}\|x\|^2 - \frac{\eta}{2}\log(1+\|x\|^2), \qquad u_0 = \nabla\Phi.$$

The potential has at most quadratic growth and its gradient has at most linear growth, so the cutoff argument in Appendix XXIII applies. By Eq. (108), all later radii are at most a fixed multiple of the source radius. The finite source second moment then gives one integrable bound on the squared cutoff error. We can therefore take the limit after expectation and time integration. For every finite-action field v satisfying the same weak continuity equation,

$$\mathcal{A}_T(v) = \mathcal{A}_T(u_0) + \int_0^T \|v_t - u_0\|^2_{L^2(p_t)}\,\mathrm{d}t.$$

The integral is nonnegative, so the base field minimizes action. Equality holds only when the fields agree almost everywhere. If the density is positive everywhere and the competing fields are continuous, they agree at every point. For the corrected field, the radial and rotational velocities are perpendicular at every point. Thus

$$\mathcal{A}_T(u_1) - \mathcal{A}_T(u_0) = \int_0^T \mathbb{E}[P(R(t,R_0))R(t,R_0)^2]\,\mathrm{d}t > 0. \tag{110}$$

Both actions are finite. In particular, $\mathcal{A}_T(u_1) \le T(a_+^2 + P_*)e^{2a_+T}m_{2,0}$.

B. Exact radial transport and improvement on every grid

The general theorem gives a second-order upper bound. To compare the errors on every grid, we use the order of the exact and numerical radii. One Euler step has radius maps

$$G_{0,h}(r) = r(1+h\alpha(r)), \qquad G_{1,h}(r) = r\sqrt{(1+h\alpha(r))^2 + h^2P(r)}.$$

Both maps preserve radius order. The base derivative is $1 + h(\alpha + q_\alpha) > 0$, while the corrected square-root factor is positive and nondecreasing, so its product with r has positive derivative.

Define

$$\mathcal{K} = \alpha^3 + 3\alpha^2 q_\alpha + \alpha q_\alpha^2 + \alpha^2 \mathcal{D}q_\alpha.$$

The exact derivatives are $R'' = PR$ and $R''' = \mathcal{K}R$. Since $\mathcal{D}q_\alpha \ge -2q_\alpha$, $\mathcal{K} \ge \alpha^3 + \alpha^2 q_\alpha + \alpha q_\alpha^2 \ge a^3 > 0$. Taylor's integral formula implies $R(h,r) > r + hf(r) + h^2P(r)r/2$ for $r, h > 0$. Also,

$$\left(1 + h\alpha + \frac{h^2P}{2}\right)^2 - [(1+h\alpha)^2 + h^2P] = h^3\alpha P + \frac{h^4P^2}{4} > 0.$$

The corrected map lies below that quadratic polynomial. Therefore

$$G_{0,h}(r) < G_{1,h}(r) < R(h,r) \quad (r,h>0). \tag{111}$$

Let $R_{i,k}$ compose the numerical radius maps through node k. Induction using monotonicity and the exact flow property proves

$$R_{0,k}(r) < R_{1,k}(r) < R(t_k, r) \quad (r > 0,\ k \ge 1).$$

All three maps send zero to zero.

The norm map is 1-Lipschitz, so spatial W_2 is at least the one-dimensional distance between the radius distributions. To achieve this lower bound, pair radii at the same quantile and give both particles the same independent uniform angle. All radius maps increase with the source radius. Using the same source radius therefore gives this quantile pairing, even when a radius has positive probability. Numerical angle changes depend only on radius and preserve radial symmetry. Thus

$$W_2(\widehat{p}_k^i, p_{t_k})^2 = \mathbb{E}[(R_{i,k}(R_0) - R(t_k,R_0))^2], \qquad i = 0, 1. \tag{112}$$

Positive radii have positive probability since $m_{2,0} > 0$. The strict pointwise inequalities prove

$$0 < W_2(\widehat{p}_k^1, p_{t_k}) < W_2(\widehat{p}_k^0, p_{t_k}) \quad (k \ge 1)$$

on every finite increasing grid, without a small-step assumption.

C. The three positive leading coefficients

With a, η, T, p_0 fixed, we refine uniform steps $h = T/N$ and calculate the radius error over all steps. We use the following quantities for the local radius errors and their effect at the final time:

$$\begin{aligned} V(T,s;r) &= \exp\left(\int_s^T f'(R(v,r))\,\mathrm{d}v\right), \\ d_0(r) &= \tfrac{1}{2}P(r)r, \\ d_1(r) &= r\left(\tfrac{1}{2}\alpha(r)P(r) + \tfrac{1}{6}\mathcal{K}(r)\right), \\ \mathcal{C}_i(T,r) &= \int_0^T V(T,s;r)d_i(R(s,r))\,\mathrm{d}s \quad (i=0,1), \\ \Theta_1(T,r) &= \tfrac{1}{2}\int_0^T \beta_a(R(s,r))\,\mathrm{d}s. \end{aligned}$$

The coefficient d_i gives the local radius error, and V describes how much the remaining exact flow increases it. Integrating their product gives $\mathcal{C}_i$, the leading radius-error coefficient at the final time. The angle-error coefficient is Θ_1. The limits below are positive, so the errors have the orders stated in Theorem S24:

$$\lim_{N\to\infty}\frac{W_2(\widehat{p}_N^0, p_T)}{h} = \|\mathcal{C}_0(T,R_0)\|_{L^2} > 0, \tag{113}$$

$$\lim_{N\to\infty}\frac{W_2(\widehat{p}_N^1, p_T)}{h^2} = \|\mathcal{C}_1(T,R_0)\|_{L^2} > 0, \tag{114}$$

$$\lim_{N\to\infty}\frac{\|\widehat{X}_N^1 - X_T^1\|_{L^2}}{h} = \|R(T,R_0)\Theta_1(T,R_0)\|_{L^2} > 0. \tag{115}$$

The last line pairs particles with the same start under the same corrected field.

a) Radius limits.: For a fixed source radius, Taylor expansion gives

$$R(h,r) - G_{0,h}(r) = h^2 d_0(r) + O(h^3), \qquad R(h,r) - G_{1,h}(r) = h^3 d_1(r) + O(h^4).$$

The corrected cubic term is $-\alpha P r h^3/2$. The exact cubic term is $\mathcal{K} r h^3/6$. Their difference is d_1 times the cubed step length.

For fixed $r > 0$, the exact and numerical radii remain in a compact interval over $[0,T]$. With $\Delta_{i,k} = R(t_k,r) - R_{i,k}(r)$, the local bounds and discrete Gronwall inequality give $\Delta_{0,k} = O(h)$ and $\Delta_{1,k} = O(h^2)$ uniformly in k. Scaling by $z_k = \Delta_{i,k}/h^{i+1}$ and expanding the one-step flow in its starting radius gives

$$z_{k+1} = z_k + h[f'(R(t_k,r))z_k + d_i(R(t_k,r))] + O(h^2).$$

For this fixed radius, the remainder is bounded uniformly in k. The recurrence is Euler's method for $z' = f'(R(t,r))z + d_i(R(t,r))$, $z(0) = 0$, with a remainder of order h^2 at each step. There are T/h steps, and the factors that increase earlier errors are bounded. Discrete Gronwall therefore shows that the total remainder tends to zero. The scaled radius error converges to the solution of this linear equation. Variation of constants gives its terminal value $\mathcal{C}_i(T,r)$.

To pass the limit through expectation, we need a bound proportional to r. Its square is integrable because the source has a finite second moment. For this rational family, every fixed-order derivative $\mathcal{D}^m\alpha$ is bounded, and $P \geq a^2$ gives the same property for $\mathcal{D}^m\beta$. With $H_1 = \alpha$ and $H_{m+1} = \alpha(H_m + \mathcal{D}H_m)$, the exact time derivatives satisfy

$$\partial_t^m R(t,r) = R(t,r)H_m(R(t,r)), \qquad |\partial_t^m R(t,r)| \leq e^{a_+ h_0}\|H_m\|_\infty r \quad (0 \leq t \leq h_0,\ m \leq 4).$$

The remainder in the exact cubic expansion is therefore bounded by $Ch^4 r$ uniformly in r. For the numerical map, set $Q_* = a_+^2 + P_*$ and $M_* = 2a_+ + 2Q_* h_0$. Since $D_h = 1 + 2h\alpha + h^2(\alpha^2 + P) \geq 1$ for $h \geq 0$,

$$\left|\partial_h^4\sqrt{D_h}\right| \leq \frac{15}{16}M_*^4 + \frac{9}{2}Q_* M_*^2 + 3Q_*^2.$$

The numerical cubic remainder is therefore also bounded by $Ch^4 r$, giving local radius-error bounds $Ch^2 r$ and $Ch^3 r$. The bounds on $\mathcal{D}\alpha$ and $\mathcal{D}P$ give $|\partial_r G_{i,h}| \leq 1 + Ch$ for $0 \leq h \leq h_0$. A local error at time t_k is at most $Ch^{i+2}e^{a_+ t_k}r$, and later steps increase it by at most a factor of e^{CT}. Summing over $N = T/h$ steps gives

$$0 \leq \Delta_{0,k}(r) \leq C_T h r, \qquad 0 \leq \Delta_{1,k}(r) \leq C_T h^2 r. \tag{116}$$

Both scaled squared errors are bounded by $C_T^2 r^2$, which is integrable under the source second moment. Dominated convergence then lets us pass the limit through the expectation in the transport formula. This proves both distribution limits.

b) Particle limit.: The optimal distribution coupling matches angles. Particles from the same starting point have an angle difference as well. One numerical step changes the angle by

$$\arctan \frac{h\beta(r)}{1+h\alpha(r)} = h\beta(r) - h^2\alpha(r)\beta(r) + O(h^3).$$

The exact increment from the same radius is

$$\int_0^h \beta(R(s,r))\,\mathrm{d}s = h\beta(r) + \frac{h^2}{2}\alpha(r)\mathcal{D}\beta(r) + O(h^3).$$

The exact increment minus the numerical increment equals $h^2\beta_a(r)/2 + O(h^3)$. The angle remainders are uniform in r. For the exact angle, its third derivative has absolute value at most $a_+(\eta/2)b_1 + a_+^2 b_2$. For the numerical angle, write the increment as $\operatorname{Im}\log(1+h[\alpha+\mathrm{i}\beta])$. Its third h derivative has absolute value at most $2(a_+^2+b_*^2)^{3/2}$, because $|1+h(\alpha+\mathrm{i}\beta)| \geq 1$. Taylor's formula therefore bounds the difference remainder by

$$\left[\frac{a_+(\eta/2)b_1 + a_+^2 b_2}{6} + \frac{(a_+^2+b_*^2)^{3/2}}{3}\right] h^3$$

for every radius.

At each step, the method evaluates angular velocity at the numerical radius. By Eq. (116), this radius has error $O(h^2 r)$. Both exact and numerical radii remain at least as large as their positive source radius. The bound $|\beta(r_1) - \beta(r_2)| \leq b_1 |\log(r_1/r_2)|$ therefore shows that using the exact radius instead changes one angle increment by $O(h^3)$. This estimate is uniform over positive source radii. We sum the local differences and use the bounded derivative of $\beta_a(R(t,r))$ to replace the sum by an integral. This gives

$$\Theta(T,r) - \Theta_N(r) = h\Theta_1(T,r) + O(h^2)$$

uniformly for $r > 0$.

The squared distance between the particles at the final time is

$$(R_{1,N} - R_T)^2 + 4R_{1,N}R_T \sin^2 \frac{\Theta_N - \Theta_T}{2}, \qquad R_T = R(T,r).$$

After division by h^2, the radial term tends to zero and the angular term tends to $R_T^2\Theta_1(T,r)^2$. The radius and angle bounds show that the scaled squared distance is at most Cr^2. The distance is zero at $r = 0$, so dominated convergence proves the particle limit.

Since $f' = \alpha + q_\alpha \geq a$, $d_0(r) \geq a^2 r/2$, $d_1(r) \geq 2a^3 r/3$, and $\beta_a \geq 2a^2$,

$$\mathcal{C}_0(T,r) \geq \frac{a^2 T}{2} e^{aT} r, \quad \mathcal{C}_1(T,r) \geq \frac{2a^3 T}{3} e^{aT} r, \quad \Theta_1(T,r) \geq a^2 T.$$

The upper bounds make all three norms finite. Since $m_{2,0} > 0$, the norms are strictly positive, so the errors have the orders stated in Theorem S24. Setting $\eta = 0$ gives the isotropic Gaussian coefficients. The transport statements still hold for every radial source with finite second moment. □

D. Sensitivity to a fixed coefficient error

For fixed $\lambda \geq 0$, let $u_\lambda(x) = \alpha(r)x + \lambda\beta(r)Jx$. Each such field generates the same continuous path, and

$$\mathbf{a}_{u_\lambda} = (1-\lambda^2)P(r)x + \lambda\beta_a(r)Jx.$$

The decomposition required for the residual bound is $w_\lambda = \lambda\beta_a(r)Jx$ and $g_\lambda = (1-\lambda^2)P(r)x$. The tangential part is globally Lipschitz and preserves the current distribution, so it has zero weighted divergence. The radial residual obeys

$$\sup_t \|g_\lambda\|_{L^2(p_t)} \leq |1-\lambda^2| P_* e^{a_+ T}\sqrt{m_{2,0}}.$$

The same bounded radial derivatives give the remaining regularity and moment bounds, uniformly for λ in a fixed bounded interval. The residual estimate therefore gives $O(H^2 + |1-\lambda^2|H)$.

For $\lambda \neq 1$, the error itself has a positive first-order coefficient. To see this, expand the radius error:

$$R(h,r) - r\sqrt{(1+h\alpha(r))^2 + h^2\lambda^2 P(r)} = (1-\lambda^2)d_0(r)h^2 + O(h^3).$$

Since α and P are nondecreasing, the numerical radius map is increasing, so the same optimal transport pairing applies. The error recurrence has local coefficient $(1-\lambda^2)d_0$. The signed scaled error therefore tends to $(1-\lambda^2)\mathcal{C}_0(T,r)$. Its absolute value is bounded by $C_T r$, so the radial transport formula and dominated convergence give

$$\lim_{N\to\infty} \frac{W_2(\widehat{p}_N^\lambda, p_T)}{h} = |1-\lambda^2| \|\mathcal{C}_0(T,R_0)\|_{L^2} > 0. \tag{117}$$

Every fixed mismatch in the nonnegative coefficient therefore gives first-order distribution error. This limiting rate can be difficult to see over a finite range of small errors. If negative coefficients are allowed, both $\lambda = \pm 1$ cancel the leading term.

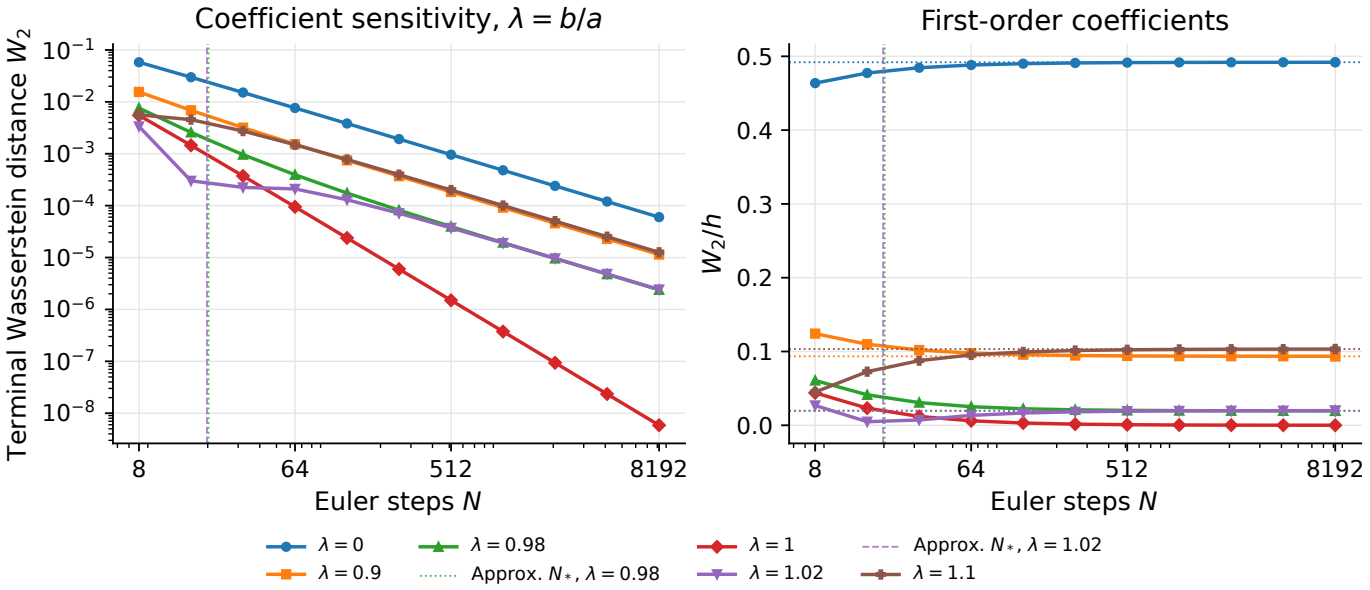


Fig. 12. A fixed rotation mismatch gives first-order Euler distribution error for these Gaussian cases, with the parameters of Figure 7. Each curve keeps $\lambda = b/a$ fixed as the grid is refined. The right panel shows error divided by h. For $\lambda \neq 1$, this ratio tends to the positive value $|1-\lambda^2|c_0^{\mathrm{G}}$ shown by a horizontal dotted line. Section XV defines c_0^{G}. With matched rotation $\lambda = 1$, the limit is zero. Vertical lines near $N = 20$ estimate where the first- and second-order terms have similar size for $\lambda = 0.98$ (dotted) and 1.02 (dashed). Markers distinguish the error curves from their limits.

E. Affine changes of coordinates, product distributions, and a dimension restriction

A fixed affine change of coordinates can give distributions that are not radial in the original coordinates. For $\mathcal{T}(x) = d_{\mathrm{aff}} + Gx$ with G invertible, set $\widetilde{u}(t,y) = Gu(t, G^{-1}(y - d_{\mathrm{aff}}))$ and $\widetilde{p}_t = \mathcal{T}_{\#}p_t$. Substitution into the differential equation and Euler update gives

$$\widetilde{F}_{s,t} \circ \mathcal{T} = \mathcal{T} \circ F_{s,t}, \qquad \widetilde{E}_{t,h} \circ \mathcal{T} = \mathcal{T} \circ E_{t,h}.$$

The transformed acceleration is $\widetilde{\mathbf{a}}_t(\mathcal{T}(x)) = G\mathbf{a}_t(x)$. For every compact smooth test ϕ,

$$\int \nabla\phi(y) \cdot \widetilde{\mathbf{a}}_t(y)\, \widetilde{p}_t(\,\mathrm{d}y) = \int \nabla(\phi \circ \mathcal{T})(x) \cdot \mathbf{a}_t(x)\, p_t(\,\mathrm{d}x) = 0.$$

Because the affine map is invertible, $\phi \circ \mathcal{T}$ has compact support. The map also preserves the required smoothness, Lipschitz, growth, and moment assumptions, with different constants.

Write $s_{\min}(G) > 0$ for the smallest singular value and $\|G\|_{\mathrm{op}}$ for the operator norm. Push forward any coupling to obtain the upper bound below, then apply that bound to $\mathcal{T}^{-1}$ to obtain the lower bound:

$$s_{\min}(G) W_2(\mu,\nu) \le W_2(\mathcal{T}_{\#}\mu, \mathcal{T}_{\#}\nu) \le \|G\|_{\mathrm{op}} W_2(\mu,\nu).$$

Differences between particles from the same starting point satisfy the same norm bounds. These factors are fixed, so the positive limits in Theorem S24 give orders $\Theta(h)$ and $\Theta(h^2)$ after the change of coordinates as well. The corrected distribution error is therefore smaller on sufficiently fine uniform grids. The estimates do not give exact coefficients or improvement on every grid. For example, $G = \mathrm{diag}(2,1)$ gives an elliptical mixture whose shape changes with time. This example is still an affine image of the symmetric one. Since $G(Du_0)G^{-1}$ need not be symmetric, the transformed base field need not minimize kinetic action.

We can also add independent stationary coordinates. This gives the product path $p_t \otimes \nu_{\mathrm{stat}}$ with field $(u_i, 0)$, where ν_{stat} has finite second moment. Projection onto the moving coordinates shows that the product W_2 is at least the original distance. Pairing each stationary coordinate with itself achieves this lower bound. Thus

$$W_2(\widehat{p}_k^i \otimes \nu_{\mathrm{stat}}, p_{t_k} \otimes \nu_{\mathrm{stat}}) = W_2(\widehat{p}_k^i, p_{t_k}).$$

The stationary coordinates also leave particle error unchanged. Taking ν_{stat} on $\mathbb{R}$ gives a three-dimensional example, and adding more coordinates gives examples in every dimension $d \ge 2$. Independent products of moving planar examples also retain both error orders and strict improvement, since their squared W_2 distances add across components.

A dimension restriction applies if a tangent correction must cancel radial acceleration everywhere on a sphere while keeping the given radial coefficient. For $u(x) = \alpha(r)x + V(x)$ with $x \cdot V(x) = 0$, differentiation in direction V gives $x \cdot (DV)V = -\|V\|^2$. Differentiating the remaining acceleration terms yields

$$x \cdot \mathbf{a}_u(x) = P(r)r^2 - \|V(x)\|^2.$$

Cancellation requires $\|V(x)\|^2 = P(r)r^2 > 0$ at every point of each positive-radius sphere. In odd-dimensional space, such a sphere has even surface dimension, and the hairy-ball theorem requires every continuous tangent field to vanish somewhere [9, Theorem 2.28]. This rules out a tangent correction with the required cancellation at every point. Other fields can still have zero weighted divergence of acceleration, as the example with stationary coordinates shows.

APPENDIX XX
ROBUSTNESS TO AN ACCELERATION RESIDUAL

When only part of acceleration preserves the distribution, we shift the starting point along that part. The remaining acceleration contributes a first-order term to the global error bound. Proposition S29 requires a bound on this remainder at every time, including Euler's evaluation times.

Proposition S29, restated. Let $T > 0$, $p_0 \in \mathcal{P}_2$, and $p_t = (F_{0,t})_{\#}p_0$. Suppose u is C^2 on a neighborhood of $[0,T] \times \mathbb{R}^d$, globally Lipschitz in space with constant L, and bounded at zero uniformly in time. Define $\mathbf{a}$ and $\mathbf{j}$ as in Appendix XVII, and assume $C_j = \sup_t \|\mathbf{j}_t\|_{L^2(p_t)} < \infty$. Suppose

$$\mathbf{a}_t = w_t + g_t, \qquad C_{w,2} = \sup_t \|w_t\|_{L^2(p_t)} < \infty, \qquad \sup_t \|g_t\|_{L^2(p_t)} \le \varepsilon.$$

For each time, require $w_t \in C^1$ to be globally Lipschitz with a common constant L_w and to have zero weighted divergence under p_t. With Euler's method and the grid defined above, put $C_w = C_j/6 + LC_{w,2}/2 + TL_wC_{w,2}/8$. Then

$$\max_k W_2(\widehat{p}_k, p_{t_k}) \le e^{LT}\left(C_w \sum_j h_j^3 + \frac{\varepsilon}{2}\sum_j h_j^2\right) \le e^{LT}T\left(C_w H^2 + \frac{\varepsilon}{2}H\right). \tag{118}$$

Proof of Proposition S29. For $X \sim p_t$, hold physical time fixed and follow the flow of w_t for auxiliary duration $h^2/2$. This motion preserves p_t. The shift has L^2 norm at most $C_{w,2}h^2/2$ and remainder at most $L_wC_{w,2}h^4/8$ after its linear term. It supplies the w_t part of the exact second-order displacement $h^2(w_t + g_t)/2$. The residual and exact-flow remainder stay in the error. The paired outputs therefore have root mean squared distance at most

$$\frac{C_j}{6}h^3 + \frac{LC_{w,2}}{2}h^3 + \frac{L_wC_{w,2}}{8}h^4 + \frac{\varepsilon}{2}h^2.$$

The Lipschitz estimate from the main proof bounds how much later Euler steps increase each error. Summing these contributions and using $\sum_j h_j^3 \le TH^2$ and $\sum_j h_j^2 \le TH$ proves the result. □

A time-dependent residual contributes $\frac{1}{2}\sum_j h_j^2 \varepsilon_{t_j}$. A bound on its time integral cannot control this sum, because the residual may be large at Euler's evaluation times. A small density-weighted divergence norm also need not give the required L^2 decomposition. Weighted L^2 projection separates acceleration into a part with zero weighted divergence and a remainder. This controls the norm, but it does not give the global Lipschitz bound required for the part that preserves the distribution.

The proposition bounds discretization error relative to the exact probability path of the approximate field. That path may differ from the intended path, so we must also estimate the error between the two paths. At the terminal time,

$$W_2(\widehat{p}_T^{\,\mathrm{approx}}, p_T) \le W_2(\widehat{p}_T^{\,\mathrm{approx}}, p_T^{\mathrm{approx}}) + W_2(p_T^{\mathrm{approx}}, p_T).$$

APPENDIX XXI
DEPENDENCE ON THE NUMERICAL SOLVER

Euler and Heun require different rotations to cancel their leading Gaussian distribution errors. We first compare their expansions at a common step length, then pair two corrected Euler steps with one base-field Heun step to compare equal numbers of field evaluations.

Proposition S33 (Euler and Heun for a fixed Gaussian rotation)**.** *Let $T > 0$, $a > 0$, and let $\Sigma_0 = C_0C_0^\top$ be positive definite in even dimension. Let $J^\top = -J$, $J^\top J = I$, $\Omega = C_0JC_0^{-1}$, $m_t = m_0 + t\xi$, and $p_t = \mathcal{N}(m_t, e^{2at}\Sigma_0)$. For a fixed real b, use*

$$v_b(t,x) = \xi + (aI + b\Omega)(x - m_t).$$

On uniform grids with $h = T/N$, Euler satisfies

$$W_2(\widehat{p}_N^{\mathrm{E},b}, p_T) = \sqrt{\operatorname{tr}\Sigma_0}\,e^{aT}\frac{T|a^2 - b^2|}{2}h + O(h^2). \tag{119}$$

Define Heun by

$$x^* = x + hv_b(t,x), \qquad x^+ = x + \frac{h}{2}[v_b(t,x) + v_b(t+h, x^*)].$$

Its terminal error satisfies

$$W_2(\widehat{p}_N^{\mathrm{H},b}, p_T) = \sqrt{\operatorname{tr}\Sigma_0}e^{aT}\frac{T|a(a^2-3b^2)|}{6}h^2 + O(h^3). \tag{120}$$

In particular, Heun's positive leading constant at $b = a$ is twice its constant at $b = 0$. The leading Heun term cancels at $b = \pm a/\sqrt{3}$, giving an $O(h^3)$ bound in this family.

a) Proof idea.: In centered whitened coordinates, each solver acts as a polynomial in $a + \mathrm{i}b$. The real part of the polynomial's logarithm gives the numerical log scale. We use this scale to calculate Gaussian distribution error.

Proof. The exact centered flow $e^{at}C_0e^{btJ}C_0^{-1}$ has the same covariance and mean for every b. Both solvers reproduce the affine mean exactly. For the centered whitened operator $B = aI + bJ$, Euler applies $I + hB$, while Heun applies $I + hB + h^2B^2/2$.

A matrix $cI + dJ$ multiplies whitened covariance by $c^2 + d^2$, the squared complex modulus of $c + \mathrm{i}d$. Thus scalar complex calculations give the distribution error. With $w = h(a + \mathrm{i}b)$, the exact, Euler, and Heun scale factors are e^{ah}, $|1 + w|$, and $|1 + w + w^2/2|$. Writing $\Delta^{\log}$ for exact minus numerical log scale after N steps, we use

$$\log(1 + w) = w - \tfrac{1}{2}w^2 + O(w^3),$$
$$\log(1 + w + w^2/2) = w - \tfrac{1}{6}w^3 + O(w^4).$$

Taking real parts and multiplying by $N = T/h$ shows that

$$\Delta_{\mathrm{E}}^{\log} = \frac{T}{2}(a^2 - b^2)h + O(h^2), \qquad \Delta_{\mathrm{H}}^{\log} = \frac{T}{6}a(a^2 - 3b^2)h^2 + O(h^3).$$

By the Gaussian scale identity, $W_2 = \sqrt{\operatorname{tr}\Sigma_0}e^{aT}|1 - e^{-\Delta^{\log}}|$. Substitution gives the stated errors. For Heun, the absolute cubic coefficients are $a^3/6$ at $b = 0$ and $a^3/3$ at $b = a$, so Euler's correction doubles Heun's leading coefficient. The cubic coefficient instead vanishes at $b^2 = a^2/3$. □

Euler's leading term cancels at $b = \pm a$, or only at $b = a$ if $b \geq 0$ is required. For fixed δ_b with $b = a(1 + \delta_b)$ and $b^2 \neq a^2$, Eq. (119) gives the positive first-order coefficient $\sqrt{\operatorname{tr}\Sigma_0}e^{aT}Ta^2|2\delta_b + \delta_b^2|/2$. Appendix XIX-D gives the corresponding result for nonlinear fields.

For this family, applying the matrix exponential of the full constant centered operator gives the exact solution. An exponential integrator that splits or approximates this operator needs a separate error analysis.

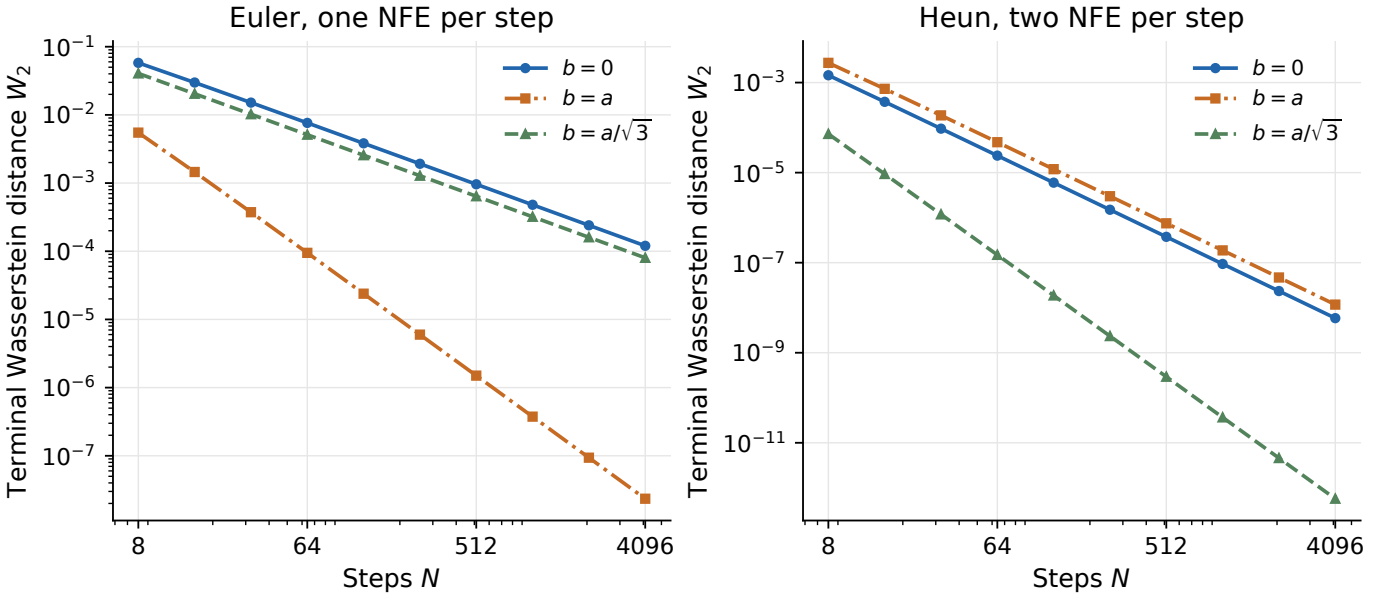


Fig. 13. Euler and Heun require different rotations to cancel their leading distribution errors. Both methods use the Gaussian path and parameters of Figure 7, with the same number of uniform steps. Rotation $b = a$ improves Euler but doubles Heun's leading coefficient relative to $b = 0$. For Heun, $b = a/\sqrt{3}$ cancels the second-order term. Heun uses two evaluations per step, so the methods have unequal NFE here. Figure 11, Table III, and Table V compare equal NFE.

A. Equal NFE for corrected Euler and base-field Heun

Proof of Proposition S28. Let N be even and $h = T/N$. In centered whitened coordinates, two corrected Euler steps of length h have scale multiplier

$$\left(\sqrt{1 + 2ah + 2a^2h^2}\right)^2 = 1 + a(2h) + \tfrac{1}{2}a^2(2h)^2.$$

This is also the positive scale multiplier of one base-field Heun step of length $2h$. Both methods reproduce the affine mean. Their covariances match, so their Gaussian distributions are identical at every shared node. Euler uses one evaluation per step and Heun uses two, giving N evaluations for each method on $[0, T]$. The same argument applies when consecutive Euler steps come in equal-length pairs on an otherwise nonuniform grid. □

B. Corrected Heun and base-field RK3 at equal NFE

At $b = a/\sqrt{3}$, Heun's second-order distribution term cancels and the remaining third-order coefficient is positive. We calculate this coefficient and compare Heun with base-field RK3 using the same number of evaluations. Put $\zeta = a + ib$, $R_2(w) = 1 + w + w^2/2$, and $C_\Sigma = \sqrt{\operatorname{tr}\Sigma_0}e^{aT}$. After $n = T/h$ uniform steps, Heun's errors are exactly

$$W_2^{\mathrm{H},b} = \sqrt{\operatorname{tr}\Sigma_0}\left|e^{aT} - |R_2(h\zeta)|^n\right|, \tag{121}$$

$$e_{\mathrm{p}}^{\mathrm{H},b} = \sqrt{\operatorname{tr}\Sigma_0}\left|R_2(h\zeta)^n - e^{T\zeta}\right|. \tag{122}$$

Here e_{p} is the root mean squared particle error from Section X. The formulas also apply to elliptical covariance, since a map difference $cI + dJ$ in transformed coordinates contributes covariance $(c^2 + d^2)\Sigma_0$ in the original coordinates.

Since $\log R_2(w) = w - w^3/6 + w^4/8 + O(w^5)$, the choice $b = a/\sqrt{3}$ cancels the real cubic term. The remaining coefficients satisfy $\operatorname{Re}\zeta^4 = -8a^4/9$ and $\operatorname{Im}\zeta^3 = 8a^3/(3\sqrt{3})$. Thus

$$W_2^{\mathrm{H},a/\sqrt{3}} = C_\Sigma \frac{a^4T}{9}h^3 + O(h^4), \tag{123}$$

$$e_{\mathrm{p}}^{\mathrm{H},a/\sqrt{3}} = C_\Sigma \frac{4a^3T}{9\sqrt{3}}h^2 + O(h^3). \tag{124}$$

Both coefficients are positive, so distribution error is third order and particle error is second order.

For comparison, Kutta's three-stage, third-order Runge–Kutta method uses

$$k_1 = v(t, x), \quad k_2 = v(t + s/2, x + sk_1/2), \quad k_3 = v(t + s, x - sk_1 + 2sk_2), \quad x^+ = x + s(k_1 + 4k_2 + k_3)/6.$$

On the base field v_0, its centered scale factor is $R_3(as) = 1 + as + (as)^2/2 + (as)^3/6$. Because $\log R_3(z) = z - z^4/24 + O(z^5)$,

$$W_2^{\mathrm{RK3},0} = C_\Sigma \frac{a^4T}{24}s^3 + O(s^4).$$

The base-field Runge–Kutta map is a positive scalar dilation. Pairing particles from the same starting point is therefore optimal, so particle error equals distribution error.

For an evaluation count N divisible by six, corrected Heun takes $N/2$ steps and base RK3 takes $N/3$. Substitution of $h = 2T/N$ and $s = 3T/N$ into the two error expansions proves

$$\lim_{N\to\infty} \frac{W_2^{\mathrm{H},a/\sqrt{3}}}{W_2^{\mathrm{RK3},0}} = \frac{64}{81}.$$

At the finite evaluation counts in Table V, corrected Heun has 18.0–20.9% smaller distribution error, while its particle error is larger.

NFE	Corrected field Heun W_2	Base field, RK3 W_2 and particle error	Corrected field Heun particle error
12	1.677×10^{-4}	2.046×10^{-4}	3.904×10^{-3}
48	2.790×10^{-6}	3.497×10^{-6}	2.581×10^{-4}
192	4.426×10^{-8}	5.588×10^{-8}	1.636×10^{-5}
384	5.547×10^{-9}	7.011×10^{-9}	4.100×10^{-6}

TABLE V

CORRECTED HEUN HAS SMALLER DISTRIBUTION ERROR BUT LARGER PARTICLE ERROR THAN BASE-FIELD RK3 AT THE LISTED EVALUATION COUNTS. THE PARAMETERS ARE THOSE OF FIGURE 7, WITH $b = a/\sqrt{3}$ FOR HEUN AND $b = 0$ FOR RK3. BOTH FIELDS STAY FIXED AS NFE INCREASES.

APPENDIX XXII
RESTRICTIONS ON THE MEAN, SECOND MOMENT, AND ACTION

The acceleration condition requires the mean to move at constant velocity. It also determines kinetic action from the path's second moment. If a path fails these restrictions, no field in the stated class can satisfy the acceleration condition. Other ways to improve numerical accuracy may still be possible.

For Proposition S25, let X_t follow a global flow on $[0,T]$. Assume its trajectories and their first two derivatives are continuous, with $\dot{X}_t = u_t(X_t)$ and $\ddot{X}_t = \mathbf{a}_t(X_t)$. One sufficient condition for the differentiations below is

$$\mathbb{E} \sup_{0\le t\le T} [\|X_t\|^2 + \|u_t(X_t)\|^2 + \|\mathbf{a}_t(X_t)\|^2] < \infty. \tag{125}$$

More generally, require that the second moment can be differentiated twice under the expectation and that $u_t, \mathbf{a}_t \in L^2(p_t)$. If $\mathbf{a}_t$ has zero weighted divergence under the current distribution, then

$$M''(t) = 2\mathbb{E}\|u_t(X_t)\|^2, \qquad M(t) = \mathbb{E}\|X_t\|^2. \tag{126}$$

a) Proof idea.: The second derivative of the second moment contains kinetic energy and an acceleration term. To remove the acceleration term, we approximate squared distance by compactly supported tests and use zero weighted divergence. The remaining identity determines energy. This energy must be at least the infimum over fields satisfying the given continuity equation.

Proof of Proposition S25. Differentiating twice shows that

$$M''(t) = 2\mathbb{E}\|u_t(X_t)\|^2 + 2\mathbb{E}[X_t \cdot \mathbf{a}_t(X_t)].$$

Condition (125) gives an integrable derivative bound, so dominated convergence and Cauchy–Schwarz justify differentiation under expectation. To test zero weighted divergence, replace squared distance by $\chi_n(x)\|x\|^2/2$, using the smooth compact cutoffs of Appendix XXIII. The gradient converges to x in $L^2(p_t)$ as the second-moment tail vanishes. Since $\mathbf{a}_t \in L^2(p_t)$, passage to the limit gives $\mathbb{E}[X_t \cdot \mathbf{a}_t(X_t)] = 0$ and proves Eq. (126).

Let $K_{\min}(t)$ be the infimum of kinetic energy over fields satisfying the weak continuity equation for the path at time t. These competitors need not satisfy the acceleration condition. The chosen field has energy at least this infimum, giving $M''(t) \ge 2K_{\min}(t)$. For a stationary distribution, $M'' = 0$ forces the field to vanish p_t-almost everywhere, or everywhere if the field is continuous and the density is positive throughout space.

For a centered radial dilation $p_t = (g(t)I)_{\#}p_0$, let p_0 be rotationally invariant and suppose $g > 0$ is C^2 and $M_0 = \int \|x\|^2 p_0(\,\mathrm{d}x) > 0$ is finite. The base field is the gradient $v_t(x) = (g'/g)x$. The cutoff argument in Appendix XXIII places this gradient in the required closure. The projection identity then implies

$$K_{\min}(t) = M_0 g'(t)^2, \qquad M''(t) = 2M_0[g'(t)^2 + g(t)g''(t)].$$

When $g''(t) < 0$, the necessary energy inequality fails. No field with the stated regularity and finite energy can then have zero weighted divergence of acceleration. For a scaling path that does admit such a field, its kinetic energy above that of the base field must equal $M_0 g(t)g''(t)$. □

When $g'' = 0$, extra energy must be zero and the base acceleration is zero. If $g'' > 0$, an even-dimensional radial distribution can satisfy this necessary condition through

$$u(t,x) = \frac{g'}{g}x + b(t)Jx, \qquad b(t)^2 = \frac{g''}{g}, \quad J^\top = -J, \quad J^\top J = I.$$

Provided the square root has the required regularity, differentiation gives $\mathbf{a}_u = [b' + 2(g'/g)b]Jx$, which has zero weighted divergence for a radial distribution. Zeros of g'' require an additional check of square-root smoothness. Terminal accuracy can still occur when the local acceleration condition fails, through cancellation across steps (Appendix XVIII).

b) Independent Gaussian interpolation at the midpoint.: Independent interpolation pairs can have different kinetic energy from their conditional-mean field. For independent two-dimensional standard Gaussians X_0, X_1, let $X_t = (1-t)X_0 + tX_1$. At $t = 1/2$, the position depends on $X_0 + X_1$, which is independent of $X_1 - X_0$. The conditional mean velocity is therefore zero, although $\mathbb{E}\|X_1 - X_0\|^2 = 4$. The path remains nonstationary, with variance decreasing before the midpoint and increasing afterwards.

Proof of Corollary S26. The same integrability assumption also lets us differentiate the mean twice under expectation. By Condition (125),

$$m'(t) = \mathbb{E}u_t(X_t), \qquad m''(t) = \mathbb{E}\mathbf{a}_t(X_t).$$

To evaluate the acceleration expectation, choose a smooth compact cutoff χ equal to one near the origin and put $\chi_n(x) = \chi(x/n)$. Testing Eq. (81) with $\phi_n(x) = \chi_n(x)x_j$ gives gradients converging pointwise to e_j, with a bound independent of n. Since $\mathbf{a}_t \in L^2(p_t) \subset L^1(p_t)$, dominated convergence gives $\mathbb{E}\mathbf{a}_{t,j}(X_t) = 0$. Hence $m''(t) = 0$ and $m(t) = m(0) + t\mathbb{E}u_0(X_0)$.

Integrating Eq. (126) gives the action identity in Eq. (98). The moment assumption makes M twice continuously differentiable, with endpoint derivatives taken from inside the interval. Thus the given path determines both the energy at each time and the total action for every field in this class.

If a gradient base field v^∇ generates the given path and satisfies the conditions in Section XIV, Eq. (99) gives

$$\int_0^T \|u_t - v_t^\nabla\|^2_{L^2(p_t)}\,\mathrm{d}t = \tfrac{1}{2}\big[M'(T) - M'(0)\big] - \mathcal{A}_T(v^\nabla).$$

The path therefore determines the additional action within this class. The identity alone does not show that such a field exists.

The assumptions of Theorem 1 also imply the required moment condition. With $U = \sup_t \|u(t,0)\|$, Gronwall's inequality gives

$$\sup_{0\le t\le T}\|X_t\| \le (\|X_0\| + TU)e^{LT}.$$

Continuity gives $C_{a,0} = \sup_t \|\mathbf{a}_t(0)\| < \infty$, and the common Lipschitz bounds give $\|u_t(X_t)\| \le U + L\|X_t\|$ and $\|\mathbf{a}_t(X_t)\| \le C_{a,0} + L_a\|X_t\|$. Each squared time supremum is therefore bounded by $C(1 + \|X_0\|^2)$ for finite C, which is integrable under the source second moment. □

APPENDIX XXIII
MINIMUM ACTION AMONG FIELDS WITH THE SAME PATH

We compare actions by using a gradient base field that can be approximated in mean square by compactly supported gradients. Such a field is orthogonal to every correction that preserves the probability path. For compactly supported potentials, zero weighted divergence gives this orthogonality directly. For the unbounded Gaussian and radial potentials, we use cutoff approximation to obtain the same result in mean square.

We use $\mathcal{V}(p)$ for fields that are jointly continuous, have finite action, and are locally Lipschitz in space uniformly on compact time-space sets. Their flows must exist on $[0,T]$ and generate the given path. The action identity also holds for the larger class of finite-action fields satisfying the same weak continuity equation.

At each time, define the space of gradients and their mean square limits by

$$\mathcal{G}_t = \overline{\{\nabla\psi : \psi \in C_c^\infty(\mathbb{R}^d)\}}^{L^2(p_t)}.$$

Suppose $v_t \in \mathcal{G}_t$ for almost every time and v has finite action. Let r have finite action with $\nabla\cdot(p_t r_t) = 0$ distributionally for almost every time. For every smooth compactly supported potential, $\mathbb{E}_{p_t}[r_t \cdot \nabla\psi] = 0$. By continuity of the inner product in weighted $L^2(p_t)$, we also have $\mathbb{E}_{p_t}[r_t \cdot v_t] = 0$. The cross term is therefore zero when we expand squared velocity and integrate, proving Eq. (99). Equal action requires $r = 0$ almost everywhere under $dt\,p_t(dx)$, which means almost surely in position for almost every time. To obtain equality everywhere, we need the continuity and positivity argument below. Differences between fields satisfying the same continuity equation have the required zero weighted divergence. The existence of their particle flows is a separate requirement.

For the gradient fields used here, a finite second moment is enough to justify approximation by compactly supported gradients. Let $v_t(x) = \nabla\varphi_t(x)$, where φ_t has at most quadratic growth and its gradient has at most linear growth, uniformly on the fixed time interval. Choose a smooth cutoff function χ_n that is one on the radius-n ball, zero outside the radius-$2n$ ball, and has $\|\nabla\chi_n\| \le C/n$. Then

$$\nabla(\chi_n\varphi_t) - \nabla\varphi_t = (\chi_n - 1)\nabla\varphi_t + \varphi_t\nabla\chi_n.$$

The squared approximation error is bounded by a constant times $\mathbb{E}_{p_t}[(1+\|X\|^2)\mathbf{1}_{\{\|X\|>n\}}]$, which vanishes by the finite second moment. For radial dilation, $\|X_t\| = g_t\|X_0\|$ with bounded g_t gives one integrable tail bound for all times. The nonlinear path satisfies the same requirement through $\|X_t\| \le e^{(a+\eta)T}\|X_0\|$ in Eq. (108). Scaling and translating a fixed Gaussian gives the corresponding bound for the Gaussian path. Compactly supported gradients therefore converge in mean square both at each time and after time integration.

When every p_t has positive density throughout space, equal action in $\mathcal{V}(p)$ implies equality everywhere. A continuous difference nonzero at one point stays nonzero on an open time-space neighborhood. Every spatial ball in that neighborhood has positive p_t mass, contradicting almost-everywhere equality under the path. The Gaussian path therefore has a unique minimizer in this continuous class.

A. *Gradient fields under the acceleration condition*

Under all assumptions of Theorem 1, suppose each u_t is a gradient on $\mathbb{R}^d$. We prove that $\mathbf{a}_t = 0$ p_t-almost everywhere and that, for p_0-almost every start, Euler's method is exact on every finite increasing grid.

Proof of Proposition S27. A gradient velocity has a symmetric spatial Jacobian, so material acceleration is also a gradient. The growth bounds let us approximate acceleration by compactly supported gradients in the weighted norm. The acceleration condition makes acceleration orthogonal to all such gradients, so its norm must be zero.

Choose the normalized potential $\psi(t,x) = \int_0^1 u(t,sx)\cdot x\,\mathrm{d}s$. Because u_t is a gradient, $\nabla_x\psi = u_t$. The C^2 regularity of u lets us differentiate in time and space. Symmetry of $D_x u_t$ then implies

$$\mathbf{a}_t = \nabla q_{\mathrm{acc,t}}, \qquad q_{\mathrm{acc,t}} = \partial_t\psi_t + \tfrac{1}{2}\|\nabla\psi_t\|^2.$$

Subtract $q_{\mathrm{acc,t}}(0)$ from this potential. With $C_{a,0} = \sup_{t\in[0,T]}\|\mathbf{a}_t(0)\| < \infty$, the common Lipschitz bound implies

$$|q_{\mathrm{acc,t}}(x)| \le C_{a,0}\|x\| + \tfrac{1}{2}L_a\|x\|^2, \qquad \|\nabla q_{\mathrm{acc,t}}(x)\| \le C_{a,0} + L_a\|x\|.$$

The growth bounds place $\mathbf{a}_t$ in $\mathcal{G}_t$ by cutoff approximation. If the cutoff potentials are not smooth enough for the tests, convolution gives smooth compactly supported potentials whose gradients converge uniformly on a fixed compact set, hence in $L^2(p_t)$. Zero weighted divergence also makes $\mathbf{a}_t$ orthogonal to $\mathcal{G}_t$, so

$$\|\mathbf{a}_t\|^2_{L^2(p_t)} = 0.$$

At each fixed time, acceleration is zero almost everywhere. We need this conclusion along complete trajectories. Choose a countable dense set of times and intersect their full-probability sets of starting points. This intersection still has full probability. By continuity of $t \mapsto \mathbf{a}_t(X_t)$, acceleration is zero at every time along each selected trajectory. Hence $X_t = X_0 + tu_0(X_0)$ and $u_t(X_t) = u_0(X_0)$. Each Euler update from an exact point uses this constant velocity and reaches the next exact point. Euler is therefore exact on every finite increasing grid for the same set of starts. □

APPENDIX XXIV
REGULAR DISPLACEMENT INTERPOLATION

Regular displacement interpolation moves each source point to its target at constant velocity along a straight line. Euler follows these trajectories exactly. Under the assumptions below, the field also minimizes action. Tao and Choi [12, Remark 7] note exactness on straight trajectories. The proposition gives sufficient assumptions on the map and proves exactness at successive grid points.

Proposition S34 (Euler exactness on regular displacement interpolation)**.** *Let $T > 0$ and let p_0 be absolutely continuous with a finite second moment. Let $\mathcal{Q} = \nabla\Phi_{\mathrm{OT}} : \mathbb{R}^d \to \mathbb{R}^d$, where $\Phi_{\mathrm{OT}} \in C^2$, and suppose*

$$0 < \ell_- I \preceq D\mathcal{Q}(x) \preceq \ell_+ I \qquad \textit{for all } x$$

for fixed finite constants ℓ_- and ℓ_+. Define

$$\mathcal{T}_t(x) = (1-t/T)x + (t/T)\mathcal{Q}(x), \qquad p_t = (\mathcal{T}_t)_{\#}p_0,$$

and define the field by

$$u(t,\mathcal{T}_t(x)) = \frac{\mathcal{Q}(x) - x}{T}.$$

The maps $\mathcal{T}_t$ are global diffeomorphisms, which means that they are continuously differentiable bijections with continuously differentiable inverses. The field is jointly continuous, globally Lipschitz in space, and has linear growth. Its trajectories are $X_t = \mathcal{T}_t(X_0)$. On every finite increasing grid, Euler's method satisfies

$$\widehat{X}_k = \mathcal{T}_{t_k}(X_0)$$

for every initial point. It is therefore exact for particles and their distributions at the grid points.

The map $\mathcal{Q}$ is the quadratic-cost optimal map from p_0 to p_T. The displayed path is its displacement interpolation, and

$$\mathcal{A}_T(u) = \frac{W_2^2(p_0,p_T)}{T}.$$

The field also minimizes action among finite-action fields satisfying this fixed continuity equation, up to $dt\,p_t(dx)$-almost-everywhere equality.

Proof. The interpolating maps need global inverses to define the field at every position. With $s = t/T$, the symmetric derivative $D\mathcal{T}_t = (1-s)I + sD\mathcal{Q}$ has eigenvalues between $\min(1,\ell_-)$ and $\max(1,\ell_+)$. The lower bound gives strong monotonicity and injectivity, as well as properness, meaning that inverse images of compact sets are compact. The inverse function theorem makes the image open, while properness makes it closed. Since the image is nonempty and $\mathbb{R}^d$ is connected, the map is onto and has a global differentiable inverse. The uniform derivative bounds and field formula give the stated continuity, spatial Lipschitz continuity, and linear growth.

Each trajectory is affine in time. In particular,

$$\mathcal{T}_{t+h}(x) = \mathcal{T}_t(x) + h\frac{\mathcal{Q}(x) - x}{T}.$$

If Euler reaches the exact point at t_k, its next evaluation gives the same constant particle velocity and its update reaches $\mathcal{T}_{t_{k+1}}(x)$. Induction proves exactness on every grid.

By the convex-gradient characterization, $\mathcal{Q}$ is the quadratic-cost optimal map [4, Section 6.2.3]. Each particle has constant velocity, giving action $\mathbb{E}\|\mathcal{Q}(X_0) - X_0\|^2/T$. Since the map is optimal, this action equals the stated Wasserstein formula.

For minimum action with the path fixed, it remains to place the velocity in the weighted gradient closure. At $y = \mathcal{T}_t(x)$, the spatial derivative is

$$D_y u(t,y) = \frac{1}{T}(D\mathcal{Q}(x) - I)\{(1-s)I + sD\mathcal{Q}(x)\}^{-1}.$$

The matrices are symmetric and commute, so $D_y u$ is symmetric and u_t is a gradient. Linear field growth gives at most quadratic potential growth, allowing cutoff approximation under finite second moments. If a cutoff potential is only C^2, convolution gives smooth potentials with uniformly convergent gradients on a compact support. Thus velocity lies in the weighted gradient closure, and the projection identity proves minimum action on the fixed path. □

These map assumptions are sufficient for both exactness and action minimization. Euler exactness alone needs only a well-defined field at evaluation times along constant-velocity straight paths. For a singular optimal-transport map, the existence of such a field needs a separate argument.

APPENDIX XXV
A SCALED ROTATED GAUSSIAN COUPLING

We scale the rotated Gaussian pairing of Liu [5, Example 3.5] to obtain the same probability path as independent Gaussian interpolation, using dependent endpoints. In even dimension, let $\sigma > 0$, $J^\top = -J$, $J^\top J = I$, and $X_0 \sim \mathcal{N}(0, I)$. With $\mathcal{T}_t = (1-t)I + \sigma t J$ on $[0,1]$, each particle follows a straight line to a scaled rotation of its start. The endpoints are dependent, but $\mathcal{T}_t\mathcal{T}_t^\top = q_{\mathrm{G}}(t)I$ with $q_{\mathrm{G}}(t) = (1-t)^2 + \sigma^2 t^2 > 0$, giving the same Gaussian marginals as independent interpolation. The velocity field is

$$u(t,x) = (\sigma J - I)\mathcal{T}_t^{-1}x = \frac{\sigma^2 t - (1-t)}{q_{\mathrm{G}}(t)}x + \frac{\sigma}{q_{\mathrm{G}}(t)}Jx. \tag{127}$$

The radial term is the conditional-mean gradient velocity of independent Gaussian interpolation, while the rotation preserves each Gaussian marginal. Since $\mathcal{T}_{t+h} = \mathcal{T}_t + h(\sigma J - I)$, an Euler update from an exact point reaches the next exact point. Induction gives particle exactness on every grid, although rotation raises action above the gradient field by $d\sigma^2 \int_0^1 q_{\mathrm{G}}(t)^{-1}\,\mathrm{d}t > 0$.

For an anisotropic endpoint CX_0 with covariance $\Sigma_1 = CC^\top$, this deterministic linear construction matches the independent-interpolation path exactly when $C + C^\top = 0$. This follows by comparing the $t(1-t)$ covariance terms. An invertible real skew-symmetric matrix has two-dimensional blocks, so each eigenvalue of $CC^\top$ has even multiplicity. Conversely, equal eigenvalues can be paired into blocks $\sqrt{\mu}J$. The restriction applies only to this linear coupling class.

APPENDIX XXVI
LIMITS OF ACTION COMPARISONS AND EVALUATION SCHEDULES

Our examples use fields with positive additional action. In a larger class of fields, the same numerical improvement can require arbitrarily little additional action. Euler uses the field only at its evaluation times, while action integrates squared velocity over time. A continuous correction can keep the same values at evaluation times and vanish outside short intervals around them.

For the Gaussian probability path in Appendix XVI, fix a finite increasing grid and $a > 0$. At step j, set

$$b_j = h_j^{-1}\sqrt{e^{2ah_j} - (1+ah_j)^2}.$$

The inequality $e^{ah_j} > 1 + ah_j$ makes b_j real and positive, with exact Euler covariance factor $(1+ah_j)^2 + b_j^2 h_j^2 = e^{2ah_j}$. Set $b(t_j) = b_j$ at each evaluation time and extend these values by continuous functions supported in disjoint intervals of width at most δ. Their sum b_δ is bounded and continuous, giving a global affine flow that preserves the Gaussian path. Shrinking the intervals keeps every Euler evaluation and output unchanged, while the added action satisfies

$$\mathrm{tr}(\Sigma_0)\int_0^T b_\delta(t)^2 e^{2at}\,\mathrm{d}t \longrightarrow 0 \quad \text{as } \delta \downarrow 0.$$

The grid is fixed and has finitely many evaluation times. The function heights are finite, so the action integral tends to zero as the total support length decreases. This gives the same numerical improvement with arbitrarily small additional action. The fields have no common bound on time derivatives.

APPENDIX XXVII
NUMERICAL SPECIFICATION

We calculate distances between full probability distributions, without using sample estimates. For the Gaussian examples, recursive matrix updates check the distance formulas and particle maps. For the nonlinear example, optimal radial transport gives the distances. We use independent reference calculations to check the nonlinear distances and their leading coefficients.

A. Gaussian checks

We evaluate the scalar Gaussian formulas at 80 decimal digits. We check the particle maps, covariances, and means against recursive Euler matrix updates in binary64 double precision. Each update starts from the preceding numerical result, with the mean moving at constant velocity. To calculate the Gaussian Wasserstein distance, we optimally align the covariance square roots by an orthogonal map. This avoids subtracting nearly equal trace terms. Quadrature along the exact flow gives an independent check of continuous action.

The nonuniform grid is $(0, 0.04, 0.17, 0.43, 0.72, 1)$.

The Gaussian calculation uses $a = 0.6$, $T = 1$, and the fixed coefficients $b = 0$ and $b = a$. In dimension two, the covariance is

$$\Sigma_0 = \begin{pmatrix} 1.5 & 0.25 \\ 0.25 & 0.75 \end{pmatrix}. \tag{128}$$

Uniform step counts are the powers of two from one through 4096. Dimensions four and eight use $1, 4, 32$ uniform steps. All three dimensions also use the nonuniform grid above. For each higher dimension d, set $\Sigma_0 = C_0 C_0^\top$, where

$$(C_0)_{ii} = 0.7 + 0.9\frac{i-1}{d-1}, \qquad (C_0)_{i,i-1} = 0.12 \ (i \geq 2),$$

and all other entries vanish. For every dimension, the mean and drift components are

$$(m_0)_i = -0.4 + \frac{i-1}{d-1}, \qquad \xi_i = 0.2 - 0.5\frac{i-1}{d-1}.$$

The matrix J uses consecutive two-dimensional quarter-turn blocks. We compare terminal squared distribution error, its maximum over grid points, and its sum over grid points. We also calculate error between particles from the same starting point and check the positive leading error coefficients in Appendix XVI.

The contraction control uses $a = b = -0.6$, the stated two-dimensional covariance, and one step on $[0, 1]$, with algebraic zeros recorded separately from floating-point residuals.

B. Nonlinear example and reference calculations

The nonlinear calculation uses $a = 0.6$, $\eta = 0.3$, $T = 1$, and source $0.6\mathcal{N}(0, 0.5^2 I_2) + 0.4\mathcal{N}(0, 1.5^2 I_2)$. The uniform step counts are powers of two from 8 through 4096. We also use the same unequal grid as above. For a spherical Gaussian component of scale s, $R_0^2/(2s^2)$ has density e^{-z} on $z \geq 0$. We therefore evaluate radius expectations by Gauss–Laguerre quadrature with 64, 128, or 256 nodes per component.

For the first reference calculation, we integrate log radius, angle, action, and the coefficient equations over time. We use SciPy 1.17.0's DOP853 ordinary differential equation solver, with primary relative and absolute tolerances 2×10^{-13} and 2×10^{-14}. For an independent reference, we separate variables in the radius equation. With $r > 0$, $a_+ = a + \eta$, and $\ell = \log(R(T, r)/r)$, this gives the implicit equation

$$\frac{\ell}{a} - \frac{\eta}{2aa_+} \log \frac{a + a_+ r^2 e^{2\ell}}{a + a_+ r^2} = T, \qquad aT \leq \ell \leq a_+ T.$$

This equation comes from integrating $1/f(r)$ between the source and terminal radii. Once we find the terminal radius, we integrate over radius to calculate angle, action, and leading coefficients. This calculation is independent of the time integration. Differentiating $\log f(R(s, r))$ gives $V(T, s; r) = f(R(T, r))/f(R(s, r))$ for $r > 0$, so

$$\mathcal{C}_i(T, r) = f(R(T, r)) \int_r^{R(T,r)} \frac{d_i(v)}{f(v)^2} \, \mathrm{d}v.$$

To separate quadrature error from reference error, we first increase the number of nodes with tolerances fixed. We then tighten the tolerances with the number of nodes fixed. The supplementary code includes all tolerances and comparison criteria.

We also compare independent particle updates in Cartesian and radius-angle coordinates, using source radii $(0.07, 0.4, 1, 3, 8)$ and angles $(0, 0.3, 1, 2.1, 4)$.

C. Solver, sensitivity, and interpolation checks

The solver calculations use $b = 0$, $b = a$, and $b = a/\sqrt{3}$ on the same grids. We check the scalar stability factors against separate matrix updates. To study Euler's sensitivity, we use fixed ratios $b/a \in \{0, 0.9, 0.98, 1, 1.02, 1.1\}$. The interpolation calculation uses $\sigma = 1.5$. Its corrected field has zero error algebraically, and we record floating-point residuals separately. These computations check the numerical evaluation of the formulas. The proofs give the convergence orders and strict comparisons.

D. Equal NFE

For the nonlinear equal-NFE comparison, we keep the same source and coefficients and use $N \in \{2^j : j = 1, \dots, 12\}$ evaluations. Euler takes N uniform steps, while base-field Heun takes $N/2$. We calculate distances using 128 Gauss–Laguerre nodes per component and compare them with results from 64 and 256 nodes. All methods use the implicit exact-radius reference. We check this reference independently by time integration at relative tolerance 2.5×10^{-14} and absolute tolerance 3×10^{-15}.

For base-field Heun, put $f(r) = \alpha(r)r$ and let $s = 2T/N$. One radius step is

$$H_s(r) = r + \tfrac{s}{2}\{f(r) + f(r + sf(r))\}.$$

Since $f' > 0$, its derivative is $1 + \frac{s}{2}\{f'(r) + f'(r + sf(r))[1 + sf'(r)]\} > 0$. The derivative is positive, so the radius map is increasing. The update also preserves angle. Pairing particles from the same starting point therefore matches angles and ordered radii. This pairing is optimal, so particle error equals W_2. We check all three methods with separate Cartesian updates. The smaller measured distribution error of corrected Euler applies to this source and these parameters.

Increasing quadrature from 128 to 256 nodes changes the errors by less than 1.8×10^{-7} relatively, while the independent reference discrepancy is below 2.6×10^{-8} of the corresponding numerical error.

The Gaussian Heun–RK3 comparison uses the parameters of Figure 7 at $12, 48, 192, 384$ field evaluations. We evaluate the complex polynomials at 50 and 80 decimal digits and check them against independent matrix-stage recurrences at 70 and 100 digits. The extra precision controls error from subtracting nearby values in the distance calculation.

E. Measured errors and refinement results

Table VI gives the errors on the finest grid and order estimates $\log_2(e_{2048}/e_{4096})$, where e_N is error after N steps. We calculate the leading coefficients from independent radius integrals, without fitting the error curves. The primary calculation uses 128 nodes per mixture component. Increasing this to 256 changes every reported terminal error, action, and coefficient by less than 8×10^{-8} relatively. The largest reference discrepancy is below 3.2×10^{-7} of the corresponding Euler error.

TABLE VI

THE NONLINEAR EXAMPLE HAS SECOND-ORDER CORRECTED DISTRIBUTION ERROR, ALTHOUGH PARTICLE ERROR REMAINS FIRST ORDER. THE CALCULATION USES THE SOURCE AND PARAMETERS OF FIGURE 10, 4,096 UNIFORM STEPS, AND 128 QUADRATURE NODES PER MIXTURE COMPONENT. OBSERVED ORDER $\log_2(e_{2048}/e_{4096})$ MEASURES THE ERROR REDUCTION WHEN THE STEP COUNT DOUBLES, WHERE e_N IS ERROR AFTER N STEPS. THE LEADING COEFFICIENTS ARE LIMITS OF ERROR DIVIDED BY h, h^2, AND h IN THE THREE ROWS.

Quantity	Error	Observed order	Leading coefficient
Base distribution	3.391×10^{-4}	0.9997	1.3894
Corrected distribution	9.687×10^{-8}	1.9995	1.6258
Corrected particle	6.551×10^{-4}	0.9998	2.6835

The base and corrected actions are approximately 4.2342 and 8.6716.

The Gaussian sensitivity calculation uses up to 8,192 steps to resolve the first-order limit for a 2% coefficient mismatch. At this step count, all displayed scaled coefficients are within 0.5% of their asymptotic predictions. We check the smallest Heun errors with independent matrix recurrences at 50 and 80 digits, including values below binary64 resolution. At 4,096 steps, the error ratio between $b = a$ and $b = 0$ is 1.9998, consistent with the limiting ratio two.